Local realizability and global completion of constant-ℓ rotating dust

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We study local realizability and global completion of stationary, axisymmetric rotating dust with a constant logarithmic derivative of the metric potential. An exact point transformation reduces the velocity equation to a linear elliptic equation. This map has a Whitney fold at the lower density boundary, while positive density selects a regular inverse sheet. Flat equatorial rotation curves admit local analytic spacetime realisations on admissible annuli. Their continuation is constrained by a condition on metric derivatives at the axis and by quantitative radial-span bounds under an invariant transverse-concavity hypothesis. These bounds are sharp for nonpositive values of the defining constant and, in the exactly flat limit, for positive values. For a nonzero defining constant, a uniformly subluminal branch along an asymptotically flat equatorial end has inverse-radius velocity decay with a positive leading coefficient. Nonnegative density near the equatorial tail then forces a locally flat vacuum region. For finite bodies, toroidal topology imposes period conditions for globally single-valued exterior potentials, although metric regularity permits a nonzero twist period. Completion without a surface layer requires the full Darmois matching data. We derive a compatibility condition for the gauge-fixed linearised Ernst equations and identify the additional data needed for spectroscopic prediction. Together, these results clarify the conditions governing the passage from local rotation profiles to a complete isolated spacetime.

I. INTRODUCTION

Nearly flat orbital-speed profiles in the outer regions of disk galaxies are a central feature of the galactic mass discrepancy [1]. Their standard interpretation invokes a substantial non-luminous matter component [2]. Relativistic models of rotating matter provide a setting in which to examine how the inferred mass distribution depends on spacetime geometry, differential rotation and the choice of observer. Connecting local kinematics to a complete gravitational configuration requires a solution of the field equations with a physically admissible matter distribution and a consistent extension to the axis or an exterior region. Comparison with observations also requires frequency shifts for specified emitters, observers and connecting null geodesics. These constraints make the relation between local realizability and global completion particularly important for pressureless rotating dust. The stationary, axisymmetric dust family described in [3, 4] can be parametrized by a scalar field $\eta(r, z)$ and a negative function $H(\eta)$. The speed measured by a zero-angular-momentum observer (ZAMO) is $v = \eta/r$, while differential rotation is encoded by $\ell = H_{,\eta}/H$. Balasin and Grumiller [5] used the rigidly rotating sector of this family in a galactic model, while Astesiano et al. [6] subsequently developed a differential-rotation construction and considered a constant value of ℓ near a galactic edge. The latter construction combines a low-velocity truncation with a near-edge replacement of the radial coefficients. Its near-edge equation admits a nonzero constant speed, motivating an examination of the corresponding exact constant-ℓ branch. The relation between this approximation and the parent equations depends, in particular, on how the prescribed equatorial profile is extended in the transverse direction.

Global restrictions on rotating dust have a substantial history. Zingg, Aste and Trautmann [7] derived finite-radius density constraints for the general non-rigid Winicour family and examined the simultaneous prescription of metric and normal data at a vacuum interface. Their analysis established the importance of treating matter support and exterior matching as linked parts of the same problem. Caporali's global result, Pfister's conditional analysis and the singularity findings for rigid van Stockum-Bonnor configurations provide further precedents [8–10]. We build on this foundation within the constant-ℓ subclass by deriving an explicit density interval, describing the geometry of the reconstruction map and establishing quantitative radial-span estimates. A complementary literature examines the magnitude and interpretation of relativistic effects in

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galactic dynamics. Calculations for physically sourced weak-field disks quantify the small gravitomagnetic corrections and emphasize the importance of consistent source modelling and approximation schemes [11–13]. Analyses of the rigidly rotating model relate its frame dragging to the asymptotic reference frame and axial structure [14]. Broader studies of stationary geometry and galactic observables provide further constraints [15, 16]. Our analysis complements these studies by following the exact constant-ℓ branch from a prescribed equatorial trace through density admissibility, geometric regularity and exterior boundary data.

The central question is whether a constant-ℓ branch can realize an exactly or approximately flat equatorial speed profile and support its continuation to a regular isolated spacetime. Because an equatorial trace does not fix the transverse derivatives, these derivatives remain available to satisfy the field equations. This freedom is essential to local realizability. Direct substitution of a spatially constant nonzero speed produces a residual in both the exact and truncated equations, while suitable transverse dependence permits local realizations of a flat equatorial trace. We begin with the exact linearizing potential and examine the inverse transformation needed to recover the physical variables. The density inequality determines the admissible range, whose lower boundary corresponds to a fold of the point transformation. Positive density selects a regular inverse sheet for reconstructing local spacetime metrics from admissible analytic equatorial data. We prove the existence of analytic, positive-density realizations of exactly flat traces on admissible annuli and establish analyticity on regular inverse sheets. The auxiliary elliptic Cauchy problem nevertheless exhibits Hadamard instability, so this analyticity does not ensure continuous dependence on smooth equatorial Cauchy data. The continuation analysis identifies geometric conditions that restrict this local freedom. Under the stated regularity and timelikeness assumptions, a metric argument at an axis of an effective rotational symmetry forces $H'(0) = 0$. Thus, a constant-ℓ branch can reach such an axis only if $\ell = 0$. For reflection-symmetric solutions on annuli, nonnegative density and an invariant transverse-concavity condition for the relative ZAMO rapidity yield quantitative bounds on the radial span of exactly or approximately flat traces. These estimates quantify how the permitted departure from flatness constrains the radial extent under the stated concavity hypothesis. We establish sharpness for nonpositive ℓ and for positive ℓ in the exactly flat limit. For a regular branch with nonzero constant ℓ extending along a standard asymptotically flat equatorial end, uniform subluminality and the exact metric relations force rv to approach a positive constant. Nonnegative density on open neighbourhoods of the equatorial tail then forces a locally flat vacuum region. Analytic continuation then implies that η is constant throughout a connected regular inverse sheet containing that region. For a finite body, completion requires an exterior realizing the boundary geometry supplied by the interior. If its meridional section is disk-shaped and separated from the axis, the resulting toroidal topology imposes period conditions for globally single-valued exterior scalar potentials. The twist period represents a matter-current flux and need not vanish for the metric to be globally regular. At the interface, matching without a surface layer requires the full Darmois data [17–19]. These conditions must still be checked when the density vanishes at the boundary. We derive a further compatibility condition from the linearized Ernst equations about Minkowski space on a fixed smooth bounded domain away from the axis. In a specified gauge, the associated Dirichlet-to-Neumann map characterizes the admissible pairs of boundary values and conormal derivatives. This provides a concrete test within the linearized problem, while the construction of a nonlinear isolated exterior remains a separate existence question. Finally, we express the observational prediction in terms of the emitter-observer frequency ratio. The local ZAMO speed enters this relation together with the emitter and observer motions and the conserved quantities of the connecting null ray. These dependencies specify the additional information needed to convert a locally realized rotation profile into a spectroscopic prediction.

The manuscript is organized as follows. Sections II–V develop the local equations and continuation bounds, providing the foundation for the analysis of asymptotic and finite completion in Sections VI–VIII. Section IX then relates the geometry to spectroscopic data. Appendices A–G present the Runge approximation and reconstruction procedure, an alternative axis proof, an explicit flat-trace solution and additional radial-span estimates and sharpness constructions. They also compare the results with the truncated field equations, describe the junction formulas and hypotheses for nonlinear exterior matching and formulate the spectroscopic prediction problem in detail.

II. THE EXACT CONSTANT-ℓ EQUATIONS

We use signature $(-,+,+,+)$, units $c = 1$ and coordinates (t, ϕ, r, z) adapted to the stationary and axial Killing fields $k = \partial_t$ and $m = \partial_\phi$. The axial orbits have period 2π. In the circular rotating-dust family

considered here, the line element is

$$ds^2 = G_{AB}dx^A dx^B + e^{\mu}(dr^2 + dz^2), \qquad G = (g_{AB})_{A,B\in\{t,\phi\}} \tag{II.1}$$

and the metric coefficients depend only on (r, z). We choose the canonical orbit-area radius $r = \sqrt{-\det G}$, so that $\det G = -r^2$. This coordinate choice is part of the exact dust representation adopted in [3, 6]. Moreover, the representation is specified by a scalar field $\eta = \eta(r, z)$ and a negative function $H = H(\eta)$ with angular velocity $\Omega = \Omega(\eta)$. The matter tensor is $T_{\mu\nu} = \rho u_\mu u_\nu$ with circular four-velocity

$$u = \frac{k + \Omega m}{\sqrt{-H}}, \qquad \frac{d\Omega}{d\eta} = \frac{H_{,\eta}}{2\eta}. \tag{II.2}$$

The defining relations $H = g(k + \Omega m, k + \Omega m) < 0$ and $\eta = g(k + \Omega m, m)$ imply $g(u, u) = -1$. In the basis $(k + \Omega m, m)$, the determinant condition becomes $Hg_{\phi\phi} - \eta^2 = -r^2$, from which the Killing-block coefficients follow, i.e.

$$g_{\phi\phi} = \frac{r^2 - \eta^2}{-H}, \qquad g_{t\phi} = \eta - \Omega g_{\phi\phi}, \qquad g_{tt} = H - 2\Omega\eta + \Omega^2 g_{\phi\phi}. \tag{II.3}$$

The remaining metric function μ is determined by the quadratures (IV.5)-(IV.6) in the local reconstruction below. For the later matching analysis, we use the convention $K_{ab} = h_a{}^\mu h_b{}^\nu \nabla_\mu n_\nu$ for a hypersurface with spacelike unit normal n, where h is the induced metric. On a region where $g_{\phi\phi} > 0$, a ZAMO has four-velocity $N^{-1}(k + \Omega_{\rm Z} m)$, with positive lapse N and angular velocity $\Omega_{\rm Z}$ given by

$$\Omega_{\rm Z} = -\frac{g_{t\phi}}{g_{\phi\phi}}, \qquad N^2 = -g_{tt} + \frac{g_{t\phi}^2}{g_{\phi\phi}} = \frac{r^2}{g_{\phi\phi}}. \tag{II.4}$$

Since $\Omega - \Omega_{\rm Z} = \eta/g_{\phi\phi}$, the dust speed relative to the ZAMO and the logarithmic derivative of H are

$$v = \frac{\sqrt{g_{\phi\phi}}(\Omega - \Omega_{\rm Z})}{N} = \frac{\eta}{r}, \qquad \ell(\eta) = \frac{H_{,\eta}}{H}. \tag{II.5}$$

We work locally on an open set $D \subset (0, \infty) \times \mathbb{R}$ in the meridional coordinates (r, z), with $v \in C^2(D)$ and $0 < v < 1$ throughout D. These inequalities imply $0 < \eta(r, z) < r$ and ensure $g_{\phi\phi} > 0$ because $H < 0$. The speed v is dimensionless, so η has dimensions of length and ℓ has dimensions of inverse length. For $\ell \equiv \ell_0 \in \mathbb{R}$, integration of $H_{,\eta} = \ell_0 H$ gives $H = -Ce^{\ell_0\eta}$ with $C > 0$. When $\ell_0 \neq 0$, integrating (II.2) yields

$$\Omega(\eta) = \Omega_* - \frac{C\ell_0}{2}\,\mathrm{Ei}(\ell_0\eta) \tag{II.6}$$

on an interval with $\eta > 0$, where Ei denotes the real exponential integral and satisfies $d\,\mathrm{Ei}(\ell_0\eta)/d\eta = e^{\ell_0\eta}/\eta$. For $\ell_0 = 0$, the angular velocity is a constant Ω_0. The constants are fixed by normalization and boundary data. With $\Delta_{rz} v = v_{rr} + v_{zz}$ and $|\nabla_{rz} v|^2 = v_r^2 + v_z^2$, the exact velocity equation is $\mathcal{E}_{\ell_0}[v] = 0$, where

$$\mathcal{E}_{\ell_0}[v] = \left[2 + \ell_0 r\left(\frac{1}{v} - v\right)\right]\Delta_{rz} v + \left[\frac{2}{r} + 3\ell_0\left(\frac{1}{v} - v\right)\right] v_r - \ell_0 r\left(\frac{1}{v^2} + 1\right)|\nabla_{rz} v|^2 + \frac{2\ell_0}{r} - \frac{2v}{r^2}. \tag{II.7}$$

For comparison, the low-velocity truncation in Eq. (4.3) of [6] is $\mathcal{T}_{\ell_0}[v] = 0$ with

$$\mathcal{T}_{\ell_0}[v] = \left(2 + \frac{\ell_0 r}{v}\right)\Delta_{rz} v + \left(\frac{2}{r} + \frac{3\ell_0}{v}\right) v_r - \frac{\ell_0 r}{v^2}|\nabla_{rz} v|^2 + \frac{2\ell_0}{r} - \frac{2v}{r^2}. \tag{II.8}$$

The exact difference is $\mathcal{E}_{\ell_0}[v] - \mathcal{T}_{\ell_0}[v] = -\ell_0\left(rv\Delta_{rz} v + 3vv_r + r|\nabla_{rz} v|^2\right)$. The subsequent near-edge equation in Eq. (4.5) of that reference also replaces the radial coefficients. All results below concern the exact equation unless the truncated equation is expressly specified. To verify the reduction of the exact velocity equation to a linear elliptic equation, we introduce an auxiliary potential and then examine its local inversion. The following expression is the constant-ℓ specialization of the auxiliary representation in Eq. (2.10) of [6]. For a fixed reference scale $\eta_* > 0$ with the same dimensions as η, we set

$$\mathcal{F} = \mathcal{F}(r, \eta) := 2\eta + \ell_0 r^2 \log\frac{\eta}{\eta_*} - \frac{\ell_0}{2}\eta^2. \tag{II.9}$$

Differentiating (II.9) with respect to η at fixed r, before substituting $\eta = \eta(r,z)$, gives the coefficient of $\Delta_{rz}v$ in (II.7),

$$\partial_\eta \mathcal{F} = 2 + \frac{\ell_0 r^2}{\eta} - \ell_0 \eta = 2 + \ell_0 r \left(\frac{1}{v} - v\right), \tag{II.10}$$

where we used the relation $v = \eta/r$. In the spatial derivatives below, $\mathcal{F}$ denotes the composite field $(r,z) \mapsto \mathcal{F}(r,\eta(r,z))$, whereas $\partial_\eta \mathcal{F}$ continues to denote the derivative of the defining function at fixed r. For that defining function, $\partial_{r\eta}\mathcal{F} = 2\ell_0 r/\eta$ and $\partial_{\eta\eta}\mathcal{F} = -\ell_0(r^2/\eta^2 + 1)$, while $(\partial_r^2 - r^{-1}\partial_r)\mathcal{F} = 0$ when η is held fixed. The chain rule therefore gives

$$\mathcal{F}_{rr} - \frac{\mathcal{F}_r}{r} + \mathcal{F}_{zz} = \left(\Delta_{rz}\eta - \frac{\eta_r}{r}\right)\partial_\eta \mathcal{F} + \frac{4\ell_0 r}{\eta}\eta_r - \ell_0\left(\frac{r^2}{\eta^2} + 1\right)|\nabla_{rz}\eta|^2. \tag{II.11}$$

Substituting $\eta = rv$, with $\eta_r = v + rv_r$, $\eta_z = rv_z$ and $\Delta_{rz}\eta = r\Delta_{rz}v + 2v_r$, yields the exact identity

$$\mathcal{L}\mathcal{F} = r\mathcal{E}_{\ell_0}[v], \qquad \mathcal{L} = \partial_r^2 - \frac{1}{r}\partial_r + \partial_z^2. \tag{II.12}$$

This identity holds for every positive $v \in C^2(D)$, including at points where $\partial_\eta \mathcal{F} = 0$. Since $r > 0$, the velocity equation is equivalent to $\mathcal{L}\mathcal{F} = 0$ for the associated composite field. Conversely, near a point satisfying (II.9) with $\partial_\eta \mathcal{F} \neq 0$, the analytic implicit-function theorem recovers η as a locally analytic function of r and the auxiliary value. Composing this local inverse with an auxiliary solution then yields $v = \eta/r$ satisfying the exact velocity equation wherever the composition is defined. Finally, the operator $\mathcal{L}$ is elliptic for $r > 0$. Its divergence form

$$\mathcal{L}f = r\left[\partial_r\left(r^{-1}\partial_r f\right) + \partial_z\left(r^{-1}\partial_z f\right)\right], \tag{II.13}$$

shows by integration by parts that it is formally self-adjoint with respect to the measure $r^{-1}drdz$ on compactly supported smooth test functions. Notice that replacing η_* by $a\eta_*$ with $a > 0$ and η fixed, changes $\mathcal{F}$ to $\mathcal{F} - \ell_0 r^2 \log a$. Since $\mathcal{L}(r^2) = 0$, this leaves $\mathcal{L}\mathcal{F}$ unchanged. The scale η_* thus fixes an auxiliary normalization and any prescribed auxiliary data must be shifted consistently to represent the same velocity field.

III. DENSITY AND THE ADMISSIBLE INVERSE SHEET

The auxiliary equation provides a local construction wherever the transformation from the velocity to the potential admits a regular inverse. Physical admissibility further requires nonnegative dust density. We first derive the density bounds and identify the lower boundary at which the point transformation has a fold. We then establish analyticity on regular inverse sheets and examine the sensitivity of auxiliary elliptic continuation to perturbations of the prescribed data. The consequences for local flat traces and extension to a regular axis are considered in Section IV.

A. Constant fields and density bounds

A spatially constant speed already illustrates the distinction between prescribing local data and solving the field equation on an open region.

Proposition 1. *Let $I \subset (0,\infty)$ and $J \subset \mathbb{R}$ be nonempty open intervals. For constant ℓ_0, no field $v(r,z) = v_0 \neq 0$ solves the exact or truncated constant-ℓ equation on $I \times J$.*

Proof. Since all derivatives of v_0 vanish, $\mathcal{E}_{\ell_0}[v_0] = \mathcal{T}_{\ell_0}[v_0] = 2r^{-2}(\ell_0 r - v_0)$. If this residual vanished throughout I, the identity $\ell_0 r = v_0$ would hold at two distinct radii, forcing $\ell_0 = 0$ and then $v_0 = 0$, contrary to the hypothesis. □

When $\ell_0 \neq 0$ and $v_0/\ell_0 > 0$, the residual vanishes only on the cylinder $r = v_0/\ell_0$. In particular, the calibration $v_0 = av_c = \ell_0 R_G$ used in [6] selects $r = R_G$. Here, v_c and R_G are reference speed and length scales, respectively, and a is dimensionless. Prescribing only the equatorial trace $v(r,0) = v_0$ leaves the transverse curvature $v_{zz}(r,0)$ available to satisfy the equation. Theorem 4 establishes the local realization of such data.

The density bounds do not require ℓ to be constant. At a regular point with $r > 0$, $0 < v < 1$ and $\eta = rv$, the density formula (see Eq. (2.6) in [6] reads

$$\rho = \frac{\eta_r^2 + \eta_z^2}{32\pi G_N \eta^2 g_{rr}} \left[\frac{\eta^2}{r^2}(2 - \eta\ell)^2 - r^2\ell^2 \right] \tag{III.1}$$

Here, $G_N > 0$ is Newton's constant and $g_{rr} = e^\mu > 0$. Writing $x = r\ell$, we express (III.1) as

$$\rho = \frac{(\eta_r^2 + \eta_z^2)\mathcal{N}_v(x)}{32\pi G_N \eta^2 g_{rr}}, \qquad \mathcal{N}_v(x) = 4v^2 - 4v^3 x - (1 - v^4)x^2. \tag{III.2}$$

The following result translates the requirement of nonnegative dust density into explicit bounds on the variables governing the inverse transformation.

Proposition 2. *At a regular point with $r > 0$, $0 < v < 1$ and finite $\ell(\eta)$, the algebraic factor $\mathcal{N}_v(x)$ is nonnegative precisely when*

$$-\frac{2v}{1 - v^2} \leq x \leq \frac{2v}{1 + v^2}. \tag{III.3}$$

At a point where $\nabla_{rz}\eta \neq 0$, condition (III.3) is equivalent to $\rho \geq 0$ and its strict form is equivalent to $\rho > 0$. At either endpoint, $\rho = 0$ irrespective of $\nabla_{rz}\eta$.

Proof. The factorization

$$\mathcal{N}_v(x) = [2v - (1 + v^2)x][2v + (1 - v^2)x] \tag{III.4}$$

has the two roots $x_\pm = \pm 2v/(1 \pm v^2)$. Its leading coefficient $-(1 - v^4)$ is negative and $x_- < 0 < x_+$. Hence, it is nonnegative exactly on $[x_-, x_+]$ and positive in the interior. The remaining factor in (III.2) is positive exactly when $\nabla_{rz}\eta \neq 0$. □

At a critical point of η, the density vanishes independently of $\mathcal{N}_v$. Continuity nevertheless restricts how the interval condition can fail within an open region of nonnegative density, as the following corollary shows.

Corollary 1. *Let U be an open regular region on which $0 < v < 1$, $\rho \geq 0$ and the factors in the first equation in (III.2) are continuous. If (III.3) fails at $p \in U$, then p has a connected open neighbourhood on which η is constant and $\rho = 0$. Hence, the density interval holds throughout U if η is not constant on any nonempty open subset of U.*

Proof. If the interval condition fails at p, continuity ensures that $\mathcal{N}_v(x) < 0$ on a sufficiently small connected neighbourhood of p. Since the density is nonnegative, the density formula forces $\nabla_{rz}\eta = 0$ throughout this neighbourhood. It follows that η is constant and ρ vanishes there. □

For an exactly flat trace the critical-point exception disappears.

Corollary 2. *Suppose $v(r,0) = v_0 \in (0,1)$ on an interval and $\ell = \ell_0$ is constant. At regular trace points, nonnegative density is equivalent to*

$$0 < r \leq R_\rho^{(+)} := \frac{2v_0}{\ell_0(1 + v_0^2)} \qquad \text{if } \ell_0 > 0, \tag{III.5}$$

$$0 < r \leq R_\rho^{(-)} := \frac{2v_0}{|\ell_0|(1 - v_0^2)} \qquad \text{if } \ell_0 < 0. \tag{III.6}$$

For $\ell_0 \neq 0$, the density vanishes at the stated radius and is negative at regular points of the prescribed trace beyond that radius, whenever such points exist. For $\ell_0 = 0$, ρ is positive at every regular trace point.

Proof. Along the trace, $\eta(r,0) = rv_0$, so $\eta_r(r,0) = v_0 \neq 0$. Thus, $\nabla_{rz}\eta$ cannot vanish there and Proposition 2 applies without the critical-point exception. Since $r > 0$, the lower inequality in (III.3) is satisfied for $\ell_0 > 0$, while the upper inequality gives (III.5). For $\ell_0 < 0$, the upper inequality is satisfied and the lower one gives (III.6). The factor $\mathcal{N}_{v_0}(r\ell_0)$ vanishes at the respective endpoint and is negative beyond it, which gives the stated density signs wherever the trace remains regular. When $\ell_0 = 0$, the identity $\mathcal{N}_{v_0}(0) = 4v_0^2 > 0$ gives strictly positive density at every regular trace point. □

These conclusions require no restriction on the transverse derivatives. Moreover, the radii $R_\rho^{(\pm)}$ locate the change in the density sign along the prescribed trace. If a material boundary is present, it must also satisfy the junction conditions in Section VII.

B. The lower density boundary and analytic reconstruction

For constant ℓ_0, equation (II.10) gives $v\partial_\eta\mathcal{F} = 2v + x(1-v^2)$ with $x = r\ell_0$. Substitution into (III.4) leads to

$$\mathcal{N}_v(x) = v[2v - (1+v^2)x]\partial_\eta\mathcal{F}. \tag{III.7}$$

To relate this identity to the recovery of the velocity, we introduce the point transformation $\Psi_{\ell_0}(r,\eta) = (r, \mathcal{F}(r,\eta))$ with $r > 0$ and $0 < \eta < r$. Here, r and η are independent arguments of the point transformation. For a physical solution, the substitution $\eta = \eta(r,z)$ is understood, and the condition $0 < \eta(r,z) < r$ expresses the assumed positivity and subluminality of the velocity. The Jacobian determinant of the point transformation is $\det D\Psi_{\ell_0} = \partial_\eta\mathcal{F}$. Wherever this determinant is nonzero, the inverse-function theorem allows η, and hence $v = \eta/r$, to be recovered locally from $(r,\mathcal{F})$.

At the upper root $x = x_+$, equation (II.10) gives $\partial_\eta\mathcal{F} = 4/(1+v^2) > 0$, so the density vanishes while the point transformation retains a smooth local inverse. At the lower root $x = x_-$, the same equation gives $\partial_\eta\mathcal{F} = 0$ and a smooth local inverse no longer exists. The following theorem identifies this singularity as a nondegenerate Whitney fold and describes reconstruction along auxiliary data that reach it.

Theorem 1. *At a point (r_0,η_0) with $r_0 > 0$, $\eta_0 = r_0v_0$, $0 < v_0 < 1$ and*

$$r_0\ell_0 = -\frac{2v_0}{1-v_0^2}, \tag{III.8}$$

the map $\Psi_{\ell_0}(r,\eta) = (r,\mathcal{F}(r,\eta))$ has a nondegenerate Whitney fold. On the inverse sheet $\partial_\eta\mathcal{F} > 0$ selected by strictly positive density, auxiliary data transverse to the fold image produce square-root behaviour on the side where a real inverse exists. For reconstruction along prescribed data to extend smoothly through the fold, the pulled-back discriminant must be the square of a smooth signed function whose sign is compatible with the chosen inverse sheet.

Proof. The Jacobian matrix and its determinant are

$$D\Psi_{\ell_0} = \begin{pmatrix} 1 & 0 \\ \partial_r\mathcal{F} & \partial_\eta\mathcal{F} \end{pmatrix}, \qquad \det D\Psi_{\ell_0} = \partial_\eta\mathcal{F}. \tag{III.9}$$

At the lower root, $\ell_0 < 0$ and the second derivative evaluated at the point in the theorem is

$$\partial_\eta^2\mathcal{F}(r_0,\eta_0) = -\ell_0\left(1 + \frac{1}{v_0^2}\right) > 0. \tag{III.10}$$

The differential has rank one, its kernel is generated by ∂_η and the determinant has a nonzero derivative in this direction. The implicit-function theorem therefore supplies a smooth critical branch $\eta_c(r)$ satisfying $\partial_\eta\mathcal{F}(r,\eta_c(r)) = 0$. Let us set $\mathcal{F}_c(r) = \mathcal{F}(r,\eta_c(r))$. The parameter-dependent Morse lemma gives a smooth source coordinate $S = S(r,\eta)$ such that

$$\mathcal{F}(r,\eta) - \mathcal{F}_c(r) = S(r,\eta)^2, \qquad \partial_\eta S > 0, \qquad S(r,\eta_c(r)) = 0. \tag{III.11}$$

With $R = r$ and target coordinate $Y = \mathcal{F} - \mathcal{F}_{\rm c}(r)$, the map becomes $(R, S) \mapsto (R, S^2)$. This is the Whitney fold normal form [20]. If $\rho > 0$, (III.2) forces $\nabla_{rz}\eta \neq 0$. Proposition 2 then places x strictly between x_- and x_+, where both linear factors in (III.4) are positive. (III.7) then gives $\partial_\eta \mathcal{F} > 0$. Since $\partial_\eta \mathcal{F} = 2S\partial_\eta S$, this sheet corresponds locally to $S > 0$. For a prescribed smooth auxiliary field $f(r, z)$ near the fold, we define the pulled-back discriminant by $\Delta(r, z) = f(r, z) - \mathcal{F}_{\rm c}(r)$. Thus, Δ measures the displacement of the prescribed auxiliary values from the fold image and reconstruction requires $S^2 = \Delta$. At a transverse zero, where $d\Delta \neq 0$, inversion gives square-root behaviour on the side $\Delta \geq 0$, while no real inverse exists on the side $\Delta < 0$. Because $(r, \eta) \mapsto (r, S)$ is a local diffeomorphism, a smooth reconstruction along the data exists precisely when $\Delta = s^2$ for a smooth signed function s compatible with the selected sheet. A root that changes sign switches sheets. This criterion concerns reconstruction along prescribed data and Ψ_{ℓ_0} itself has no smooth local inverse at the fold. □

If the discriminant $\Delta(t)$ restricted to a smooth one-dimensional slice has a zero of finite even order $2m$ with $m \geq 1$ and a positive first nonzero Taylor coefficient, then

$$\Delta(t) = (t - t_0)^{2m} h(t), \qquad s(t) = (t - t_0)^m \sqrt{h(t)}, \qquad h(t_0) > 0, \tag{III.12}$$

where h is smooth and positive near t_0. The function s is a smooth signed square root. If m is odd, it changes sign across t_0 and switches sheets. Remaining on the $\partial_\eta \mathcal{F} \geq 0$ side requires the nonnegative root, which is then not smooth at t_0. Thus, finite even order alone does not ensure smooth reconstruction on a prescribed sheet. In several variables, the corresponding requirement is a factorization $\Delta = s^2$ with s smooth and compatible with the sheet choice. Metric regularity must also be checked in the reconstruction formulas. Away from the fold, the regular inverse allows elliptic regularity to pass from the auxiliary field to the physical variables. The following theorem establishes analyticity and its consequence for regions on which η is constant.

Theorem 2. *Let $D \subset (0, \infty) \times \mathbb{R}$ be connected and open. Moreover, let $\eta \in C^2(D)$ be positive. Suppose $v = \eta/r$ satisfies $\mathcal{E}_{\ell_0}[v] = 0$ for a fixed constant ℓ_0 with $\mathcal{E}_{\ell_0}$ defined in (II.7) and that $\partial_\eta \mathcal{F} \neq 0$ throughout D. Then, η, v and all locally reconstructed metric coefficients are real analytic on D. If η is constant on a nonempty open subset, it is constant throughout D.*

Proof. By (II.12), the equation $\mathcal{E}_{\ell_0}[v] = 0$ implies $\mathcal{L}[\mathcal{F}(r, \eta(r, z))] = r\mathcal{E}_{\ell_0}[v] = 0$, where the spatial derivatives act on the composite auxiliary field. For $r > 0$, the coefficients of $\mathcal{L} = \partial_r^2 - r^{-1}\partial_r + \partial_z^2$ are real analytic and its principal part is $\partial_r^2 + \partial_z^2$. Analytic elliptic regularity therefore makes the composite field real analytic on D [21]. The defining function $\mathcal{F}(r, \eta)$ is itself real analytic for $r > 0$ and $\eta > 0$. Hence, the condition $\partial_\eta \mathcal{F} \neq 0$ allows the analytic implicit-function theorem to recover η locally as a real-analytic function of (r, z). It follows that η and $v = \eta/r$ are real analytic throughout D. Analyticity passes to $H(\eta)$ and $\Omega(\eta)$, hence to the Killing block and then to the closed one-form defining the meridional quadrature, whose local primitives are analytic. Let us suppose now that $\eta = \eta_0 > 0$ on a nonempty open subset $U_0 \subset D$ and define $\mathcal{F}_0(r) = \mathcal{F}(r, \eta_0)$ using (II.9). Since η_0 is constant, $\mathcal{F}_0$ is a linear combination of 1 and r^2 with constant coefficients. Both functions are $\mathcal{L}$-harmonic, so $\mathcal{F} - \mathcal{F}_0$ is also $\mathcal{L}$-harmonic and vanishes on U_0. This difference is real analytic on the connected domain D and the identity theorem therefore gives $\mathcal{F} = \mathcal{F}_0$ throughout D. Now let us define the level set $E = \{(r, z) \in D \mid \eta(r, z) = \eta_0\}$. At a point of E, the given field $\eta(r, z)$ and the constant value η_0 solve the same implicit relation $\mathcal{F}(r, \eta) = \mathcal{F}_0(r)$. Local uniqueness of the inverse, ensured by $\partial_\eta \mathcal{F} \neq 0$, makes them equal on a sufficiently small neighbourhood of that point. Hence, E is open in D. It is also closed by continuity of η and nonempty because it contains U_0. Connectedness of D gives $E = D$, so $\eta = \eta_0$ throughout D. □

Strictly positive density ensures $\partial_\eta \mathcal{F} > 0$, so the theorem's nonvanishing-derivative condition is satisfied. The theorem also shows that, within a connected domain satisfying its hypotheses, constancy of η on any nonempty open subset implies constancy throughout the domain. The density formula then gives $\rho = 0$ everywhere on that domain.

C. Continuation and the elliptic Cauchy problem

Having established analyticity, we now examine the stability of continuation from prescribed equatorial Cauchy data. We construct a sequence of exact solutions of the auxiliary equation whose Cauchy data

tend to zero while their amplitudes grow away from the equatorial plane, thus demonstrating the failure of continuous dependence. For the following theorem and its proof, we fix a reference length $L_0 > 0$ and denote the dimensionless variables r/L_0, z/L_0 and $\mathcal{F}/L_0$ again by r, z and $\mathcal{F}$. Thus, k is a dimensionless mode index with physical wave number k/L_0.

Theorem 3. *For positive integers k, let*

$$\mathcal{F}_k(r,z) = re^{-\sqrt{k}} J_1(kr)\cosh(kz). \tag{III.13}$$

Here, J_1 is the Bessel function of the first kind of order one. These functions satisfy $\mathcal{L}\mathcal{F}_k = 0$ for $r > 0$ and $z \in \mathbb{R}$. On any compact radial interval $I = [r_1, r_2]$ with $0 < r_1 < r_2 < \infty$, their Cauchy data on the equatorial plane satisfy $(\mathcal{F}_k(\cdot,0), \partial_z\mathcal{F}_k(\cdot,0)) \longrightarrow (0,0)$ as $k \to \infty$. This convergence holds in $C^m(I) \times C^m(I)$ for every fixed integer $m \geq 0$ and in $H^s(I) \times H^s(I)$ for every fixed real $s \geq 0$, where $H^s(I)$ denotes the L^2-based Sobolev space on I. At every fixed height $z_0 > 0$, however, $\|\mathcal{F}_k(\cdot,z_0)\|_{L^2(I)} \longrightarrow \infty$. Thus, even among data admitting analytic solutions, continuation to any fixed positive height is discontinuous at zero when the Cauchy data are equipped with the $C^\infty(I) \times C^\infty(I)$ topology and the solution trace is measured in $L^2(I)$.

Proof. The Bessel equation implies $\left(\partial_r^2 - r^{-1}\partial_r\right)[rJ_1(kr)] = -k^2 rJ_1(kr)$, while $\partial_z^2 \cosh(kz) = k^2\cosh(kz)$. Substitution into (III.13) therefore gives $\mathcal{L}\mathcal{F}_k = 0$. The Cauchy traces on the equatorial plane are $\mathcal{F}_k(r,0) = e^{-\sqrt{k}} rJ_1(kr)$ and $\partial_z\mathcal{F}_k(r,0) = 0$. Let us recall that the derivative recurrence for Bessel functions expresses each fixed derivative of J_1 as a finite linear combination of Bessel functions of fixed integer order (see Eq. 10.6.7 in [22]). These functions are bounded on the positive real axis. Leibniz's rule consequently gives, for every fixed integer $m \geq 0$

$$\|\mathcal{F}_k(\cdot,0)\|_{C^m(I)} \leq C_{I,m}(1+k)^m e^{-\sqrt{k}} \longrightarrow 0, \tag{III.14}$$

where $C_{I,m}$ is independent of k. Together with the vanishing normal trace, this proves convergence of the Cauchy data in $C^\infty(I) \times C^\infty(I)$. For any fixed real $s \geq 0$, we choose an integer $m \geq s$. On the bounded interval I, the continuous inclusions $C^m(I) \subset H^m(I) \subset H^s(I)$ then give convergence of both traces in $H^s(I)$. To examine the fields at positive height, we use the large-argument expansion provided by Eq. 10.17.3 in [22], i.e.

$$rJ_1(kr) = \sqrt{\frac{2r}{\pi k}}\cos\left(kr - \frac{3\pi}{4}\right) + O\left(\frac{1}{k^{3/2}}\right), \tag{III.15}$$

which is uniform for $r \in I$ because $r_1 > 0$. Squaring and integrating this expression, using $\cos^2\theta = (1+\cos 2\theta)/2$, gives

$$\|rJ_1(kr)\|^2_{L^2(I)} = \frac{1}{\pi k}\int_I r\, dr + O\left(\frac{1}{k^2}\right). \tag{III.16}$$

Indeed, integration by parts shows that $\int_I r\cos(2kr - 3\pi/2)\, dr = O(k^{-1})$, so its contribution to the squared norm is $O(k^{-2})$. Since $\int_I r\, dr = (r_2^2 - r_1^2)/2 > 0$, the norm has a strictly positive leading coefficient. For each fixed $z_0 > 0$, $\cosh(kz_0) \sim e^{kz_0}/2$, and hence

$$\|\mathcal{F}_k(\cdot,z_0)\|_{L^2(I)} = e^{-\sqrt{k}}\cosh(kz_0)\|rJ_1(kr)\|_{L^2(I)} \sim \sqrt{\frac{r_2^2 - r_1^2}{8\pi k}} e^{kz_0 - \sqrt{k}} \longrightarrow \infty. \tag{III.17}$$

The growth of kz_0 dominates both $\sqrt{k}$ and $\log k$. Thus, the Cauchy data converge to zero in the stated smooth topology, whereas the solution traces at $z = z_0$ have unbounded $L^2(I)$ norms. Continuity of continuation at zero would require these traces to converge to the zero solution in $L^2(I)$. This contradiction proves the asserted discontinuity, even when the data are restricted to those admitting analytic solutions. □

Runge approximation addresses the separate question of approximating a local $\mathcal{L}$-harmonic solution, defined near a compact interior region, by fields on a larger domain. Such approximation is possible provided that removing a compact neighbourhood of the chosen region from the larger domain leaves no connected component with compact closure contained in that domain. Under the reconstruction hypotheses detailed in Appendix A, sufficiently accurate approximation of the field and its derivatives through second order preserves the strict bounds ensuring a regular inverse, subluminality and positive density on that region. The resulting control of the reconstructed geometry is confined to the chosen interior region, so global completion remains a separate problem.

IV. LOCAL FLAT TRACES AND REGULAR-AXIS COMPLETION

We now examine whether a speed profile that is constant along an equatorial interval can arise from a solution with nontrivial transverse dependence. Using the auxiliary potential, we construct such traces on admissible annuli and reconstruct the corresponding analytic dust metrics with positive density. We then investigate whether these local realizations can extend to a regular rotation axis. Under the stated axis hypotheses, regularity forces $\ell_0 = 0$. Analyticity then makes an exactly flat trace constant along the connected equatorial interval reaching the axis, in conflict with the required vanishing of the speed there. This leads us to consider continuation within annular domains, where an additional transverse-maximum condition yields an explicit finite radial window for an exactly flat trace.

A. Analytic realization and metric reconstruction

We begin by showing that an exactly flat equatorial trace can be realized by a local analytic dust spacetime whenever the prescribed data lie strictly within the admissible density interval.

Theorem 4. *Let $r_* > 0$, $0 < v_0 < 1$, $\ell_0 \in \mathbb{R}$ and*

$$\partial_\eta \mathcal{F}(r_*, r_* v_0) = 2 + \ell_0 r_* \left(\frac{1}{v_0} - v_0 \right) \neq 0. \tag{IV.1}$$

There is a neighbourhood $D_* = (r_* - \delta, r_* + \delta) \times (-z_*, z_*)$, where $0 < \delta < r_*$, on which the exact constant-ℓ equation has a unique real-analytic solution with Cauchy data $v(r,0) = v_0$ and $v_z(r,0) = 0$. The solution is even in z and satisfies

$$v(r,z) = v_0 + \frac{v_0 - \ell_0 r}{r^2 \left[2 + \ell_0 r \left(\frac{1}{v_0} - v_0 \right) \right]} z^2 + O(z^4). \tag{IV.2}$$

The remainder in this expansion is locally uniform in r. With $H = -Ce^{\ell_0 \eta}$, the angular velocity Ω is obtained by integrating (II.2), while μ follows from the quadratures (IV.5)-(IV.6). Fixing $C > 0$, the additive constant in Ω, and the reference value $\mu(r_*, 0)$ therefore determines a unique real-analytic local metric of the exact rotating-dust family. If

$$-\frac{2v_0}{1 - v_0^2} < \ell_0 r_* < \frac{2v_0}{1 + v_0^2}, \tag{IV.3}$$

then D_* can be chosen so that $0 < v < 1$ and $\rho > 0$ throughout it.

Proof. Let $f(r,z)$ denote the auxiliary field, to distinguish it from the defining function $\mathcal{F}(r,\eta)$. We prescribe the analytic Cauchy data

$$f(r,0) = \mathcal{F}(r, r v_0) = 2 r v_0 + \ell_0 r^2 \log\left(\frac{r v_0}{\eta_*} \right) - \frac{\ell_0}{2} r^2 v_0^2, \qquad f_z(r,0) = 0. \tag{IV.4}$$

Since $r_* > 0$, the equation $f_{zz} = -f_{rr} + r^{-1} f_r$ has analytic coefficients near $(r_*, 0)$ and $z = 0$ is noncharacteristic. The Cauchy-Kowalevski theorem, therefore, gives a unique local analytic solution [23]. Reflection in z preserves the equation and the Cauchy data, so uniqueness makes f even. Condition (IV.1) allows the analytic implicit-function theorem to recover a unique local field $\eta(r,z)$ satisfying $\mathcal{F}(r, \eta(r,z)) = f(r,z)$ and $\eta(r_*, 0) = r_* v_0$. Local uniqueness of this inverse gives $\eta(r,0) = r v_0$ and preserves reflection symmetry. The exact identity (II.12) then shows that $v = \eta / r$ solves $\mathcal{E}_{\ell_0}[v] = 0$ with the provided Cauchy data. Any other analytic solution with the same velocity data produces the same auxiliary Cauchy data. Uniqueness for the auxiliary equation, followed by local uniqueness of the inverse, therefore establishes uniqueness of the reconstructed solution. After shrinking the neighbourhood, we have $\eta > 0$ and $0 < v < 1$. The functions $H = -Ce^{\ell_0 \eta}$ and Ω, obtained from $\Omega_{,\eta} = H_{,\eta} / (2\eta)$ with the prescribed additive constant, are analytic there. Hence, (II.3) defines

an analytic Killing block with $g_{\phi\phi} > 0$ and $\det G = -r^2$. The remaining metric function is determined by the exact quadratures [3, 6]

$$M_r = \mu_r = \frac{g_{tt,r}g_{\phi\phi,r} - g_{tt,z}g_{\phi\phi,z} - (g_{t\phi,r})^2 + (g_{t\phi,z})^2}{2r} = \frac{(\partial_\eta \mathcal{F})^2}{8r}(\eta_z^2 - \eta_r^2), \tag{IV.5}$$

$$M_z = \mu_z = \frac{g_{tt,z}g_{\phi\phi,r} + g_{tt,r}g_{\phi\phi,z} - 2g_{t\phi,r}g_{t\phi,z}}{2r} = -\frac{(\partial_\eta \mathcal{F})^2}{4r}\eta_r\eta_z. \tag{IV.6}$$

Here, a comma followed by r or z denotes partial differentiation with respect to that coordinate. Moreover, $\partial_\eta \mathcal{F}$ is evaluated at $(r, \eta(r, z))$. To verify the simplification at a chosen point p, we make the constant change of angular coordinate $\bar{\phi} = \phi - \Omega(p)t$. The transformed angular velocity vanishes at p, although its derivatives need not vanish. Evaluating the metric differentials in this frame gives $dg_{tt} = (H_{,\eta} - 2\eta\Omega_{,\eta})d\eta = 0$ and $dg_{t\phi} = 2^{-1}(\partial_\eta \mathcal{F})d\eta$. The quadratic combinations of metric derivatives in the two quadratures are invariant under this constant change of Killing basis, whose determinant is one, so the reduced expressions hold in the original frame as well. Their compatibility follows from the identity

$$\partial_z M_r - \partial_r M_z = \frac{\eta_z \partial_\eta \mathcal{F}}{4r}\mathcal{L}f = 0. \tag{IV.7}$$

Thus, $M_r dr + M_z dz$ is a closed analytic one-form on a sufficiently small, simply connected rectangle. Integrating it with the assigned value $\mu(r_*, 0)$ determines a unique analytic μ. Since $\det G < 0$ and $e^\mu > 0$, the reconstructed metric is Lorentzian. Together with the density formula, these relations satisfy the equations of the exact dust representation. Along the trace, we have $v_r = v_{rr} = v_z = 0$. The field equation gives

$$\left[2 + \ell_0 r\left(\frac{1}{v_0} - v_0\right)\right] v_{zz}(r, 0) + \frac{2\ell_0}{r} - \frac{2v_0}{r^2} = 0. \tag{IV.8}$$

Solving for $v_{zz}(r, 0)$ yields

$$v_{zz}(r, 0) = \frac{2(v_0 - \ell_0 r)}{r^2\left[2 + \ell_0 r\left(\frac{1}{v_0} - v_0\right)\right]}. \tag{IV.9}$$

Taylor expansion in z, together with evenness and (IV.9), gives (IV.2). To establish positivity of the density, we observe that the prescribed data yield $\eta_r(r_*, 0) = v_0 > 0$ and $\eta_z(r_*, 0) = 0$. The strict bounds (IV.3) therefore imply $\rho(r_*, 0) > 0$ by Proposition 2. Finally, continuity preserves this inequality and subluminality on a smaller patch. □

Once the stated constants are fixed, this construction determines the local metric. Notice that extending the annular patch requires additional conditions at an axis or material boundary and, where appropriate, an exterior solution. We first examine the axis.

B. The metric condition at a regular axis

Let k and m be the stationary and axial Killing fields introduced in Section II. We recall that a regular axis is a fixed component of the effective $U(1)$ action generated by m with angular period 2π. We choose local coordinates (t, x, y, z) near the rotation axis, with x and y being Cartesian coordinates transverse to it. Within this coordinate neighbourhood U, $m = -y\partial_x + x\partial_y$ and the axis is the set of points satisfying $x = y = 0$ which we denote by $\mathcal{A} := \{p \in U \mid x(p) = 0,\ y(p) = 0\}$. The next result combines metric regularity at the axis with the exact dust identities to establish $H'(0) = 0$, which forces $\ell_0 = 0$ in the constant-ℓ class.

Theorem 5. *Let a stationary Lorentz metric extend as a C^2 metric across a regular axis component $\mathcal{A}$. We assume that k commutes with m and is timelike on $\mathcal{A}$. Let a nontrivial (η, H) branch with η of class C^1 away from the axis meet $\mathcal{A}$ and satisfy the exact relations (II.2)-(II.3) off the axis with $0 < |\eta| < r$ and $\eta \to 0$ at the axis. If H is C^1 at zero and $H(0) = -C < 0$, then $H'(0) = 0$. In particular, a constant-ℓ branch satisfying these hypotheses has $\ell_0 = 0$.*

Proof. Let $\varrho = \sqrt{x^2 + y^2}$ be the transverse Cartesian radius in a smooth chart adapted to the axial action. Rotational invariance and the C^2 Cartesian Taylor expansion of the metric give

$$g(m,m) = a\varrho^2 + o(\varrho^2), \qquad g(k,m) = O(\varrho^2), \qquad g(k,k) = -D + O(\varrho^2), \tag{IV.10}$$

The coefficients $a, D > 0$ may vary along the axis and the estimates are uniform near any fixed axis point. The determinant identity $r^2 = g(k,m)^2 - g(k,k)g(m,m)$ therefore yields $r^2 = Da\varrho^2 + o(\varrho^2)$ with $r \asymp \varrho$. Here, $r \asymp \varrho$ means that $c_1\varrho \le r \le c_2\varrho$ locally for some constants $c_1, c_2 > 0$. Subluminality therefore gives $\eta \to 0$ along the branch. The same C^2 Taylor expansion gives the radial derivative estimates

$$\partial_\varrho g(k,k) = O(\varrho), \qquad \partial_\varrho g(k,m) = O(\varrho), \qquad \partial_\varrho g(m,m) = 2a\varrho + o(\varrho). \tag{IV.11}$$

The derivatives are taken at fixed coordinates along the axis and these estimates are again locally uniform. By contradiction, let us suppose that $h_1 = H'(0) \neq 0$. Since H is C^1, we have $H_{,\eta} = h_1 + o(1)$ as $\eta \to 0$. Integrating $\Omega_{,\eta} = H_{,\eta}/(2\eta)$ then gives

$$\Omega(\eta) = \frac{h_1}{2} \log\left(\frac{|\eta|}{\eta_*}\right) + o\left(\left|\log\left(\frac{|\eta|}{\eta_*}\right)\right|\right), \tag{IV.12}$$

where $\eta_* > 0$ is a fixed reference scale with the same dimensions as η. Hence, $|\Omega| \to \infty$ as the axis is approached through the branch. On the other hand, if we differentiate $H = g(k,k) + 2\Omega g(k,m) + \Omega^2 g(m,m)$ and use $g(k,m) + \Omega g(m,m) = \eta$, we end up with

$$dg(k,k) + 2\Omega\ dg(k,m) + \Omega^2\ dg(m,m) = dH - 2\eta\ d\Omega = 0. \tag{IV.13}$$

Taking the transverse radial component of (IV.13) and dividing by $\varrho\Omega^2$ gives, by (IV.11), $0 = O(\Omega^{-2}) + O(\Omega^{-1}) + 2a + o(1)$. As the axis is approached, the first two terms tend to zero whereas $2a > 0$, which is a contradiction. Thus, $H'(0) = 0$. For $H = -Ce^{\ell_0\eta}$, this reads $-C\ell_0 = 0$ and $C > 0$ gives $\ell_0 = 0$. □

The quadratic axis expansion is essential here because elementary flatness on a punctured neighbourhood alone does not supply it. After a time normalization with $H(0) = -1$, an analytic H must consequently satisfy $H(\eta) = -1 + O(\eta^2)$. The vanishing linear coefficient in an analytic expansion of H was also obtained under additional assumptions on rotation at the axis in [24]. A complementary proof based on the regularity of the dust congruence is given in Appendix B. Having established that regular-axis completion requires $\ell_0 = 0$, we now determine the rate at which the ZAMO speed must vanish as the axis is approached.

Lemma 1. *Under the metric and axis hypotheses of Theorem 5, a branch with $\ell_0 = 0$ satisfies $\eta = O(r^2)$ and $v = O(r)$ at the axis.*

Proof. The angular velocity is a finite constant Ω_0 and $\eta = g(k,m) + \Omega_0 g(m,m)$. Both terms on the right are $O(\varrho^2)$ by (IV.10). Since $r \asymp \varrho$, division by r gives the asserted decay. □

C. A complete regular-axis branch

To study global rigidity on the $\ell_0 = 0$ branch, we recast the auxiliary equation $\mathcal{L}\mathcal{F} = 0$ as a harmonic problem. The following proposition shows that the substitution $\mathcal{F} = r^2 U$ yields the five-dimensional Laplace equation away from the axis, providing the basis for the global argument.

Proposition 3. *Let $\mathcal{F} \in C^2((0,\infty) \times \mathbb{R})$ with $\mathcal{F} = r^2 U$. Furthermore, we define $\widetilde{U}(y,z) = U(|y|,z)$ for $y \in \mathbb{R}^4 \setminus \{0\}$ and $z \in \mathbb{R}$. Then*

$$\mathcal{L}\mathcal{F} = r^2\left(U_{rr} + \frac{3}{r}U_r + U_{zz}\right) = r^2 \Delta_{\mathbb{R}^5}\widetilde{U}. \tag{IV.14}$$

Thus, away from the lifted axis $y = 0$, $\mathcal{L}$-harmonic fields correspond to five-dimensional harmonic fields invariant under rotations of y.

Proof. Differentiating r^2U gives $\partial_r(r^2U) = 2rU + r^2U_r$ and $\partial_r^2(r^2U) = 2U + 4rU_r + r^2U_{rr}$. Combining these radial derivatives with $\partial_z^2(r^2U) = r^2U_{zz}$ gives the first identity. The radial Laplacian in $\mathbb{R}^4_y$ is $\partial_r^2 + 3r^{-1}\partial_r$, which gives the second. □

The next theorem combines regularity along the complete axis with decay at infinity to show that the lifted harmonic field vanishes, leaving a flat vacuum metric.

Theorem 6. *Let us suppose the $\ell_0 = 0$ branch is defined for all $r > 0$ and $z \in \mathbb{R}$ and extends regularly to the complete axis. Assume there is no finite boundary, puncture, additional end, horizon or singular axis segment. For every compact interval $K \subset \mathbb{R}$, we assume*

$$|\eta(r,z)| \le C_K r^2 \quad \text{for } z \in K, \quad 0 < r < r_K \tag{IV.15}$$

with suitable positive constants C_K and r_K. Suppose also that

$$|\eta(r,z)| \le C_\infty \frac{r}{R} \tag{IV.16}$$

with $R = \sqrt{r^2+z^2} \ge R_0$. Then $\eta = 0$. After stationary and meridional normalization, the metric takes the Minkowski form and the dust density vanishes.

Proof. For $\ell_0 = 0$, we have $\mathcal{F} = 2\eta$, so $\widetilde{U}(y,z) = 2\eta(|y|,z)/|y|^2$ is harmonic away from the line $\Gamma = \{(0,z) \in \mathbb{R}^4 \times \mathbb{R}\}$. The axis estimate bounds $\widetilde{U}$ near every compact segment of Γ. To show that Γ is removable, we choose a smooth radial cutoff $\chi_\varepsilon(|y|)$ that is zero for $|y| < \varepsilon$ and one for $|y| > 2\varepsilon$. On any fixed compact set, the tube $|y| < 2\varepsilon$ has volume $O(\varepsilon^4)$, while $|\nabla\chi_\varepsilon| = O(\varepsilon^{-1})$ and $|\Delta_{\mathbb{R}^5}\chi_\varepsilon| = O(\varepsilon^{-2})$. For a test function $\varphi \in C_c^\infty(\mathbb{R}^5)$, harmonicity away from Γ gives

$$\int_{\mathbb{R}^5} \widetilde{U}\Delta_{\mathbb{R}^5}(\chi_\varepsilon\varphi)\, dX = 0. \tag{IV.17}$$

Local boundedness of $\widetilde{U}$ makes the cutoff error terms tend to zero with largest order $O(\varepsilon^2)$. Letting $\varepsilon \to 0$ therefore gives $\int \widetilde{U}\Delta_{\mathbb{R}^5}\varphi\, dX = 0$. Thus, $\widetilde{U}$ is distributionally harmonic across Γ and Weyl's lemma supplies a harmonic extension to all of $\mathbb{R}^5$ [23]. On a sphere of radius $S \ge R_0$, we write $r = S\sin\theta$. The outer estimate gives

$$|\widetilde{U}| \le \frac{2C_\infty}{S^2\sin\theta}. \tag{IV.18}$$

Writing the surface measure as $d\sigma = S^4\sin^3\theta\, d\theta\, d\omega_3$, where $d\omega_3$ is the measure on the unit three-sphere and and $|\mathbb{S}^4_S|$ denotes its total surface area, shows that the bound is integrable at both poles. Indeed, the remaining angular integral is proportional to $\int_0^\pi \sin^2\theta\, d\theta < \infty$, so

$$\frac{1}{|\mathbb{S}^4_S|}\int_{\mathbb{S}^4_S} |\widetilde{U}|\, d\sigma \le \frac{C_{\rm av}}{S^2}. \tag{IV.19}$$

Here $C_{\rm av} > 0$ is independent of S. Let us fix $X \in \mathbb{R}^5$ and choose $S > \max\{2|X|, R_0\}$. For $\xi \in \mathbb{S}^4_S$, the inequality $|X - \xi| \ge S/2$ bounds the Poisson kernel by $32/|\mathbb{S}^4_S|$. The Poisson formula and (IV.19) therefore give

$$|\widetilde{U}(X)| \le \frac{32}{|\mathbb{S}^4_S|}\int_{\mathbb{S}^4_S} |\widetilde{U}|\, d\sigma \le \frac{32C_{\rm av}}{S^2}. \tag{IV.20}$$

Letting $S \to \infty$ yields $\widetilde{U} = 0$, hence $\eta = 0$. The remaining relations give $H = -C$, $\Omega = \Omega_0$ and $\mu = \mu_0$, all constant, so the line element becomes

$$ds^2 = -Cdt^2 + \frac{r^2}{C}(d\phi - \Omega_0 dt)^2 + e^{\mu_0}(dr^2 + dz^2). \tag{IV.21}$$

Regularity of the axis with angular period 2π requires $e^{\mu_0} = 1/C$, as follows from the circumference-to-radius ratio. The coordinates $T = \sqrt{C}t$, $\bar{r} = r/\sqrt{C}$, $\bar{z} = z/\sqrt{C}$ and $\bar{\phi} = \phi - \Omega_0 t$ then put the metric in the Minkowski form

$$ds^2 = -dT^2 + d\bar{r}^2 + \bar{r}^2 d\bar{\phi}^2 + d\bar{z}^2. \tag{IV.22}$$

Its Einstein tensor vanishes, so the dust density is zero. □

The theorem concerns the complete orbit half-plane with the stated axis and decay estimates. A finite body matched to a vacuum exterior requires separate analysis. Within the strictly positive-speed class, the theorem excludes a branch satisfying all these global hypotheses. The following nonexistence result for an exactly flat trace requires only analyticity along a connected equatorial interval and regularity at its axis endpoint. It does not use the preceding theorem's assumptions on the full domain or asymptotic decay.

Theorem 7. *Let an exact constant-ℓ solution be defined on an open meridional domain $\mathcal{U} \subset (0,\infty) \times \mathbb{R}$ and satisfy the hypotheses of Theorem 5. Moreover, we assume that its equatorial radial domain $\mathcal{I}_0 = \{r > 0 \mid (r,0) \in \mathcal{U}\}$ is a connected open interval with an endpoint at $r = 0$, where the branch meets the regular axis. Then, no nonempty open interval $I \subset \mathcal{I}_0$ supports $v(r,0) = v_0$ with $0 < v_0 < 1$.*

Proof. The metric axis condition forces $\ell_0 = 0$, so $\mathcal{F} = 2\eta$ and $\eta_{rr} - r^{-1}\eta_r + \eta_{zz} = 0$. This equation is elliptic with analytic coefficients on $r > 0$. Interior analytic regularity makes η, and therefore $v = \eta/r$, real analytic there [21]. If $v(r,0) = v_0$ on a nonempty open subinterval, the one-dimensional identity theorem applied to the analytic function $r \mapsto v(r,0) - v_0$ gives $v(r,0) = v_0$ throughout $\mathcal{I}_0$. It follows that $v(r,0) \to v_0 > 0$ as $r \downarrow 0$. Lemma 1 instead gives $v(r,0) = O(r)$, which proves the claim. □

Connectedness ensures that the flat interval and the axis endpoint belong to the same analytic equatorial branch. A matching interface that changes the governing branch, or a singularity, can interrupt the required continuation. A region with $\rho = 0$ alone does not interrupt analyticity within the same regular constant-ℓ branch. Notice that the theorem requires no reflection symmetry, density sign or vertical-curvature assumption. For $\ell_0 = 0$, the reflection-symmetric local continuation is $v = v_0\sqrt{r^2 + z^2}/r$. Its direct verification and finite luminal height are presented in Appendix B. The next result shows that, for an annular trace separated from the axis, nonnegative density and an equatorial maximum in the transverse direction together restrict the radial interval over which an exactly flat speed profile can persist.

Theorem 8. *Let an exact constant-ℓ solution be C^2 on $I \times (-z_*, z_*)$, where $I \subset (0,\infty)$ is a nonempty open interval. Further suppose $v(r,0) = v_0 \in (0,1)$ and $\rho(r,0) \geq 0$ on I. Assume that $z = 0$ is a local vertical maximum of v at every $r \in I$. Then $\ell_0 > 0$ and*

$$\frac{v_0}{\ell_0} \leq r \leq \frac{2v_0}{\ell_0(1 + v_0^2)}. \tag{IV.23}$$

Moreover, strictly positive density and $v_{zz}(r,0) < 0$ give strict inequalities.

Proof. Flatness gives $v_r = v_{rr} = 0$ on the trace, while the vertical-maximum condition gives $v_z = 0$ and $v_{zz} \leq 0$. Thus, the trace balance (IV.8) applies. Moreover, $\eta_r = v_0 > 0$ and $\eta_z = 0$, so every trace point is noncritical. Setting $x = \ell_0 r$, Proposition 2 yields $x_- = -2v_0/(1 - v_0^2) \leq x \leq 2v_0/(1 + v_0^2) = x_+$. The coefficient in the trace equation is

$$\partial_\eta \mathcal{F}(r, rv_0) = 2 + \frac{1 - v_0^2}{v_0} x. \tag{IV.24}$$

At $x = x_-$ it vanishes, whereas (IV.8) would then require $x_- = v_0$. This is impossible because $x_- < 0 < v_0$. Hence, $x > x_-$ and $\partial_\eta \mathcal{F}(r, rv_0) > 0$, so

$$v_{zz}(r,0) = \frac{2(v_0 - x)}{r^2 \partial_\eta \mathcal{F}(r, rv_0)} \leq 0. \tag{IV.25}$$

It follows that $x \geq v_0 > 0$ and $\ell_0 > 0$. Combining this with $x \leq x_+$ proves the window. Positive density excludes $x = x_+$, while strictly negative vertical curvature gives $x > v_0$. □

Hence, any two radii $r_1 < r_2$ in the trace interval satisfy

$$\frac{r_2}{r_1} \leq \frac{2}{1+v_0^2} < 2. \tag{IV.26}$$

Notice that the bound uses only equatorial data and the sign of the transverse curvature. In the next section, near-flat traces lead to quantitative bounds on the same radial extent.

V. FINITE RADIAL SPANS OF EQUATORIAL TRACES

We now extend the analysis of exactly flat traces to profiles with a bounded logarithmic slope. For reflection-symmetric solutions, we formulate transverse concavity in terms of the relative dust-ZAMO rapidity and derive an exact equatorial identity relating this condition to the variation of the slope with radius. Combining this identity with nonnegative density yields finite radial-span bounds for each sign of ℓ_0. Analytic constructions establish sharpness for the nonpositive class and, in the exactly flat limit, for the positive class. For positive ℓ_0 and nonzero slope tolerance, the exact supremum of admissible spans remains open, although the bounds obtained here exclude a semi-infinite approximately flat trace under the stated hypotheses.

A. Transverse shape and the logarithmic identity

Let $\mathsf{q} = e^{\mu}(dr^2 + dz^2)$ denote the orbit metric and let $r = \sqrt{-\det G}$ be the orbit-area radius. Where $dr \neq 0$, we define the unit vector field tangent to the level curves of r and the relative dust-ZAMO rapidity by

$$\mathsf{n} = \frac{J \operatorname{grad}_{\mathsf{q}} r}{\left|\operatorname{grad}_{\mathsf{q}} r\right|_{\mathsf{q}}}, \qquad \psi = \operatorname{artanh} v. \tag{V.1}$$

Here, J denotes rotation by $\pi/2$ in each tangent plane with respect to the orbit metric and the chosen orientation. In the Weyl coordinates, we choose the orientation for which $J(\partial_r) = \partial_z$ and $J(\partial_z) = -\partial_r$. Hence, the unit vector field n points along the curves of constant r. On the equatorial reflection surface, n is transverse to that surface. We impose there the weak transverse-shape condition $\mathsf{n}(\mathsf{n}\psi) \leq 0$, where $\mathsf{n}(\mathsf{n}\psi)$ is the second derivative of ψ along the unit-speed integral curves of n. This condition is unchanged by reversing the orientation and imposes no independent requirement on the transverse density profile. Reflection symmetry relates it to the transverse second derivative of the speed as follows.

Proposition 4. *For a reflection-symmetric field with $0 < v < 1$, the identity*

$$\mathsf{n}(\mathsf{n}\psi) = \frac{e^{-\mu} v_{zz}}{1-v^2} \tag{V.2}$$

holds on the equatorial plane. Hence, the transverse-shape condition is equivalent to $v_{zz}(r, 0) \leq 0$.

Proof. In the Weyl chart, $\operatorname{grad}_{\mathsf{q}} r = e^{-\mu}\partial_r$ and its norm is $e^{-\mu/2}$. With the chosen orientation, we therefore have $\mathsf{n} = e^{-\mu/2}\partial_z$ and

$$\mathsf{n}(\mathsf{n}\psi) = e^{-\mu}\psi_{zz} + e^{-\mu/2}\psi_z \partial_z(e^{-\mu/2}). \tag{V.3}$$

Reflection symmetry gives $v_z = \psi_z = 0$ on the equator, so the term involving the derivative of the metric factor vanishes there. The identity $\psi_{zz} = v_{zz}/(1-v^2) + 2vv_z^2/(1-v^2)^2$ then gives (V.2). Its prefactor is positive because $0 < v < 1$, which proves the equivalence. □

Strict negativity gives a strict local maximum along the transverse curve at fixed r. If the second derivative vanishes, however, the weak condition alone does not determine whether there is a local maximum, because this second-order condition does not control the higher-order transverse behaviour. This distinction is relevant to sharpness since the solutions attaining the nonpositive-ℓ_0 span bound have zero transverse curvature. We quantify departures from an exactly flat trace by bounding the logarithmic slope of the equatorial speed.

Definition 1. Let $u(r) = v(r,0) > 0$ be a C^1 trace on $[R_1, R_2]$, where $0 < R_1 < R_2$. For $0 \le \varepsilon < 1$, the trace is ε-flat if

$$\left| \frac{d\log u}{d\log r} \right| \le \varepsilon. \tag{V.4}$$

Here, $d/d\log r$ means $r\,d/dr$, equivalently differentiation with respect to the dimensionless variable $\log(r/R_1)$. Its logarithmic radial span is $\Lambda = \log(R_2/R_1)$. Notice that this definition is unchanged by a constant rescaling of the radial coordinate. Integrating the slope bound between $R_1 \le r_1 < r_2 \le R_2$ gives

$$\left(\frac{r_1}{r_2}\right)^{\varepsilon} \le \frac{u(r_2)}{u(r_1)} \le \left(\frac{r_2}{r_1}\right)^{\varepsilon}. \tag{V.5}$$

Let $\sigma = \log(r/R_1) \in [0, \Lambda]$ and introduce the dimensionless quantities

$$s = \frac{d\log u}{d\sigma}, \quad \dot{s} = \frac{ds}{d\sigma}, \quad y = \frac{\ell_0 r}{u}, \quad Z = \frac{r^2}{u} v_{zz}(r,0), \quad \partial_\eta \mathcal{F}(r, ru(r)) = 2 + y(1-u^2). \tag{V.6}$$

Throughout this section, a dot denotes differentiation with respect to σ, and $\partial_\eta \mathcal{F}$ is evaluated at $(r, ru(r))$ unless its arguments are displayed. This is the same principal coefficient introduced in (II.10). For an ε-flat trace, $|s| \le \varepsilon$, and differentiation gives

$$u_r = \frac{us}{r}, \qquad u_{rr} = \frac{u}{r^2}(\dot{s} + s^2 - s), \qquad \dot{y} = (1-s)y. \tag{V.7}$$

Because $\varepsilon < 1$, we also have $\eta_r(r,0) = u(1+s) > 0$. Thus, the trace contains no critical point of η and nonnegative density imposes the interval in Proposition 2 at every point. The next lemma combines this restriction with the exact velocity equation and the transverse-shape condition.

Lemma 2. *Let v be a C^2 reflection-symmetric solution of the exact constant-ℓ_0 equation (II.7) on an open neighbourhood of $[R_1, R_2] \times \{0\}$ and $0 < u < 1$. Its equatorial data satisfy*

$$(Z + \dot{s})\partial_\eta \mathcal{F} + 2s^2(1 - yu^2) + 2ys(1-u^2) + 2y - 2 = 0. \tag{V.8}$$

If the trace is ε-flat and $\rho(r,0) \ge 0$, then

$$-\frac{2}{1-u^2} \le y \le \frac{2}{1+u^2}, \qquad \partial_\eta \mathcal{F} > 0. \tag{V.9}$$

If the transverse-shape condition also holds, then

$$\dot{s}\partial_\eta \mathcal{F} \ge 2(1-y) - 2s^2(1-yu^2) - 2ys(1-u^2). \tag{V.10}$$

Proof. Reflection symmetry gives $v_z(r,0) = 0$. Substituting (V.7) and $v_{zz}(r,0) = uZ/r^2$ into $\mathcal{E}_{\ell_0}[v] = 0$, multiplying by r^2/u and collecting the coefficients of s^2 and s yields (V.8). Since $\eta_r = u(1+s) > 0$, Proposition 2 gives the stated interval for y. This interval implies $\partial_\eta \mathcal{F} \ge 0$ with equality possible only at its lower endpoint $y = -2/(1-u^2)$. At that endpoint, the logarithmic identity reduces, after division by the positive factor $2/(1-u^2)$, to $(s+1)[(1+u^2)s - (3-u^2)] = 0$. The resulting slopes are $s = -1$ and $s = (3-u^2)/(1+u^2) > 1$, both incompatible with $|s| \le \varepsilon < 1$. Hence, nonnegative density and the slope bound imply $\partial_\eta \mathcal{F} > 0$, so the trace lies on the regular inverse sheet. Finally, the transverse-shape condition gives $Z \le 0$, and rearranging the identity yields

$$\dot{s}\partial_\eta \mathcal{F} = 2(1-y) - 2s^2(1-yu^2) - 2ys(1-u^2) - Z\partial_\eta \mathcal{F}. \tag{V.11}$$

The last term is nonnegative, which proves the inequality. □

The density and transverse-shape assumptions play distinct roles. Density confines y to the admissible interval and, together with the slope bound, excludes the fold. Transverse concavity then gives a lower bound on the change of s with logarithmic radius. The resulting span estimates require no prescribed value of the inner-edge parameter $y(R_1) = \ell_0 R_1/u(R_1)$. A density-only estimate, with an explicitly calibrated inner-edge parameter, is given in Appendix C.

B. Universal bounds and their sharp cases

We now integrate the slope inequality, treating the two signs of ℓ_0 separately.

Theorem 9. *Let v be a C^2 reflection-symmetric solution of the exact constant-ℓ_0 equation on an open neighbourhood of $[R_1, R_2] \times \{0\}$, where $0 < R_1 < R_2$. Further suppose that, throughout the equatorial interval, $0 < u < 1$, $\rho(r, 0) \geq 0$, $\mathsf{n}(\mathsf{n}\psi) \leq 0$ and $|d \log u / d \log r| \leq \varepsilon$ for some $0 \leq \varepsilon < 1$. Let us set*

$$\underline{v} = \min_{[R_1, R_2]} u > 0. \tag{V.12}$$

If $\ell_0 \leq 0$, then

$$\Lambda \leq 2 \operatorname{artanh} \varepsilon = \log\left(\frac{1+\varepsilon}{1-\varepsilon}\right). \tag{V.13}$$

If $\ell_0 > 0$, then every $\alpha \in (0, 1-\varepsilon)$ gives $\Lambda \leq \mathcal{B}_+(\varepsilon, \alpha, \underline{v})$ with

$$\mathcal{B}_+(\varepsilon, \alpha, \underline{v}) = \frac{\varepsilon(2+\alpha)}{(1+\varepsilon)(1-\varepsilon-\alpha)} + \frac{1}{1-\varepsilon} \log \frac{2}{\alpha(1+\underline{v}^2)}. \tag{V.14}$$

Proof. The preceding lemma gives $\partial_\eta \mathcal{F} > 0$ and (V.10). For $\ell_0 \leq 0$, we have $y \leq 0$ and the right-hand side of that inequality satisfies the exact factorization

$$2(1-y) - 2s^2(1-yu^2) - 2ys(1-u^2) - (1-s^2)\partial_\eta \mathcal{F} = -y(1+s)[3 - u^2 - s(1+u^2)]. \tag{V.15}$$

Here, $-y \geq 0$ and $1 + s > 0$. Since $s < 1$, the last factor exceeds $2(1-u^2) > 0$ and is therefore positive. Hence, division by $\partial_\eta \mathcal{F}$ gives $\dot{s} \geq 1 - s^2$. As $|s| < 1$, we have

$$\frac{d}{d\sigma} \operatorname{artanh} s = \frac{\dot{s}}{1-s^2} \geq 1. \tag{V.16}$$

Integrating over $[0, \Lambda]$ gives $\Lambda \leq \operatorname{artanh} s(\Lambda) - \operatorname{artanh} s(0) \leq 2 \operatorname{artanh} \varepsilon$. This proves the first bound. When $\varepsilon = 0$, it excludes every nondegenerate interval. For $\ell_0 > 0$, we instead have $y > 0$ and

$$\frac{d \log y}{d\sigma} = 1 - s \geq 1 - \varepsilon > 0. \tag{V.17}$$

Thus, y increases strictly, while the density interval gives the uniform upper bound

$$y \leq Y_{\max} = \frac{2}{1+\underline{v}^2} < 2. \tag{V.18}$$

Let us fix $\alpha \in (0, 1-\varepsilon)$. Monotonicity divides the interval into an initial portion where $y \leq \alpha$ and a remaining part where $y \geq \alpha$. Either may be empty. On the initial portion, $0 < 1 - yu^2 \leq 1$ and $0 < \partial_\eta \mathcal{F} \leq 2 + \alpha$. Using $|s| \leq \varepsilon$ in the slope inequality gives $\dot{s}\partial_\eta \mathcal{F} \geq 2(1-\alpha) - 2\varepsilon^2 - 2\alpha\varepsilon = 2(1+\varepsilon)(1-\varepsilon-\alpha) > 0$. Dividing by the positive coefficient and using its upper bound yields

$$\dot{s} \geqslant c_\alpha := \frac{2(1+\varepsilon)(1-\varepsilon-\alpha)}{2+\alpha}. \tag{V.19}$$

The slope increases by at most 2ε while remaining in $[-\varepsilon, \varepsilon]$, so this portion has logarithmic length at most $2\varepsilon/c_\alpha$. On the remaining part, the initial value of y is at least α and its final value is at most $Y_{\max}$. If we integrate (V.17), we can bound the logarithmic length of this part by $(1-\varepsilon)^{-1} \log(Y_{\max}/\alpha)$. Notice that this bound is positive since $Y_{\max} > 1 > \alpha$. The same estimates remain valid when either portion is empty. Adding them proves (V.14). □

Both bounds are finite for every fixed $\varepsilon < 1$. For $\ell_0 \leq 0$, the permitted span tends to zero with the slope tolerance. For $\ell_0 > 0$, the optimized bound has a nonzero exactly flat limit, as shown below. We first establish sharpness for the nonpositive class by constructing an analytic solution with $\ell_0 = 0$ that attains the bound. For $\ell_0 < 0$, every solution satisfies a strict inequality on each nondegenerate interval.

Corollary 3. *For every $0 < \varepsilon < 1$, equality in (V.13) is attained by a real-analytic solution with $\ell_0 = 0$, positive density and $v_{zz} = 0$. For $\ell_0 < 0$, the inequality is strict on every nondegenerate interval.*

Proof. Let us choose $r_0 > 0$, set $\tau = \log(r/r_0)$ and prescribe $u(r) = a\cosh\tau$ on $|\tau| \le \operatorname{artanh}\varepsilon$. Its logarithmic slope is $s = \tanh\tau$. The corresponding field

$$\eta(r,z) = ru(r) = \frac{a}{2}\left(\frac{r^2}{r_0} + r_0\right) \tag{V.20}$$

is independent of z. Both 1 and r^2 are $\mathcal{L}$-harmonic, so $\mathcal{L}(2\eta) = 0$. Since $\mathcal{F} = 2\eta$ and $\partial_\eta\mathcal{F} = 2$ when $\ell_0 = 0$, the auxiliary identity gives an exact velocity solution on a regular inverse sheet and the local metric reconstruction applies. Moreover, $\eta_r = ar/r_0 > 0$, so the density is positive. Choosing $0 < a < \sqrt{1-\varepsilon^2}$ ensures $u < 1$ on the entire interval, since $\cosh(\operatorname{artanh}\varepsilon) = (1-\varepsilon^2)^{-1/2}$. These strict inequalities persist in a neighbourhood of the interval and $v_{zz} = 0$ because the field is independent of z. Its endpoint radii are $R_1 = r_0 e^{-\operatorname{artanh}\varepsilon}$ and $R_2 = r_0 e^{\operatorname{artanh}\varepsilon}$, so the logarithmic span is exactly $2\operatorname{artanh}\varepsilon$. For $\ell_0 < 0$, we have $y < 0$ and the right-hand side of (V.15) is strictly positive. Hence, $\dot{s} > 1 - s^2$ and integration over any nondegenerate interval gives a strict span inequality. □

Appendix C also constructs strictly negative branches approaching the same supremum as $\ell_0 \uparrow 0$. These approximating fields are analytic solutions of the full two-dimensional equation. For $\ell_0 > 0$, the comparison function $\mathcal{B}_+$ can be minimized over α. An explicit minimizer and the small-ε expansion are given in Appendix C. The exactly flat limit can also be seen directly. Let us hold $\underline{v}$ fixed and choose $\alpha = 1 - \sqrt{\varepsilon}$ for $0 < \varepsilon < 1$. Then $0 < \alpha < 1 - \varepsilon$, the first term in $\mathcal{B}_+$ tends to zero as $\varepsilon \downarrow 0$ and the second tends to $\log[2/(1+\underline{v}^2)]$. For $\varepsilon = 0$, the speed is constant, $u = v_0$ and taking $\alpha \uparrow 1$ directly in the theorem gives

$$\frac{R_2}{R_1} \le \frac{2}{1+v_0^2}. \tag{V.21}$$

This agrees with the annular bound in Theorem 8. The next proposition shows that the limiting endpoint ratio is approached by local analytic solutions with strict density and transverse-concavity margins.

Proposition 5. *Let us fix $0 < v_0 < 1$ and $\ell_0 > 0$. Among reflection-symmetric real-analytic local solutions with $u = v_0$, strictly positive density and $v_{zz}(r,0) < 0$ on a closed equatorial interval $[R_1, R_2]$, the endpoint ratio has supremum*

$$\sup\frac{R_2}{R_1} = \frac{2}{1+v_0^2}. \tag{V.22}$$

The strict inequalities exclude attainment on such an interval.

Proof. For a constant trace, $s = \dot{s} = 0$, so (V.8) becomes $Z\partial_\eta\mathcal{F} = 2(1-y)$. Since $\partial_\eta\mathcal{F} > 0$, strict transverse concavity is equivalent to $y > 1$. Also, $\eta_r = v_0 > 0$, so strictly positive density gives $y < 2/(1+v_0^2)$. Every admissible closed interval therefore lies in the open radial window

$$\frac{v_0}{\ell_0} < r < \frac{2v_0}{\ell_0(1+v_0^2)}, \tag{V.23}$$

which proves the upper bound and excludes equality. To show that this bound is the supremum, we choose closed intervals whose endpoints approach the two boundaries of the window from within. On each interval, we prescribe the auxiliary Cauchy data $f(r,0) = \mathcal{F}(r, rv_0)$ and $f_z(r,0) = 0$ for $\mathcal{L}f = 0$ where $f(r,z)$ denotes the auxiliary field. The analytic construction in Theorem 4 applies at every point of the interval. Local uniqueness identifies the solutions on overlapping neighbourhoods and compactness provides a neighbourhood of the entire interval. Because of the positivity of $\partial_\eta\mathcal{F}$ along the prescribed trace, the local inverses agree on the sheet containing $\eta(r,0) = rv_0$. They therefore reconstruct a reflection-symmetric analytic solution. Notice that the strict density and transverse inequalities hold along the trace and moreover, compactness and continuity preserve subluminality and positive density after the neighbourhood is reduced (if necessary). The endpoint ratios tend to $2/(1+v_0^2)$, proving the stated supremum. □

For positive ℓ_0 and fixed $\varepsilon > 0$, the comparison bound in Theorem 9 is not sharp. In Appendix C, we obtain a strictly smaller bound by minimizing the lower bound for $\dot{s}$ over the relevant ranges of (u, y, s). The exact supremum of admissible spans remains undetermined because these three variables must satisfy their coupled differential equations. Thus, optimizing the explicit comparison function does not resolve the sharpness problem in this regime. Nevertheless, its uniformity in the outer endpoint is enough to exclude a semi-infinite approximately flat trace.

Corollary 4. *For fixed $R_1 > 0$ and $0 \le \varepsilon < 1$, no single solution can satisfy the hypotheses of Theorem 9 on every interval $[R_1, R_2]$ with $R_2 > R_1$.*

Proof. For $\ell_0 \le 0$, the first bound is independent of R_2. For $\ell_0 > 0$, we fix any $\alpha \in (0, 1-\varepsilon)$. Although $\underline{v}$ may decrease as R_2 increases, the explicit formula gives $\mathcal{B}_+(\varepsilon, \alpha, \underline{v}) \le \mathcal{B}_+(\varepsilon, \alpha, 0)$, where the value at zero is obtained by the same formula and is finite. Thus, in either case a constant independent of R_2 bounds $\log(R_2/R_1)$, contradicting its divergence as $R_2 \to \infty$. □

These bounds give a finite radial limitation within the constant-ℓ_0 class under the stated density, slope and rapidity-shape assumptions, despite the freedom to realize traces locally on annular neighbourhoods. They concern the speed measured by ZAMOs. Relating this local speed to spectroscopic data requires the emitter–observer relation discussed in Section IX.

VI. ASYMPTOTIC DECAY AND VACUUM CONTINUATION

We now examine whether a locally realized trace can extend along an unbounded equatorial end satisfying asymptotic flatness conditions. For $\ell_0 \neq 0$, uniform subluminality and the exact metric relations force rv to approach a positive constant, thereby fixing the leading inverse-radius decay of the speed. If the density is nonnegative on an open neighbourhood of the equatorial tail, the same relations force η to be constant there and the metric to be locally flat. Analyticity extends this constant value throughout any connected regular inverse sheet containing the tail. We then examine how the angular identification of the resulting vacuum metric restricts regular-axis completion and affects causality.

Let (t, x, y, z) be asymptotically Cartesian coordinates in an asymptotically nonrotating frame and set $\mathcal{R} = \sqrt{x^2+y^2+z^2}$ and $\varrho = \sqrt{x^2+y^2}$. We choose the equatorial surface to be $z = 0$ in both the canonical and asymptotically Cartesian charts. An equatorial tail is a set of points $(r, 0)$ with $r > R$ for some $R > 0$. With spatial indices $i, j \in \{x, y, z\}$, the metric falloff is

$$g_{tt} = -1 + O(\mathcal{R}^{-1}), \qquad g_{ti} = O(\mathcal{R}^{-1}), \qquad g_{ij} = \delta_{ij} + O(\mathcal{R}^{-1}). \tag{VI.1}$$

For the equatorial results below, these estimates are required only along the equatorial tail. Notice that this directional hypothesis does not assert the existence of a complete regular isolated exterior with uniform falloff in every spatial direction. Moreover, no decay assumptions on spatial derivatives are needed for the arguments below. Such conditions enter the formulation of the full isolated-body problem [25–27]. We first translate the Cartesian estimates into control of the azimuthal metric components and the orbit-area radius.

Lemma 3. *Let us suppose (VI.1) holds along the equatorial tail with stationary field $k = \partial_t$ and axial field $m = \partial_\phi = -y\partial_x + x\partial_y$. Then, $g_{t\phi} = O(1)$, $g_{\phi\phi} = \varrho^2 + O(\varrho)$ and $\Omega_{\mathrm{Z}} = O(\varrho^{-2})$. The orbit-area radius r satisfies $r^2 = \varrho^2 + O(\varrho)$ and $r/\varrho = 1 + O(\varrho^{-1})$. The same estimates hold on fixed cones separated from the axis whenever (VI.1) holds there.*

Proof. Contracting the Cartesian metric components with m gives $g_{t\phi} = -yg_{tx} + xg_{ty} = O(\varrho/\mathcal{R})$ and $g_{\phi\phi} = \varrho^2 + (g_{ij} - \delta_{ij})m^i m^j = \varrho^2 + O(\varrho^2/\mathcal{R})$. On the equator, $\mathcal{R} = \varrho$, which proves the first two estimates. The third follows from $\Omega_{\mathrm{Z}} = -g_{t\phi}/g_{\phi\phi}$. Substituting these estimates and $g_{tt} = -1 + O(\varrho^{-1})$ into the exact identity $r^2 = -g_{tt}g_{\phi\phi} + g_{t\phi}^2$ gives $r^2 = \varrho^2 + O(\varrho)$. Taking the positive square root yields the stated ratio. On any fixed cone separated from the axis, $\varrho/\mathcal{R}$ is bounded below by a positive constant, so the same estimates apply. □

For the constant-ℓ_0 family, the azimuthal metric component gives

$$\frac{g_{\phi\phi}}{r^2} = \frac{1 - v^2}{Ce^{\ell_0 rv}} \tag{VI.2}$$

with $C > 0$. If the speed approached a limit strictly between zero and one, this ratio would tend to zero for $\ell_0 > 0$ and to infinity for $\ell_0 < 0$, whereas the asymptotic estimates require it to tend to one. The next theorem determines the actual decay without assuming in advance that the speed has a limit.

Theorem 10. *Let a C^2 solution of the exact constant-ℓ_0 equation with $\ell_0 \neq 0$ have a regular reconstructed metric on an open neighbourhood of an equatorial tail satisfying (VI.1). Further suppose there is a constant $\delta \in (0,1)$ such that $0 < v(r,0) \leq 1-\delta$ for all sufficiently large r. Then, along the equatorial tail*

$$\lim_{r\to\infty} \eta(r,0) = \lim_{r\to\infty} rv(r,0) = q, \qquad q = -\frac{\log C}{\ell_0} > 0. \tag{VI.3}$$

At this limiting value, $\Omega(q) = 0$ and the equatorial angular velocity and speed satisfy $v = O(r^{-1})$ and $\Omega = O(r^{-2})$. More precisely,

$$\eta = q + O\left(\frac{1}{r^2}\right), \qquad v = \frac{q}{r} + O\left(\frac{1}{r^3}\right), \qquad H = -1 + O\left(\frac{1}{r^2}\right). \tag{VI.4}$$

In particular, such an end requires $C \neq 1$.

Proof. Lemma 3 gives $g_{\phi\phi}/r^2 = 1 + O(r^{-1})$. Taking the logarithm of (VI.2) yields $\ell_0\eta = \log(1-v^2) - \log C - \log(g_{\phi\phi}/r^2)$. Uniform subluminality gives $\delta(2-\delta) \leq 1 - v^2 < 1$, so every term on the right is bounded. Since $\ell_0 \neq 0$, it follows that $\eta = O(1)$ and therefore $v = \eta/r \to 0$. Returning to the same identity now gives $\eta \to q = -(\log C)/\ell_0$ with $q \geq 0$ because $\eta > 0$. The exact Killing-block relation also gives $\Omega = (\eta - g_{t\phi})/g_{\phi\phi}$. Its numerator is bounded and its denominator is asymptotic to r^2, so $\Omega = O(r^{-2})$. To exclude $q = 0$, we suppose that $\eta \to 0^+$. The identity $\Omega_{,\eta} = H_{,\eta}/(2\eta)$ with $H = -Ce^{\ell_0\eta}$ then gives $\Omega_{,\eta} = -C\ell_0/(2\eta) + O(1)$. Integration yields the logarithmic behaviour also visible in (II.6), i.e.

$$|\Omega| = \frac{C|\ell_0|}{2}\left|\log\left(\frac{\eta}{\eta_*}\right)\right|[1 + o(1)] \longrightarrow \infty. \tag{VI.5}$$

Here the fixed reference scale $\eta_* > 0$ makes the logarithm dimensionless. This divergence contradicts $\Omega = O(r^{-2})$, so $q > 0$. The function $\Omega(\eta)$ is smooth near q and its decay along the tail implies $\Omega(q) = 0$. Since $Ce^{\ell_0 q} = 1$, we have $H(q) = -1$ and $\Omega_{,\eta}(q) = -\ell_0/(2q) \neq 0$. The inverse-function theorem therefore gives a local bound $|\eta - q| \leq K|\Omega(\eta)|$ for some constant $K > 0$. Along the tail this implies $\eta - q = O(r^{-2})$. Division by r gives $v = q/r + O(r^{-3})$ and expansion of $H = -Ce^{\ell_0\eta}$ about q gives $H = -1 + O(r^{-2})$. Finally, $C = 1$ would imply $q = 0$, which has been excluded. □

The decay argument uses only the exact metric relations and requires no sign condition on the density. In the logarithmic low-velocity expansion used in [6], the relevant parameter is $\ell_0\eta$. Its magnitude grows without bound along a fixed nonzero-speed trace, so extrapolation to arbitrarily large radius requires a uniform remainder estimate. Appendix D gives the corresponding conditional asymptotic test and explains this loss of uniformity. We now impose nonnegative density on an open neighbourhood of the tail. This forces an open region with constant η and vanishing density. The regular inverse-sheet hypothesis is then used to extend that constant value by analyticity.

Theorem 11. *Under the hypotheses of Theorem 10, let us suppose there is $R_0 > 0$ such that every equatorial point $(r,0)$ with $r > R_0$ has an open neighbourhood on which $\rho \geq 0$. Then there is a connected open neighbourhood U of an equatorial tail on which $\eta = q$ and $\rho = 0$. If U is contained in a connected open domain D of the same solution on which $\partial_\eta \mathcal{F}(r,\eta(r,z)) \neq 0$, these identities hold throughout D. After asymptotic normalization, the metric on this connected vacuum plateau is*

$$ds^2 = -(dt - qd\phi)^2 + r^2 d\phi^2 + dr^2 + dz^2. \tag{VI.6}$$

The coordinate t is single-valued and $\phi \sim \phi + 2\pi$ identifies angular coordinates differing by 2π at fixed t, r, z, so the axial orbits are closed.

Proof. Theorem 10 gives $v = O(r^{-1})$. Hence, both endpoints of the density interval (III.3) tend to zero along the tail, while $|r\ell_0| \to \infty$. For all sufficiently large equatorial points, $r\ell_0$ therefore lies strictly outside the interval and the algebraic density factor $\mathcal{N}_v(r\ell_0)$ is negative. At each such point, we choose a small connected

open neighbourhood contained in the region where $\rho \geq 0$ and where this factor remains negative. (III.2) then forces $\nabla_{rz}\eta = 0$ throughout that neighbourhood, so η is constant and $\rho = 0$ there. These neighbourhoods cover the tail and each meets it. Since the tail is connected, their union U is connected. The local constant values therefore agree and the limit $\eta \to q$ fixes their common value as q. If $U \subset D$ with D as in the statement, Theorem 2 gives $\eta = q$ throughout D and the density formula gives $\rho = 0$ there. On the plateau, $H = H(q) = -1$ and $\Omega = \Omega(q) = 0$. The quadratures (IV.5)-(IV.6) give $\mu = \mu_0$ for a constant μ_0. To determine this constant from the stated falloff, let L denote length in the orbit metric along the equatorial radial curve, measured from a fixed point on the tail. In canonical coordinates, $L = e^{\mu_0/2}(r - r_*)$ for a fixed reference radius r_*. In the asymptotically Cartesian chart, the orbit-metric radial component is $1 + O(\varrho^{-1})$. Indeed, the spatial metric has this falloff and projecting out the stationary and axial directions changes that component only by $O(\varrho^{-2})$. Integration therefore gives $L/\varrho \to 1$. Combining this with $r/\varrho \to 1$ yields $e^{\mu_0/2} = 1$, so $\mu_0 = 0$ without any assumptions on derivatives of the asymptotic remainder. Finally, (II.3) gives $g_{tt} = -1$, $g_{t\phi} = q$ and $g_{\phi\phi} = r^2 - q^2$, which proves (VI.6). □

Thus, a connected regular inverse sheet containing a point of positive density cannot also contain the nonnegative-density equatorial end described above. Completing a finite dust body therefore requires a different vacuum solution unless at least one hypothesis of the continuation theorem fails. The plateau itself also has a restriction on continuation towards a regular axis, as the following corollary shows.

Corollary 5. *If the plateau in* (VI.6) *extends to a connected flat region carrying the same stationary and axial Killing fields and containing a regular fixed point of the axial action, then* $q = 0$. *Consequently, the positive-speed end of Theorem 10 admits no regular-axis completion of this kind.*

Proof. On the plateau, $g(k, m) = q$. In local flat coordinates, Killing fields are affine functions of the coordinates, so their scalar products are real analytic. The identity $g(k, m) = q$ therefore continues throughout the connected flat extension. At a regular axial fixed point, $m = 0$, which gives $q = 0$. This contradicts the positive value obtained in Theorem 10. □

To describe the global effect of the angular identification, we unwrap the periodic coordinate so that $\phi \in \mathbb{R}$. On this angular cover, $T = t - q\phi$ puts the metric into the flat cylindrical form

$$ds^2 = -dT^2 + dr^2 + r^2 d\phi^2 + dz^2. \tag{VI.7}$$

The original periodicity becomes $(T, \phi) \mapsto (T - 2\pi q, \phi + 2\pi)$. Thus, a complete turn at fixed t changes T by $-2\pi q$ and T is not single-valued on the original spacetime. The locally flat metric therefore retains a joint identification of time and angle. The reconstructed subluminal branch satisfies $v = q/r < 1$ and therefore lies in $r > q$. Throughout this region, the single-valued coordinate t has timelike gradient because $g^{tt} = -1 + q^2/r^2 < 0$. Thus, t is a global time function and the quotient is stably causal. If the vacuum metric is extended to $0 < r < q$, the axial field becomes timelike because $g_{\phi\phi} = r^2 - q^2 < 0$ and its periodic orbits are closed timelike curves. At $r = q$ those orbits are null. These latter regions belong to an extension of the vacuum metric beyond the subluminal branch. Finally, the plateau satisfies the stated falloff along the equator and on fixed cones separated from the axis, but extension by the same formula does not give a complete regular isolated end. In Cartesian coordinates $x = r\cos\phi$, $y = r\sin\phi$, we have $d\phi = (-ydx + xdy)/r^2$. Hence, we find

$$g_{tx} = -\frac{qy}{r^2}, \qquad g_{ty} = \frac{qx}{r^2}, \qquad \sqrt{g_{tx}^2 + g_{ty}^2} = \frac{q}{r}. \tag{VI.8}$$

The spatial corrections are $O(r^{-2})$, so (VI.1) holds where r and $\mathcal{R}$ are comparable. At fixed $r > q$, however, the mixed-component norm remains nonzero as $|z| \to \infty$ and therefore fails uniform $O(\mathcal{R}^{-1})$ falloff. The neighbourhood supplied by Theorem 11 need not contain such a vertical sequence. Its existence consequently leaves the construction of a complete isolated exterior as a separate problem.

VII. FINITE BOUNDARIES AND SHELL-FREE MATCHING

For a finite dust body, completion requires matching the interior geometry to a vacuum exterior across a timelike boundary. We first identify candidate boundaries on which the density vanishes and compute their

induced metric and extrinsic curvature. These data determine whether a proposed exterior can be joined to the interior without a surface layer. If the induced metrics agree but the extrinsic curvatures differ, the jump determines the surface stress. We also derive the boundary values and normal derivatives of the orbit-area radius that any such exterior must realize.

Let $D \subset (0,\infty)\times\mathbb{R}$ be an open regular meridional region on which $0 < v < 1$. The density factorization (III.2) gives the decomposition $\{(r,z)\in D \mid \rho(r,z)=0\} = \mathcal{C}\cup\mathcal{Z}_-\cup\mathcal{Z}_+$ with

$$\mathcal{C} = \big\{(r,z)\in D \mid \nabla_{rz}\eta(r,z)=0\big\}, \quad \mathcal{Z}_\pm = \left\{(r,z)\in D \;\middle|\; r\ell_0 = \pm\frac{2v}{1\pm v^2}\right\}. \tag{VII.1}$$

Here v is evaluated at (r,z). The critical set $\mathcal{C}$ may intersect either algebraic branch, while $\mathcal{Z}_\pm$ are disjoint because their defining values of $r\ell_0$ have opposite signs. These sets supply candidates when the density is required to approach zero at the boundary. Vanishing boundary density is not, however, a general requirement for shell-free dust-vacuum matching, as the normal matter constraints below will show. A smooth embedded curve γ in the zero-density set generates the symmetry-preserving hypersurface $\Sigma = \mathbb{R}_t\times S^1_\phi\times\gamma$. In a regular region with $r>0$, the negative determinant of the Killing block gives one timelike and one spacelike tangent direction, while the meridional tangent supplies a second spacelike direction. Thus, Σ is timelike. Notice that singular points and degenerate axial orbits require separate analysis. Let us parametrize γ by Euclidean meridional arclength λ with a dot denoting $d/d\lambda$ so that $\dot r^2+\dot z^2=1$. we further define the Euclidean unit tangent $T_{\rm E}$ and unit normal $N_{\rm E}$ by

$$T_{\rm E} = \dot r\partial_r + \dot z\partial_z, \qquad N_{\rm E} = -\dot z\partial_r + \dot r\partial_z, \qquad \frac{dT_{\rm E}}{d\lambda} = \kappa_{\rm E}N_{\rm E}, \tag{VII.2}$$

where $\kappa_{\rm E} = \dot r\ddot z - \dot z\ddot r$ is the signed Euclidean curvature. For the physical orbit metric $\mathsf{q} = e^\mu(dr^2+dz^2)$, proper arclength $\lambda_{\rm p}$ and the corresponding unit fields are $d\lambda_{\rm p} = e^{\mu/2}d\lambda$, $e_s = e^{-\mu/2}T_{\rm E}$ and $n = e^{-\mu/2}N_{\rm E}$. We denote the induced metric by h and keep the convention $K(X,Y) = g(\nabla_X n, Y)$ for tangent vectors X, Y to Σ. The normal orientation fixes the sign of both K and $\kappa_{\rm E}$. With these conventions, the two fundamental forms take the following reduced form.

Lemma 4. *Let us consider the intrinsic basis $(\partial_t,\partial_\phi,e_s)$ on Σ, where $e_s = d\gamma/d\lambda_{\rm p} = e^{-\mu/2}T_{\rm E}$. Then, the first and second fundamental forms are*

$$h_{AB} = G_{AB}, \qquad h_{As} = 0, \qquad h_{ss} = 1, \tag{VII.3}$$

$$K_{AB} = \frac{1}{2}n^\nu\partial_\nu G_{AB}, \qquad K_{As} = 0, \qquad K_{ss} = e^{-\mu/2}\left[\frac{1}{2}N_{\rm E}(\mu) - \kappa_{\rm E}\right]. \tag{VII.4}$$

with $A,B\in\{t,\phi\}$ Here, $N_{\rm E}(\mu)$ denotes the directional derivative of μ along the Euclidean unit normal.

Proof. Restriction of the block metric to Σ gives (VII.3). The meridional normal is invariant under the two Killing flows, so $[n,\partial_A]=0$. The identity $2K=\mathcal{L}_n g$ on tangent vectors then gives $2K_{AB} = n^\nu\partial_\nu G_{AB}$. Block orthogonality and independence of t and ϕ give $K_{As}=0$. For the meridional component, we set $\omega=\mu/2$ and let $\delta = dr^2+dz^2$ denote the Euclidean metric with Levi-Civita connection $\nabla^{\rm E}$. The conformal connection of $\mathsf{q} = e^{2\omega}\delta$ satisfies $\nabla^{\mathsf{q}}_X Y = \nabla^{\rm E}_X Y + X(\omega)Y + Y(\omega)X - \delta(X,Y)\nabla^{\rm E}\omega$. Here $\nabla^{\rm E}\omega$ denotes the Euclidean gradient. Using $e_s = e^{-\omega}T_{\rm E}$, $n = e^{-\omega}N_{\rm E}$ and $\nabla^{\rm E}_{T_{\rm E}}N_{\rm E} = -\kappa_{\rm E}T_{\rm E}$ gives $\nabla^{\mathsf{q}}_{e_s}n = e^{-2\omega}[(N_{\rm E}(\omega)-\kappa_{\rm E}]T_{\rm E}$. Taking the q inner product with e_s yields $K_{ss} = e^{-\omega}[N_{\rm E}(\omega)-\kappa_{\rm E}]$, which is the stated expression. □

For a proposed matching, let $-$ denote the dust interior and $+$ a stationary, axisymmetric, circular vacuum exterior. We use local conformal meridional coordinates on each side, with orbit metrics $\mathsf{q}_\pm = e^{\mu_\pm}(dr_\pm^2+dz_\pm^2)$. Moreover, we fix a boundary identification preserving the stationary and axial Killing basis and the axial period [28]. Both unit normals point from the dust towards the vacuum and $[Q] = Q^+ - Q^-$ denotes the jump after pulling the two quantities back to the common boundary. The meridional chart orientations are chosen so that (VII.2) gives this normal direction on both sides. Let $\gamma_\pm(\xi) = (r_\pm(\xi), z_\pm(\xi))$, where the same parameter value ξ labels identified boundary points and a prime denotes $d/d\xi$. This parameter need not be Euclidean arclength in either chart. An identification preserving the Killing basis could also include shifts $t_+ = t_- + f(\xi)$ and $\phi_+ = \phi_- + g(\xi)$. Their mixed induced-metric components would be $h^+_{A\xi} = G^+_{At}f' + G^+_{A\phi}g'$. Since the interior mixed components vanish and G^+ is invertible, equality of the induced metrics forces $f' = g' = 0$.

Absorbing the remaining constant shifts into the coordinate origins, we may therefore use the same intrinsic coordinates (t, ϕ, ξ) on both sides. Equality of the remaining induced-metric components requires

$$G^{+}_{AB} = G^{-}_{AB}, \tag{VII.5}$$

$$e^{\mu_+}[(r'_+)^2 + (z'_+)^2] = e^{\mu_-}[(r'_-)^2 + (z'_-)^2]. \tag{VII.6}$$

Only after these conditions hold does the common induced metric determine a common proper arclength and unit tangent e_s. In evaluating the second fundamental forms, $N_{\rm E}$ and $\kappa_{\rm E}$ are computed separately on each side using its own Euclidean arclength and the prescribed common normal direction. The next proposition states the resulting junction conditions.

Proposition 6. *Let a rotating-dust interior and a circular vacuum exterior have smooth timelike symmetry-preserving boundaries away from the axis with each metric C^2 up to its boundary. Moreover, we fix a boundary identification and the common normal orientation described above. If $[h] \neq 0$, this identification does not define a standard Darmois-Israel matching. If $[h] = 0$, the matching has no surface layer precisely when $[K_{ab}] = 0$. In the common proper-arclength basis, the complete conditions are (VII.5)-(VII.6) together with*

$$[n^\nu \partial_\nu G_{AB}] = 0, \qquad \left[e^{-\mu/2}\left(\tfrac{1}{2}N_{\rm E}(\mu) - \kappa_{\rm E}\right)\right] = 0. \tag{VII.7}$$

If $[h] = 0$ but $[K_{ab}] \neq 0$, the resulting continuous metric has a surface stress-energy tensor

$$S_{ab} = -\frac{1}{8\pi G_{\rm N}}\left([K_{ab}] - h_{ab}[K]\right), \qquad [K] = h^{ab}[K_{ab}]. \tag{VII.8}$$

In particular, vanishing density on a candidate boundary does not replace any of these geometric matching conditions.

Proof. Equality of the first fundamental forms is the continuity requirement for the standard junction construction [17–19]. Once it holds, the absence of a surface layer is equivalent to equality of the second fundamental forms. Lemma 4 gives exactly the two normal matching conditions above, since the mixed components of K vanish on both sides. When the second fundamental forms differ, the distributional Einstein equation gives (VII.8) with the stated normal and jump conventions. Taking the trace with the common induced metric and using $h^{ab}h_{ab} = 3$, gives

$$h^{ab}S_{ab} = \frac{[K]}{4\pi G_{\rm N}}. \tag{VII.9}$$

Thus, $S_{ab} = 0$ implies first $[K] = 0$ and then $[K_{ab}] = 0$. The converse follows by direct substitution. To clarify the role of the boundary density, we recall that the circular dust four-velocity is given by (II.2). Both Killing fields therein are tangent to a symmetry-preserving boundary, so $u \cdot n = 0$ and $T_{\mu\nu}n^\nu = 0$ regardless of ρ. The Gauss and Codazzi equations determine the normal Einstein projections from the boundary data (h, K). Hence, these projections agree at a Darmois boundary and match their zero vacuum values without imposing $\rho = 0$. A finite jump in the volume density is therefore not excluded by the junction constraints when the metric is C^2 on each side. Requiring $\rho = 0$ is an additional restriction that makes the full matter tensor vanish at the boundary. In either case, a vacuum exterior realizing the required metric and normal data must still be found. □

There is also a distinction between cutting an existing vacuum region and constructing a new exterior. Within a C^2 vacuum neighbourhood, the two sides of a smooth cut inherit the same metric, so their fundamental forms agree. For a smooth candidate curve in $\mathcal{C} \cup \mathcal{Z}_- \cup \mathcal{Z}_+$, the formulas above instead specify data that an exterior must realize. The existence of that exterior remains a separate field-equation problem. The matching conditions also determine the orbit-area radius and its normal derivative, as follows.

Corollary 6. *Let $\varpi = \sqrt{-\det G} > 0$ in the common stationary and axial Killing basis. Every shell-free matching satisfies $[\varpi] = 0$ and $[n^\nu \partial_\nu \varpi] = 0$. For the dust interior, $\varpi^- = r$. Whenever the exterior admits canonical Weyl coordinates, their orbit-area radius must therefore realize the boundary data*

$$\varpi^+|_\Sigma = r|_\Sigma, \qquad n^\nu \partial_\nu \varpi^+|_\Sigma = n^r_\Sigma. \tag{VII.10}$$

Here, each normal derivative is taken on its own side and n^r_Σ denotes the interior normal component $(n^-)^r$ evaluated on the boundary.

Proof. Continuity of the induced metric gives $[G] = 0$, so the determinants agree and $[\varpi] = 0$. Since $2K_{AB} = n^\nu \partial_\nu G_{AB}$, continuity of the second fundamental form also gives $[n^\nu \partial_\nu G] = 0$. Applying Jacobi's formula to $\det G = -\varpi^2$ on each side yields

$$\frac{2}{\varpi} n^\nu \partial_\nu \varpi = \operatorname{tr}\left(G^{-1} n^\nu \partial_\nu G\right). \tag{VII.11}$$

Both G and its normal derivative agree across the boundary, so the right-hand side has the same value on each side. Since ϖ is positive and continuous, its normal derivative must also agree. Substituting $\varpi^- = r$ and $n^\nu \partial_\nu r = n^r$ gives (VII.10). □

Appendix E presents the surface tensor in a symmetry-adapted basis and gives explicit formulas for a cylindrical boundary. The conditions derived here specify the full local boundary geometry required for shell-free matching. The next section addresses the topology and compatibility requirements for a vacuum exterior realizing these data.

VIII. TOPOLOGY AND COMPATIBILITY OF THE EXTERIOR

The boundary data derived above must be realized by a vacuum metric on the entire exterior domain. We first examine the role of topology. A body whose meridional section is a disk separated from the axis is, under the assumptions below, a solid torus. The resulting exterior has a noncontractible cycle, so closed reconstruction forms need not admit single-valued potentials. In particular, a global scalar twist potential requires a vanishing twist period, whereas a globally defined metric may admit local twist scalars differing by additive constants. We then study the compatibility of the boundary values and normal derivatives supplied by the interior. Linearization about Minkowski space gives a Dirichlet-to-Neumann criterion for the scalar Ernst equations on a fixed bounded domain. This criterion resolves that linear boundary problem, while the nonlinear isolated completion requires further reconstruction, regularity and asymptotic conditions.

A. Toroidal bodies and period conditions

The relevant period measures the integral of a closed one-form around a loop surrounding the body in the orbit surface. The following theorem identifies the topology and the periods needed for global Weyl and twist potentials.

Theorem 12. *Let $\mathscr{S}$ be a smooth spacelike section of the stationary flow, invariant under an effective axial $U(1)$ action, with a regular axis and no exceptional orbits in the region considered. The chosen stationary slicing identifies $Q = \mathscr{S}/U(1)$ with the two-Killing orbit surface. Let π be the axial quotient projection and suppose Q is diffeomorphic to a closed half-plane. We further assume that a compact connected invariant body $\Omega_{\mathrm{mat}} \subset \mathscr{S}$ projects onto a smoothly embedded closed disk B in the interior of Q. Then, $\Omega_{\mathrm{mat}} \simeq D^2 \times S^1$ is a solid torus and $\partial\Omega_{\mathrm{mat}} \simeq T^2 = S^1 \times S^1$. The exterior orbit surface $Q^+ = Q \setminus \operatorname{int} B$ has fundamental group $\pi_1(Q^+) \simeq \mathbb{Z}$ and first de Rham cohomology $H^1_{\mathrm{dR}}(Q^+) \simeq \mathbb{R}$. Suppose the exterior metric is circular and vacuum with a Lorentzian Killing block away from the axis. On the regular interior of Q^+, let q be the orbit metric, $\varpi = \sqrt{-\det G}$ and $*_{\mathsf{q}}$ be the Hodge star for a chosen orientation. The one-forms $*_{\mathsf{q}} d\varpi$ and the stationary twist form ω_k descended to this surface are closed. If γ is a smooth closed curve representing a generator of the first homology group of Q^+ with integer coefficients, single-valued potentials satisfying $d\zeta = *_{\mathsf{q}} d\varpi$ and $d\chi = \omega_k$ exist precisely when their respective periods vanish, i.e.*

$$P_\zeta = \oint_\gamma *_{\mathsf{q}} d\varpi = 0, \qquad P_\chi = \oint_\gamma \omega_k = 0. \tag{VIII.1}$$

Where $d\varpi \neq 0$, (ϖ, ζ) are local Weyl coordinates. They give a global chart on the regular interior only if $d\varpi$ is nonzero throughout that domain and the coordinate map is injective. Axis values are treated by regular extension to the boundary of the orbit surface.

Proof. Invariance of the body gives $\Omega_{\mathrm{mat}} = \pi^{-1}(B)$. Since B lies away from the axis and there are no exceptional orbits, the axial action is free over B. Thus, $\pi^{-1}(B)$ is a principal circle bundle over a contractible

disk and is therefore trivial. This gives the solid torus and its toroidal boundary. Removing the interior disk from the half-plane leaves a surface with the homotopy type of a circle, which proves the statements about π_1 and H^1_{dR}. Passing to the regular interior does not change this homotopy type. For a circular vacuum metric, the orbit-area function ϖ is harmonic with respect to q, so $d(*_{\mathsf{q}} d\varpi) = 0$. The stationary twist one-form is defined by $(\omega_k)_\tau = \epsilon_{\tau\nu\rho\sigma} k^\nu \nabla^\rho k^\sigma$, where ϵ is the spacetime volume tensor. The Killing identity gives $d\omega_k = 0$ in vacuum. The form is invariant under both Killing flows and annihilates their tangent directions, the latter property following from circularity. It therefore descends to a closed one-form on the regular orbit surface [4, 29]. A closed one-form is exact precisely when its periods vanish. Here, every homology class is an integer multiple of $[\gamma]$, so a single period determines exactness. Applying this criterion to the two closed forms proves (VIII.1). Finally, $d\zeta = *_{\mathsf{q}} d\varpi$ implies $d\varpi \wedge d\zeta = |d\varpi|^2_{\mathsf{q}} dA_{\mathsf{q}}$, where dA_{q} is the oriented area form. Thus, the two differentials are independent wherever $d\varpi \neq 0$, giving a local chart. Nonvanishing throughout the domain and injectivity make this map a global chart onto its image. Notice that the regular axis is a boundary of the orbit surface, where the Killing block degenerates, and is not part of this nondegenerate coordinate argument. □

For n disjoint bodies with disk-shaped meridional sections, the exterior has n independent cycles. Each closed reconstruction form then requires one vanishing period around each removed disk to admit a global potential. Twist periods should instead be retained as data whenever a single-valued twist scalar is not imposed. A separate period condition enters the reconstruction of the meridional conformal factor. In a fixed global Weyl chart, let us write the vacuum quadratures as $d\mu = \alpha_\mu$, where α_μ is the globally defined one-form constructed from compatible vacuum fields. The field equations give $d\alpha_\mu = 0$, ensuring local primitives. A single-valued μ exists only when its periods vanish. For one body, this is the condition

$$P_\mu = \oint_\gamma \alpha_\mu = 0. \tag{VIII.2}$$

These reconstruction conditions accompany the Darmois data at the material boundary and are not supplied by equality of the boundary fundamental forms alone. The twist period also has a direct interpretation as a matter-current flux. Let us write $k^\flat = g(k, \cdot)$ and $R(k)^\flat = R_{\mu\nu} k^\nu dx^\mu$. By $*$, we denote the four-dimensional Hodge star. The Killing identity gives [29]

$$d\omega_k = 2 * \left(k^\flat \wedge R(k)^\flat \right). \tag{VIII.3}$$

For dust, $T_{\mu\nu} = \rho u_\mu u_\nu$ and $g(u,u) = -1$, so Einstein's equation gives $R_{\mu\nu} = 8\pi G_N \rho (u_\mu u_\nu + \frac{1}{2} g_{\mu\nu})$. Substituting $u = U(k + \Omega m)$ with $U = (-H)^{-1/2} > 0$ yields

$$R(k)^\flat = 8\pi G_N \rho \left[(k \cdot u) u^\flat + \frac{1}{2} k^\flat \right], \qquad d\omega_k = 16\pi G_N \rho (k \cdot u) U \Omega * \left(k^\flat \wedge m^\flat \right). \tag{VIII.4}$$

The term proportional to $k^\flat$ drops out of the wedge product. On a meridional section of the body, $* \left(k^\flat \wedge m^\flat \right) = \pm r dA_{\mathsf{q}}$ with the sign fixed by orientation. Let us choose γ homologous to ∂B in the exterior and orient them consistently. Closedness in the vacuum, continuity of the twist form under Darmois matching in the common Killing basis and Stokes' theorem give, in succession

$$\oint_\gamma \omega_k^+ = \oint_{\partial B} \omega_k^+ = \oint_{\partial B} \omega_k^- = \int_B d\omega_k^-. \tag{VIII.5}$$

The continuity used here follows because the twist depends on the metric and its first derivatives, which agree in coordinates adapted to a Darmois matching. Hence, we have

$$P_\chi = \pm 16\pi G_N \int_B \rho (k \cdot u) U \Omega r \; dA_{\mathsf{q}}. \tag{VIII.6}$$

If the asymptotically normalized k and the dust velocity u are future timelike in the body, then $k \cdot u < 0$. If $\rho \geq 0$ is positive on a set of positive area and Ω has one strict sign wherever $\rho > 0$, the integral is nonzero. Thus, requiring $P_\chi = 0$ restricts the matter current and does not follow from metric regularity or asymptotic flatness. A weak-field example illustrates the distinction. A smooth compact toroidal azimuthal current of one sign admits a globally defined vector potential, while the associated gravitomagnetic field has nonzero

circulation around a meridional loop linking the current. In this weak-field example, continuation around a loop linking the current changes the local twist scalar by an additive constant. Relativistic Dyson rings provide a separate numerical comparison with nonlinear toroidal perfect-fluid equilibria [30]. The period interpretation used here follows from (VIII.6) and makes no claim that those perfect-fluid solutions solve the present dust problem.

Theorem 12 characterizes the existence of global scalar potentials. A metric completion may use local twist scalars χ_i satisfying $d\chi_i = \omega_k$ with constant differences $\chi_i - \chi_j$ on connected overlaps. Continuation around γ changes the scalar by P_χ, while its differential remains single-valued. Requiring a global twist scalar therefore selects the additional subclass $P_\chi = 0$. Whether the twist scalar is global or defined only locally, the metric coefficients must agree on overlaps and satisfy all reconstruction conditions.

B. The global completion problem

We now collect the conditions for an isolated vacuum completion. Unlike the directional falloff used in the equatorial decay argument, this definition requires a full asymptotically flat end.

Definition 2. An admissible isolated vacuum completion across Σ is a connected stationary, axisymmetric, circular exterior $(\mathcal{M}^+, g^+)$ with commuting Killing fields k^+ and m^+ satisfying the following conditions.

1. The metric is C^2 up to Σ, smooth in the exterior and satisfies $R_{\mu\nu}[g^+] = 0$. Axial orbits have period 2π.

2. The boundary is smooth, connected, timelike and symmetry-preserving. Its identification with the dust boundary preserves the Killing basis and axial period and the first and second fundamental forms agree.

3. The exterior has no additional boundary, horizon, curvature singularity or conical defect. Every axis segment in its closure is regular.

4. The metric is globally defined. Coefficients reconstructed from local potentials agree on overlaps and satisfy their integrability and period conditions. Local twist scalars may differ by additive constants, so P_χ need not vanish. Wherever a global Weyl chart is used on the regular orbit surface, $P_\zeta = 0$, $d\varpi$ is nowhere zero and the coordinate map is injective. Reconstruction of a single-valued meridional conformal factor in that chart also requires $P_\mu = 0$. Each required period condition is imposed around every independent cycle.

5. There is one asymptotically flat end satisfying (VI.1) uniformly in all spatial directions. The stationary field is normalized to unit time translation and the angular coordinate is asymptotically nonrotating.

A density cutoff can identify a candidate boundary within the interior solution. A shell-free completion must realize the full data $(G_{AB},\ n^\nu\partial_\nu G_{AB},\ K_{ss})\big|_\Sigma$ together with the common meridional line element. In a local Ernst formulation, these data determine boundary values and normal derivatives of the potentials up to their normalization constants. For nonzero twist periods, the local potentials also require compatible constants on overlaps or prescribed jumps across cuts. Since the vacuum equations relate boundary values to their normal derivatives, prescribing both requires a compatibility test. We formulate such a test on a connected bounded open domain $D \subset \mathbb{R}^2$ with smooth boundary and $\overline{D} \subset (0,\infty)\times\mathbb{R}$. We work in a fixed Weyl chart (r, z), where $r = \varpi$, and fix the stationary and axial Killing basis. On a region where $f = -g(k,k) > 0$, the Ernst variables are $X = (f, \chi)$. For the scalar problem below, χ is globally single-valued, so we restrict to the zero twist-period sector. We retain the additive constant mode when stating the boundary-space theorem and choose a consistent twist normalization when applying it to particular data. The Ernst equations follow from the action

$$I_{\mathrm{E}}[f,\chi] = \frac{1}{2}\int_D \frac{r}{f^2}\left(|\nabla f|^2 + |\nabla\chi|^2\right)\, dr dz. \tag{VIII.7}$$

Here, ∇ and its norm refer to the Euclidean metric $dr^2 + dz^2$. Linearization about $(f,\chi) = (1,0)$ gives two decoupled scalar equations. Their Dirichlet-to-Neumann map associates each boundary value with the conormal derivative of its unique solution. The graph of this map is therefore the set of compatible Cauchy data.

Theorem 13. *Let D be as above. Each scalar perturbation $w \in \{\delta f, \delta\chi\}$ at the Minkowski solution satisfies*

$$\partial_r(r\partial_r w) + \partial_z(r\partial_z w) = 0. \tag{VIII.8}$$

Let ν denote the outward Euclidean unit normal to ∂D. For the pair $\mathbf{w} = (\delta f, \delta\chi)$, we define $\Lambda_{\mathrm{E}}\varphi = r\partial_\nu \mathbf{w}|_{\partial D}$ using the unique $H^1(D, \mathbb{R}^2)$ weak solution with trace $\varphi \in H^{1/2}(\partial D, \mathbb{R}^2)$. The conormal trace is understood weakly when necessary, so Λ_{E} maps $H^{1/2}$ continuously into $H^{-1/2}$. This map is symmetric under boundary duality and its natural realization on $L^2(\partial D, \mathbb{R}^2)$ is nonnegative and self-adjoint with domain $H^1(\partial D, \mathbb{R}^2)$. Its principal symbol is $\sigma_1(\Lambda_{\mathrm{E}})(x, \xi) = r(x)|\xi| \operatorname{Id}_{\mathbb{R}^2}$. Here, $x \in \partial D$, ξ is a nonzero tangential covector and $|\xi|$ is its norm in the induced Euclidean boundary metric. The graph of Λ_{E} is the full Cauchy-data space of these scalar equations and is an exact Lagrangian subspace of $H^{1/2}(\partial D, \mathbb{R}^2) \oplus H^{-1/2}(\partial D, \mathbb{R}^2)$ with its canonical symplectic form. Thus, prescribed linearized scalar boundary values φ and conormal data p extend over D precisely when $p = \Lambda_{\mathrm{E}}\varphi$.

Proof. Let us write $f = 1 + \epsilon w_1$ and $\chi = \epsilon w_2$. The coefficient of ϵ^2 in (VIII.7) is

$$I_{\mathrm{E}}^{(2)}[w_1, w_2] = \frac{1}{2}\int_D r\left(|\nabla w_1|^2 + |\nabla w_2|^2\right)\, drdz. \tag{VIII.9}$$

Variation gives (VIII.8) for each component. Since r is bounded above and away from zero on $\overline{D}$, the weighted Dirichlet form is coercive on $H_0^1(D)$ by Poincare's inequality. Extending an $H^{1/2}$ boundary value to $H^1(D)$ and solving for a zero-trace correction gives existence by the Lax-Milgram theorem. Testing the difference of two solutions with the same trace against itself gives $\int_D r|\nabla w|^2 = 0$, which proves uniqueness. For boundary values φ and ψ, let $\mathbf{w}_\varphi$ and $\mathbf{w}_\psi$ denote their solutions. Green's identity gives

$$\langle \varphi, \Lambda_{\mathrm{E}}\psi\rangle = \int_D r\nabla\mathbf{w}_\varphi \cdot \nabla\mathbf{w}_\psi\, drdz = \langle \psi, \Lambda_{\mathrm{E}}\varphi\rangle. \tag{VIII.10}$$

The brackets pair $H^{1/2}$ boundary values with $H^{-1/2}$ conormal data and the dot product sums over the two scalar components. This proves symmetry and nonnegativity. For the canonical boundary form $\sigma\left((\varphi, p), (\psi, q)\right) = \langle\varphi, q\rangle - \langle\psi, p\rangle$, symmetry makes the graph isotropic. If (φ, p) is symplectically orthogonal to every $(\psi, \Lambda_{\mathrm{E}}\psi)$, then $\langle\varphi, \Lambda_{\mathrm{E}}\psi\rangle = \langle\psi, p\rangle$ for all ψ. Symmetry therefore gives $p = \Lambda_{\mathrm{E}}\varphi$. The graph equals its symplectic orthogonal and is Lagrangian. The functional $W(\varphi) = \frac{1}{2}\langle\varphi, \Lambda_{\mathrm{E}}\varphi\rangle$ satisfies $dW(\varphi)[\psi] = \langle\psi, \Lambda_{\mathrm{E}}\varphi\rangle$, so the graph is the graph of dW and is exact. To obtain the principal symbol, we freeze r at a boundary point and flatten the boundary. The principal equation is then the half-space Laplace equation. Its decaying tangential Fourier mode has outward normal derivative $|\xi|$ times its boundary value. Multiplication by $r(x)$ gives the stated symbol for each component. The nonnegative closed boundary energy form defines the self-adjoint L^2 realization and elliptic regularity for this first-order boundary operator gives its domain $H^1(\partial D, \mathbb{R}^2)$. □

The theorem concerns the two globally single-valued scalar fields on the fixed domain D, including their constant modes. A normalization of the twist scalar can be chosen consistently when comparing data because adding a constant does not change its conormal derivative. For nonzero twist periods, one must instead use local potentials with constant transitions, a domain cut along selected curves with prescribed jumps or the closed twist one-form itself. Even in the zero twist-period sector, scalar solvability does not settle every metric-reconstruction condition. At the Minkowski background, reconstruction of the rotational metric coefficient involves the one-form $r *_{\mathrm{E}} d(\delta\chi)$, up to an orientation sign, where $*_{\mathrm{E}}$ is the Euclidean meridional Hodge star. The linearized twist equation makes this form closed, but its periods need not vanish on a multiply connected domain. Those periods must also be checked if the reconstructed coefficient is to be single-valued. Applying the scalar criterion to interior data also requires the correct normal convention. The map uses the outward Euclidean normal to D, whereas the Darmois data use the physical unit normal pointing from dust to vacuum. At a material inner boundary of the exterior, these directions are opposite. Their normalization must also be converted using the meridional conformal factor before comparing conormal data. At the Minkowski background this factor is one, so the first-order comparison retains only the orientation sign. A complete isolated exterior additionally requires conditions at infinity and on the axis, the twist periods fixed by the interior current, and the independent orbit-area and meridional metric data. Appendix F sets out the hypotheses for the nonlinear formulation. These compatibility questions have several precedents. Zingg, Aste and Trautmann examined the simultaneous prescription of metric and normal data for non-rigid dust [7]. Moreover, Dirichlet existence and uniqueness results are available for spherical boundary geometries

with suitably bounded or perturbatively small data [31–33]. Such results do not by themselves ensure that an exterior realizes the full Darmois data of a prescribed interior. More specialized constructions include Marvan's matchings of rigid van Stockum dust to Papapetrou vacuum across suitable boundaries [34]. These provide exact matching examples without establishing the isolated completion required here. Existence must also be distinguished from uniqueness. Mars and Senovilla prove uniqueness of the isolated exterior under their assumed matching and asymptotic hypotheses, with the stationary Killing field timelike throughout the exterior and using global Weyl coordinates and a global Ernst potential [35]. Their result also controls the additive twist-constant ambiguity, but its application to nonzero twist periods requires a separate check of the global assumptions. Hauser and Ernst give a construction procedure whose exterior need not be asymptotically flat [36]. MacCallum, Mars and Vera derive necessary and sufficient perturbative compatibility conditions through second order about static isolated bodies, with explicit tests for boundaries of spherical topology [37]. We underline that these results address complementary formulations and do not supply the nonlinear toroidal completion we have studied here. The present analysis leaves that nonlinear existence problem unresolved. Our local theorems provide positive-density realizations of prescribed traces, while the radial-span estimates restrict their extent under the stated shape assumptions. Continuing the same connected regular sheet with nonzero ℓ_0 to an asymptotically flat equatorial end under the density and subluminality hypotheses would force a vacuum plateau throughout that sheet. Completion of a finite body therefore requires a separate vacuum exterior realizing the full Darmois data and the global conditions above. Last but not least, its twist period must agree with the interior matter-current flux and need not vanish.

IX. FROM LOCAL ROTATION PROFILES TO SPECTROSCOPIC OBSERVABLES

To relate a local speed profile to spectroscopic measurements, we must specify the emitter, the observer and the null geodesic connecting them. For related local observer constructions and photon frequency-shift calculations in stationary axisymmetric dust spacetimes, we refer to [38]. Let $x_{\rm e}$ and $x_{\rm o}$ denote the emission and reception events in a stationary, axisymmetric spacetime. Moreover, we choose a future-directed null geodesic connecting them. In the geometric-optics description, the photon wave vector p is a nonzero tangent to this affinely parametrized ray. The stationary and axial Killing fields determine the conserved quantities $E = -g(p,k)$ and $L = g(p,m)$. We use the same Killing fields at both endpoints. For a circular emitter and observer, let us assume that $k + \Omega_a m$ is future-directed and timelike at x_a. Their normalized four-velocities are then

$$u_a = U_a(k + \Omega_a m), \qquad U_a = [-g(k + \Omega_a m, k + \Omega_a m)|_{x_a}]^{-1/2} \tag{IX.1}$$

with $a \in \{\mathrm{e}, \mathrm{o}\}$. The frequency measured by each observer is obtained by contracting p with that observer's four-velocity. Conservation of E and L then relates the two measurements.

Proposition 7. *For $E \neq 0$, we define the conserved impact parameter $b = L/E$. The positive measured frequencies $\omega_a = -g(p, u_a)$ determine the observed redshift through*

$$1 + z_{\rm obs} = \frac{\omega_{\rm e}}{\omega_{\rm o}} = \frac{U_{\rm e}}{U_{\rm o}} \frac{1 - \Omega_{\rm e} b}{1 - \Omega_{\rm o} b}. \tag{IX.2}$$

For an emitter comoving with the dust, $U_{\rm e} = (-H_{\rm e})^{-1/2}$, where $H_{\rm e}$ is the value of H at emission.

Proof. The affine geodesic equation $\nabla_p p = 0$ and the Killing equation imply $\nabla_p[g(p,k)] = p^\mu p^\nu \nabla_\mu k_\nu = 0$. The same argument applies to m, so E and L have the same values at both endpoints. Substituting the circular four-velocities gives $\omega_a = U_a(E - \Omega_a L)$. Since p and u_a are future-directed, $\omega_a > 0$ and hence $E - \Omega_a L > 0$. The denominator of the frequency ratio therefore cannot vanish. Dividing the endpoint expressions and using $L = bE$ yields (IX.2). The expressions in terms of E and L remain valid when $E = 0$, although b is then undefined. For a dust emitter, (II.2) gives the stated normalization. □

Let us suppose now that the chosen ray escapes to an asymptotically flat end of a completed spacetime. We normalize k to the future unit time translation at infinity, so that $g(k,k) \to -1$. Along this ray, $E > 0$ and b remains fixed. We further consider circular observers at reception events whose asymptotic radius tends

to infinity, with $U_{\rm o} \to 1$ and $\Omega_{\rm o} b \to 0$. These conditions hold, in particular, for observers at rest relative to k, for whom $\Omega_{\rm o} = 0$ and $U_{\rm o} = (-g_{tt})^{-1/2} \to 1$. For a dust emitter, the frequency ratio then reduces to

$$1 + z_{\rm obs} \longrightarrow \frac{1 - \Omega_{\rm e} b}{\sqrt{-H_{\rm e}}}. \tag{IX.3}$$

To evaluate this expression, we must determine the impact parameter of the ray actually received. This requires solving a null-geodesic boundary-value problem for the completed metric and the specified source and viewing geometry. If several connecting rays exist, the frequency-ratio formula applies separately to each. The relation between the ray parameter and the local speed becomes explicit in the ZAMO frame at emission. For a dust emitter in the subluminal region considered here, $g_{\phi\phi} > 0$ and the ZAMO four-velocity and unit azimuthal vector are $u_{\rm Z} = (k + \Omega_{\rm Z} m)/N$ and $e_{\hat{\phi}} = m/\sqrt{g_{\phi\phi}}$. All quantities in the following local relations are evaluated at $x_{\rm e}$. The dust four-velocity and photon wave vector decompose as $u_{\rm e} = \Gamma(u_{\rm Z} + v e_{\hat{\phi}})$ and $p = \omega_{\rm Z}(u_{\rm Z} + \widehat{\nu})$, where $\Gamma = (1 - v^2)^{-1/2}$ and $\omega_{\rm Z} = -g(p, u_{\rm Z})$. The propagation direction $\widehat{\nu}$ lies in the ZAMO rest space and satisfies $g(\widehat{\nu}, u_{\rm Z}) = 0$ and $g(\widehat{\nu}, \widehat{\nu}) = 1$. Contracting these decompositions gives $\omega_{\rm e} = \Gamma\omega_{\rm Z}(1 - v\cos\alpha)$ with $\cos\alpha = g(\widehat{\nu}, e_{\hat{\phi}})$. Thus, α is the photon propagation angle relative to the positive azimuthal direction in the ZAMO frame. Contracting p with m and using $k = N u_{\rm Z} - \Omega_{\rm Z} m$ also gives $L = \omega_{\rm Z}\sqrt{g_{\phi\phi}}\cos\alpha$ and $E = \omega_{\rm Z}(N + \Omega_{\rm Z}\sqrt{g_{\phi\phi}}\cos\alpha)$. Hence, for $E \neq 0$, we have

$$b = \frac{\sqrt{g_{\phi\phi}}\cos\alpha}{N + \Omega_{\rm Z}\sqrt{g_{\phi\phi}}\cos\alpha}. \tag{IX.4}$$

This identity links the local emission direction to the conserved impact parameter. The direction of the ray that reaches the observer must still be determined from its propagation through the spacetime. Finally, the span theorems constrain the equatorial ZAMO-speed trace $v(r, 0)$. A spectroscopic prediction also depends on the completed metric, the emitter and observer motions and the connecting rays selected by the viewing geometry. Applying the span bounds to a spectroscopically inferred rotation curve therefore requires a quantitative comparison with $v(r, 0)$, together with control of the relation between observed position and emission radius. Appendix G gives a detailed formulation of this prediction problem.

X. CONCLUSIONS AND OUTLOOK

The constant-ℓ dust family allows local realizability and global completion to be studied within the exact field equations. The auxiliary representation reduces the velocity equation to a linear elliptic problem. The density factor identifies a fold at the lower endpoint of the density interval, while strictly positive density selects a regular inverse sheet. This structure permits analytic realizations of exactly flat equatorial traces with positive density on admissible annuli. Fixing the reconstruction constants then determines the corresponding local dust metric. More generally, the reconstructed fields are real analytic on regular inverse sheets. The explicit instability sequence for the auxiliary Cauchy problem nevertheless demonstrates failure of continuous dependence on smooth equatorial data, distinguishing regularity of individual solutions from stability of continuation. Continuation beyond these local neighbourhoods is subject to geometric restrictions. Under the stated axis regularity and timelikeness hypotheses, a constant-ℓ branch reaching the axis must have $\ell_0 = 0$. Analyticity also excludes a positive exactly flat trace on a connected equatorial branch extending to a regular axis. For reflection-symmetric annular solutions, nonnegative density and invariant transverse concavity of the relative dust-ZAMO rapidity yield finite radial-span bounds at every fixed logarithmic-slope tolerance below unity. For positive tolerance, the bound for the nonpositive class is attained at $\ell_0 = 0$, while every strictly negative branch satisfies a strict inequality. For positive ℓ_0, analytic local solutions with positive density and strict transverse concavity approach the exactly flat supremum. The asymptotic analysis imposes a different restriction. For nonzero ℓ_0, the stated metric falloff along the equatorial tail and uniformly subluminal positive speed force $rv \to q > 0$. If density is nonnegative on an open neighbourhood of the tail, then sufficiently far along that tail there is an open region on which $\eta = q$ and $\rho = 0$. Analyticity makes these identities hold throughout any connected regular inverse sheet of the same solution containing that region. The resulting metric is locally flat and retains a joint identification of time and angle. Its subluminal region $r > q$ is stably causal, but a regular axis cannot be added while preserving flatness and the same Killing fields. We underline that the directional falloff used in this argument does not establish a complete isolated end. For a finite body,

completion requires a vacuum exterior realizing the full Darmois data. Zero-density surfaces provide one class of candidate boundaries, but density vanishing alone does not ensure matching of the induced metric and extrinsic curvature. For toroidal bodies satisfying the stated orbit-space hypotheses, topology imposes period conditions on single-valued exterior potentials. A globally defined metric may nevertheless carry a nonzero twist period fixed by the interior matter-current flux. In the sector with vanishing twist period, the linearized Ernst analysis characterizes compatible boundary values and conormal derivatives about Minkowski space on a fixed smooth bounded domain away from the axis. This provides a boundary test for the scalar equations, with metric reconstruction periods and the conditions at the axis and infinity remaining separate requirements.

The central unresolved question is whether the local solutions constructed here can form a finite dust body admitting an isolated vacuum exterior. A possible approach is to determine the interior data, material boundary and exterior geometry together as a coupled matching problem. An approach based on the implicit-function theorem would require a suitable matched reference configuration and the necessary solvability properties of its linearized matching problem. Throughout this construction, one must enforce the metric reconstruction periods, match the twist periods to the interior current and impose axis regularity and asymptotic normalization. Local twist potentials with additive monodromy allow nonzero twist periods to be included in this formulation. A further open question concerns the optimal radial span for positive ℓ_0 at nonzero flatness tolerance. Resolving it requires sharper control of the coupled equatorial equations and analytic constructions approaching the proposed sharp bound. The auxiliary Cauchy instability also motivates regularized reconstruction methods with explicit assumptions on the data and error estimates compatible with positive density and inverse-sheet regularity. These questions concern the constant-ℓ class studied here. Allowing ℓ to vary would make it possible to investigate whether a nearly constant annular regime can connect to a regular axis with $H'(0) = 0$. Under the axis timelikeness assumption $H(0) < 0$, this also requires $\ell(0) = 0$. Other matter models and asymptotic geometries require their own existence and continuation analyses. For any completed spacetime, specified emitter and observer motions and the connecting null geodesics determine the frequency ratio. A forward spectroscopic calculation would then allow a quantitative comparison between the inferred rotation curve and the prescribed equatorial ZAMO-speed trace.

CONFLICT OF INTEREST

The authors declare that they have no financial or non-financial interests relevant to the content of this article.

DATA AVAILABILITY

No observational or experimental datasets were used in this theoretical study. The analytical derivations are provided in the main text and appendices. Supporting symbolic verification scripts and density-boundary data are available in the repository at

https://github.com/dutykh/Constant-Ell-Rotating-Dust/.

OVERVIEW OF THE APPENDICES

They provide the constructions and calculations supporting the analysis of local realizability and global completion. The principal theorems and their proofs are given in the main text, whose notation and conventions we retain. Appendices A-C develop the reconstruction and continuation arguments. The Runge approximation result is stated with the hypotheses needed for metric reconstruction. An alternative axis argument based on the dust congruence and an explicit flat-trace solution for $\ell_0 = 0$ complement this analysis. The radial-span discussion then gives further estimates, including the refinement for $\ell_0 > 0$ together with analytic constructions that establish the stated sharpness results. To assess the scope of the retained edge approximation, Appendix D compares it with the exact equations and identifies the remainder hypothesis needed for a large-radius test. Appendices E-G connect the completion problem to its observational interpretation. The junction formulas make the surface stresses and cylindrical boundary data explicit. Building on these data, the nonlinear exterior

discussion sets out the additional hypotheses needed beyond the main text's linearized Ernst criterion. The spectroscopic formulation then specifies the emitter, observer and ray data required to predict a frequency shift once the geometry is given.

Appendix A: Runge approximation and metric reconstruction

Runge approximation allows a local auxiliary solution to be approximated on a compact interior region by solutions defined on a larger domain. The result below states the topological condition that permits this approximation and the hypotheses that preserve a local physical reconstruction. We fix $\ell_0 \in \mathbb{R}$ and the reference scale $\eta_* > 0$. As in the main text, $f(r,z)$ denotes the auxiliary field, while $\mathcal{F}(r,\eta)$ is the defining function of the point transformation. Thus, reconstruction requires $f(r,z) = \mathcal{F}(r,\eta(r,z))$ and the auxiliary equation is $\mathcal{L}f = 0$ with $\mathcal{L}$ defined as in (II.12).

Theorem A.1. *Let D_1 be a connected open subset of the meridional half-plane $(0,\infty)\times\mathbb{R}$ with compact closure contained in that half-plane. Let K and K' be compact sets satisfying $K \subset \operatorname{int} K'$ and $K' \subset D_1$. we further suppose that f_0 is a real $\mathcal{L}$-harmonic function on a neighbourhood of K' and that no connected component of $D_1 \setminus K'$ has compact closure contained in D_1. For every nonnegative integer m, there are real $\mathcal{L}$-harmonic functions f_j on D_1 such that $f_j \to f_0$ in $C^m(K)$. Here, this convergence means uniform convergence on K of all derivatives in (r,z) through order m. For the reconstruction statement, we take $m \geq 2$. Let us choose finitely many simply connected open sets V_a with $\overline{V}_a \subset \operatorname{int} K'$ and compact sets $K_a \subset V_a$ such that $K = \bigcup_a K_a$. On a neighbourhood of each $\overline{V}_a$, we require that a positive inverse branch η_a satisfies $\mathcal{F}(r,\eta_a(r,z)) = f_0(r,z)$ and $\partial_\eta \mathcal{F}(r,\eta_a) \neq 0$. We fix $C > 0$ in $H = -Ce^{\ell_0\eta}$, the additive integration constant in Ω for all reconstructions and on each V_a, the meridional primitive μ_a is given by its value at a chosen point $p_a \in V_a$. Let us write $v_a = \eta_a/r$ and ρ_a be the corresponding density. Moreover, we assume that, on every K_a*

$$\partial_\eta \mathcal{F}(r,\eta_a) \geq a_0 > 0, \qquad 0 < v_a \leq 1-\delta, \qquad \rho_a \geq \rho_0 > 0, \tag{A.1}$$

where $a_0, \rho_0 > 0$ and $0 < \delta < 1$ are independent of a. The approximants can be chosen so that, for all sufficiently large j, each selected branch admits a reconstruction $\eta_{j,a}$ on V_a. Using the same constants in H and Ω and the normalization $\mu_{j,a}(p_a) = \mu_a(p_a)$, the reconstructed quantities satisfy

$$\partial_\eta \mathcal{F}(r,\eta_{j,a}) \geq \frac{a_0}{2}, \qquad 0 < v_{j,a} \leq 1-\frac{\delta}{2}, \qquad \rho_{j,a} \geq \frac{\rho_0}{2} \tag{A.2}$$

on every K_a, where $v_{j,a} = \eta_{j,a}/r$. These conclusions concern the local metrics on the individual sets V_a. Agreement on overlaps requires the additional compatibility conditions discussed below.

Proof. The operator $\mathcal{L}$ is elliptic with analytic coefficients on D_1. Its formal adjoint with respect to $drdz$ is $\mathcal{L}^* = \partial_r^2 + r^{-1}\partial_r + \partial_z^2 - r^{-2}$, which is also elliptic with analytic coefficients. Analytic elliptic regularity therefore gives unique continuation for solutions of the adjoint equation. The complement hypothesis is the compact Runge condition. It permits a Runge neighbourhood of K' within the domain of f_0, so the Lax–Malgrange approximation theorem supplies real solutions f_j on D_1 converging uniformly to f_0 on K' [39]. Since $\mathcal{L}(f_j - f_0) = 0$ near K', interior elliptic estimates give, for every compact $Q \subset \operatorname{int} K'$ and every fixed m

$$\|f_j - f_0\|_{C^m(Q)} \leq C_{Q,m}\|f_j - f_0\|_{C^0(K')} \longrightarrow 0, \tag{A.3}$$

where $C_{Q,m}$ is independent of j. This proves the first assertion and also gives the required convergence on each $\overline{V}_a$. Regularity of the selected inverse and compactness of $\overline{V}_a$ allow the implicit-function theorem to be applied uniformly along its graph. For sufficiently large j, this yields a positive inverse $\eta_{j,a}$ on the same branch with $\eta_{j,a} \to \eta_a$ in $C^m(\overline{V}_a)$. The fixed constants in H and Ω then give convergence of the Killing-block coefficients. By the quadratures (IV.5)-(IV.6), the corresponding meridional one-forms $\alpha_{\mu,j,a}$ converge to $\alpha_{\mu,a}$ in $C^{m-1}(\overline{V}_a)$. These forms are closed because their auxiliary fields solve $\mathcal{L}f = 0$. Simple connectedness therefore gives primitives and the common base-point normalization implies

$$\mu_{j,a}(p) - \mu_a(p) = \int_{p_a}^{p} (\alpha_{\mu,j,a} - \alpha_{\mu,a}), \tag{A.4}$$

where $p \in V_a$ and the integral is taken along any path in V_a. To control these integrals uniformly on K_a, we cover K_a by finitely many balls with closures in V_a and join their centres to p_a by fixed paths. The resulting paths to points of K_a have uniformly bounded length. Uniform convergence of the one-forms consequently gives uniform convergence of the normalized primitives on K_a. The identities $d\mu_{j,a} = \alpha_{\mu,j,a}$ then give convergence of their derivatives through order m. It follows that the metric coefficients converge on every K_a. The density formula (III.1) depends continuously on η, its first derivatives and μ wherever $\eta > 0$, so $\rho_{j,a} \to \rho_a$ uniformly there. Compactness also supplies a positive lower bound for each v_a. The finitely many compact sets K_a therefore admit a common sufficiently large index beyond which positivity of the speed and all three displayed margins are preserved. □

The compact sets K_a make the local conclusion precise. Their union is K, but the primitives normalized independently on the sets V_a need not agree on overlaps. Even when the selected inverses agree, two local meridional primitives can differ by a constant on each connected overlap. Hence, a single metric on the union requires compatible choices of these constants. The associated period condition follows directly from the interior quadrature. For a reconstructed field η, we write $\alpha_\mu = M_r dr + M_z dz$ with M_r and M_z given by (IV.5)-(IV.6). Since $d\alpha_\mu = (\partial_z M_r - \partial_r M_z) dz \wedge dr$, the compatibility identity is

$$d\alpha_\mu = \frac{\eta_z \partial_\eta \mathcal{F}(r,\eta)}{4r} \mathcal{L} f \; dz \wedge dr, \qquad f(r,z) = \mathcal{F}(r, \eta(r,z)). \tag{A.5}$$

Thus, an auxiliary solution makes α_μ closed. On a connected reconstruction domain where the inverse branches agree and α_μ is globally defined, a single-valued primitive exists precisely when its integral around every homology generator vanishes. When these periods vanish, the local primitives can be adjusted by additive constants to agree on overlaps. Previously prescribed base-point values are compatible only if they are the values of one such global primitive. Runge approximation alone does not impose these period or normalization conditions. The theorem preserves the strict physical inequalities on the compact interior region through the specified local reconstructions. It does not establish physical admissibility throughout D_1 or impose exact equality of a prescribed equatorial trace. The result concerns approximation of an already given local solution and does not resolve the instability of continuation from Cauchy data because an asymptotic end and a material matching surface still require separate constructions.

Appendix B: Further axis arguments and an explicit flat trace

The first part complements the metric argument in the main text by using regularity of the dust four-velocity at the axis. The second gives an explicit flat-trace solution for $\ell_0 = 0$ and identifies the region in which its continuation away from the equator remains subluminal.

1. Regularity of the dust congruence at the axis

The metric argument in the main text derives $H'(0) = 0$ from quadratic axis expansions and the timelikeness of the stationary Killing field. The argument below assumes only that this field is nonzero at the axis, but requires a C^1 timelike extension of the dust four-velocity. Its key ingredient is that a regular azimuthal vector field cannot have an unbounded angular coefficient. The following lemma applies even when the region occupied by the branch does not contain a full punctured neighbourhood of the axis.

Lemma B.1. *Let $\mathcal{V}$ be an open region invariant under the axial rotations and disjoint from a regular axis $\mathcal{A}$. Let $X = \omega m$ be an axisymmetric vector field on $\mathcal{V}$. If X admits a C^1 extension to a neighbourhood of an axis point $q \in \mathcal{A} \cap \overline{\mathcal{V}}$, then ω has a finite limit as $p \in \mathcal{V}$ approaches q.*

Proof. we choose regular coordinates (t, x, y, ζ), with x, y transverse to the axis and t, ζ along it, such that $m = -y\partial_x + x\partial_y$. Let $q = (t_0, 0, 0, \zeta_0)$ and use X also for its C^1 extension. Any point of $\mathcal{V}$ sufficiently near q can be rotated to $(t, \varrho, 0, \zeta)$, where $\varrho = \sqrt{x^2 + y^2} > 0$, without changing ω. The opposite point $(t, -\varrho, 0, \zeta)$ belongs to the same orbit. Since $X^y = \omega x$ on that orbit, we find

$$\omega(t, \varrho, 0, \zeta) = \frac{X^y(t, \varrho, 0, \zeta) - X^y(t, -\varrho, 0, \zeta)}{2\varrho} = \frac{1}{2\varrho} \int_{-\varrho}^{\varrho} \partial_x X^y(t, s, 0, \zeta) \; ds. \tag{B.1}$$

The integral uses the extension, so the transverse diameter need not lie in $\mathcal{V}$. Continuity of $\partial_x X^y$ makes this average tend to $\partial_x X^y(q)$ as $(t, \varrho, \zeta) \to (t_0, 0, \zeta_0)$. Axisymmetry gives the same limit for an arbitrary approach through $\mathcal{V}$. □

For $\ell_0 \neq 0$, integrating $\Omega_{,\eta} = H_{,\eta}/(2\eta)$ with $H = -Ce^{\ell_0 \eta}$ gives the logarithmic behaviour

$$\Omega(\eta) = -\frac{C\ell_0}{2} \log\left(\frac{|\eta|}{\eta_*}\right) + O(1) \tag{B.2}$$

as $\eta \to 0$ within a connected interval on which $\eta \neq 0$. Here, $\eta_* > 0$ is a fixed reference scale with the same dimensions as η. The expansion follows from $\mathrm{Ei}(x) = \gamma_\mathrm{E} + \log|x| + O(|x|)$ for real $x \to 0$, where γ_E is Euler's constant (see Eqs. 6.2.3 and 6.2.7 in [22]). It holds from either sign of η and changing η_* changes only the bounded term. We now combine this divergence with the regularity established by the lemma.

Theorem B.1. *Let an exact stationary, axisymmetric rotating-dust solution of the (η, H) family be defined on a connected invariant open region $\mathcal{U}$ whose closure meets a regular axis segment $\mathcal{A}$. We assume that the metric extends as a C^2 Lorentz metric across $\mathcal{A}$, that the stationary Killing field k is nonzero there and that the axial field m has period 2π. Let the dust four-velocity admit a C^1 timelike extension to a neighbourhood of every regular limit point of the branch, including any interior limit point where η tends to zero. we further suppose that $H < 0$, $|\eta|/r < 1$ away from the axis and $H_{,\eta}/H = \ell_0$ on the connected range of η. If $\eta \not\equiv 0$, then $\ell_0 = 0$.*

Proof. At a regular axis point, m vanishes and the metric remains finite. The identity $r^2 = g(k,m)^2 - g(k,k)g(m,m)$ therefore gives $r \to 0$ without requiring k to be timelike. Subluminality then gives $\eta \to 0$ because $|\eta| < r$. Let us choose a connected component $\mathcal{U}_*$ of the nonempty open set $\{p \in \mathcal{U} \mid \eta(p) \neq 0\}$. If $\mathcal{U}_* = \mathcal{U}$, its closure meets $\mathcal{A}$. Otherwise, connectedness of $\mathcal{U}$ implies that $\mathcal{U}_*$ has a nonempty relative boundary inside $\mathcal{U}$. Continuity of η makes it zero at every such boundary point. In either case, there is a sequence $p_j \in \mathcal{U}_*$ approaching a regular axis point or a regular interior point,with $\eta(p_j) \to 0$. On $\mathcal{U}_*$, the constant-ℓ law gives $H = -Ce^{\ell_0\eta}$ with $C > 0$. Thus, in the decomposition $u = U(k + \Omega m)$, the factor $U = (-H)^{-1/2}$ satisfies $U(p_j) \to C^{-1/2} > 0$. Let us suppose that $\ell_0 \neq 0$. Then, (B.2) implies $|\Omega(p_j)| \to \infty$. At an interior limit point, $r > 0$ and $\det G = -r^2$ ensures that k and m are linearly independent. A regular local one-form ϑ can therefore be chosen with $\vartheta(k) = 0$ and $\vartheta(m) = 1$. The identity $\vartheta(u) = U\Omega$ contradicts continuity of u, since its right-hand side is unbounded along the sequence. At an axis limit point, the commuting Killing fields and the nonvanishing of k permit a local flow coordinate t that is invariant under axial rotations and satisfies $k(t) = 1$. Hence, $\tau = dt$ obeys $\tau(k) = 1$ and $\tau(m) = 0$. On $\mathcal{U}_*$, we have $U = \tau(u)$, so this identity supplies a C^1 extension of U. This signalizes that $X = u - Uk$ extends as a C^1 field and equals $U\Omega m$ on the branch. The component $\mathcal{U}_*$ is invariant under axial rotations, so Lemma B.1 gives a finite limit of $U\Omega$. Since $U \to C^{-1/2} > 0$, this contradicts the divergence of Ω. Both possible limits therefore exclude $\ell_0 \neq 0$. □

For the resulting $\ell_0 = 0$ branch, $H = -C$ and the angular velocity is a finite constant Ω_0. At the axis, the extended four-velocity is $C^{-1/2}k$, so its unit timelike normalization gives $g(k,k) = -C < 0$. The canonical radius r is consequently comparable to the transverse Cartesian radius. The quadratic axis expansions from the main text then give $g(k,m) = O(r^2)$ and $g(m,m) = O(r^2)$, and therefore $\eta = g(k,m) + \Omega_0 g(m,m) = O(r^2)$ and $v = O(r)$. If a connected equatorial interval reaches this axis segment, analyticity for $r > 0$ and the limit $v \to 0$ exclude a positive exactly flat trace on any nonempty open subinterval. This recovers the axis-connected flat-trace exclusion using regularity of the material congruence.

2. The explicit continuation for $\ell_0 = 0$

For $\ell_0 = 0$, the defining potential is $\mathcal{F} = 2\eta$. The identity (II.12) therefore reduces the exact velocity equation to $\mathcal{L}\eta = 0$. This linear equation allows the reflection-symmetric continuation of a flat equatorial trace to be written explicitly.

Corollary B.1. *Let $\ell_0 = 0$, $0 < v_0 < 1$ and $r_* > 0$. The unique local analytic solution of (II.7) with data $v(r,0) = v_0$ and $v_z(r,0) = 0$ for r near r_* is the restriction of*

$$v(r,z) = v_0 \frac{\sqrt{r^2 + z^2}}{r}. \tag{B.3}$$

This expression is real analytic for all $r > 0$ and is even in z. At each fixed r, the equator is a strict minimum, with

$$v_{zz}(r,0) = \frac{v_0}{r^2} > 0. \tag{B.4}$$

The continuation satisfies $0 < v < 1$ precisely in the region consisting of all points $(r,z) \in (0,\infty)\times\mathbb{R}$ such that $|z| < r\sqrt{v_0^{-2}-1}$ and reaches $v = 1$ at its two boundary heights for each fixed $r > 0$.

Proof. We set $R = \sqrt{r^2+z^2}$. For any twice differentiable function f of R, the identities $\partial_r f(R) = f'(R)r/R$ and $\partial_{rr}f(R) + \partial_{zz}f(R) = f''(R) + f'(R)/R$ give $\mathcal{L}[f(R)] = \left(\partial_r^2 - r^{-1}\partial_r + \partial_z^2\right) f(R) = f''(R)$. Thus, $\eta = v_0 R$ solves $\mathcal{L}\eta = 0$. Since $r > 0$, its equatorial data are $\eta(r,0) = v_0 r$ and $\eta_z(r,0) = 0$, which are exactly the prescribed data after substituting $\eta = rv$. The coefficients of $\mathcal{L}$ are analytic near $(r_*, 0)$ and the coefficient of η_{zz} is one, so $z = 0$ is noncharacteristic. Uniqueness for the analytic Cauchy problem therefore identifies this candidate with the local solution. Dividing by r gives (B.3). Direct differentiation yields $v_z = v_0 z/(rR)$ and $v_{zz} = v_0 r/R^3$, proving (B.4) and strict growth with $|z|$ away from the equator. Finally, $v < 1$ is equivalent to $v_0^2(r^2+z^2) < r^2$, which gives the stated region and boundary heights. □

These heights delimit the subluminal region of the explicit continuation. At $v = 1$, the identity $g_{\phi\phi} = (r^2-\eta^2)/C$ gives $g_{\phi\phi} = 0$, so the axial orbits become null and the ZAMO description ceases to apply. The growth away from the equator therefore limits the physical range of this flat-trace construction even though the formula for v remains analytic for $r > 0$.

Appendix C: Further radial-span estimates and trace controls

We develop the radial-span analysis of the main text. We first derive a bound from nonnegative density and the slope constraint, without imposing transverse concavity, and identify its dependence on the inner-edge calibration. We then optimize the positive-ℓ_0 comparison bound and improve it by minimizing the full slope drift. A formulation in terms of equatorial trace data explains the remaining sharpness question and the conditions under which such data admit analytic local realizations. Finally, we construct strictly negative branches whose spans approach the sharp bound for $\ell_0 \le 0$. We use the same equatorial variables of the main text. For $u(r) = v(r,0)$ on $[R_1, R_2]$, where $0 < R_1 < R_2$, we set $\sigma = \log(r/R_1)$ and $\Lambda = \log(R_2/R_1)$. A dot denotes differentiation with respect to σ. The dimensionless quantities used below are those defined in (V.6). Unless its arguments are displayed, $\partial_\eta\mathcal{F}$ is evaluated at $(r, ru(r))$. It is the principal coefficient defined in (II.10). The estimates involving transverse concavity use the following common hypotheses.

Assumption C.1. Let v be a C^2 reflection-symmetric solution of the exact constant-ℓ_0 equation (II.7) on an open neighbourhood of $[R_1, R_2]\times\{0\}$. For some $0 \le \varepsilon < 1$, we assume throughout the equatorial interval that

$$0 < u < 1, \qquad \rho(r,0) \ge 0, \qquad v_{zz}(r,0) \le 0, \qquad |s| \le \varepsilon. \tag{C.1}$$

Reflection symmetry makes the transverse condition equivalent to the invariant rapidity-concavity condition in the main text. The slope bound gives $\eta_r(r,0) = u(1+s) > 0$. Together with nonnegative density and the exact equatorial identity (V.8), it also ensures $\partial_\eta\mathcal{F} > 0$, as established there. The first estimate needs no transverse-concavity assumption.

Corollary C.1. *Let v be a C^2 reflection-symmetric solution of the exact constant-ℓ_0 equation near $[R_1, R_2]\times\{0\}$ with $\ell_0 > 0$. We further suppose that its equatorial trace satisfies $0 < u < 1$, $\rho(r,0) \ge 0$ and $|s| \le \varepsilon$ for some $0 \le \varepsilon < 1$. Let us define $\gamma_1 = \ell_0 R_1/u(R_1)$ and $\underline{v}$ as in (V.12). Then, we have*

$$\frac{R_2}{R_1} \le \left[\frac{2}{\gamma_1(1+\underline{v}^2)}\right]^{1/(1-\varepsilon)}. \tag{C.2}$$

Proof. Since $y > 0$, the identity $\dot{y} = (1-s)y$ and the slope bound give

$$\log\frac{y(R_2)}{\gamma_1} = \int_0^\Lambda (1-s)\, d\sigma \ge (1-\varepsilon)\Lambda. \tag{C.3}$$

Moreover, $\eta_r = (1+s)u > 0$, so nonnegative density bounds y by the upper endpoint of the density interval at every point of the trace. In particular, $y(R_2) \le 2/[1+u(R_2)^2] \le 2/(1+\underline{v}^2)$. Substitution into (C.3) and exponentiation give the stated ratio bound. □

To compare with the near-edge parameterization in Eqs. (4.1) and (4.5) of [6], we impose the calibration $R_1 = R_{\mathrm{G}}$, $\ell_0 = av_{\mathrm{c}}/R_{\mathrm{G}}$ and $u(R_{\mathrm{G}}) = av_{\mathrm{c}}$ on the exact branch. Here, $R_{\mathrm{G}} > 0$ is the reference radial scale, $v_{\mathrm{c}} > 0$ is a reference speed and $a > 0$ is dimensionless with $0 < av_{\mathrm{c}} < 1$. This choice gives $\gamma_1 = 1$, so

$$\frac{R_2}{R_{\mathrm{G}}} \le \left(\frac{2}{1+\underline{v}^2}\right)^{1/(1-\varepsilon)}. \tag{C.4}$$

For an exactly flat trace, $u = av_{\mathrm{c}}$ and the bound becomes $R_2/R_{\mathrm{G}} \le 2/(1+a^2v_{\mathrm{c}}^2)$. Its dependence on the calibration is essential to this argument. The right-hand side of (C.2) diverges as $\gamma_1 \downarrow 0$, so this estimate alone gives no bound uniform in the inner-edge parameter. we recall that in the main text, we obtain a bound independent of that parameter by imposing transverse concavity.

1. Optimization of the positive-ℓ comparison

For $0 < \varepsilon < 1$, $0 < \alpha < 1-\varepsilon$ and $0 < \underline{v} < 1$, the positive-ℓ_0 comparison bound is $\mathcal{B}_+(\varepsilon, \alpha, \underline{v})$ given by (V.14). The parameter α specifies where the proof separates the initial region of small y from the remaining interval. Optimizing this choice gives the following explicit bound.

Corollary C.2. *For $0 < \varepsilon < 1$, we define*

$$K_\varepsilon = \frac{\varepsilon(3-\varepsilon)(1-\varepsilon)}{1+\varepsilon}, \quad \alpha_\varepsilon = 1-\varepsilon+\frac{K_\varepsilon}{2} - \frac{1}{2}\sqrt{K_\varepsilon[K_\varepsilon + 4(1-\varepsilon)]}. \tag{C.5}$$

Then, $\alpha_\varepsilon \in (0, 1-\varepsilon)$ is the unique minimizer of $\mathcal{B}_+(\varepsilon, \alpha, \underline{v})$ for every $\underline{v} \in (0,1)$. Under the hypotheses of Assumption C.1 with $\ell_0 > 0$ and $\underline{v} = \min_{[R_1,R_2]} u$, it follows that

$$\frac{R_2}{R_1} \le e^{\mathcal{B}_+(\varepsilon, \alpha_\varepsilon, \underline{v})}. \tag{C.6}$$

For fixed $\underline{v}$, the optimized logarithmic bound satisfies

$$\inf_{0<\alpha<1-\varepsilon} \mathcal{B}_+(\varepsilon, \alpha, \underline{v}) = \log\frac{2}{1+\underline{v}^2} + 2\sqrt{3\varepsilon} + O(\varepsilon) \tag{C.7}$$

as $\varepsilon \downarrow 0$.

Proof. For fixed ε and $\underline{v}$, the function $\mathcal{B}_+$ diverges at both endpoints of $(0, 1-\varepsilon)$. Differentiating with respect to α gives

$$\partial_\alpha \mathcal{B}_+ = \frac{\varepsilon(3-\varepsilon)}{(1+\varepsilon)(1-\varepsilon-\alpha)^2} - \frac{1}{(1-\varepsilon)\alpha}. \tag{C.8}$$

Thus, a stationary point satisfies $(1-\varepsilon-\alpha)^2 = \alpha K_\varepsilon$. The quotient $(1-\varepsilon-\alpha)^2/\alpha$ decreases strictly from $+\infty$ to zero, because its derivative is $1-(1-\varepsilon)^2/\alpha^2 < 0$ on this interval. There is therefore exactly one stationary point. Solving the quadratic and taking its root in $(0, 1-\varepsilon)$ gives (C.5) and the endpoint divergence makes it the global minimizer. Applying the main-text span bound at this value of α proves (C.6). To obtain the small-ε behaviour, we expand $K_\varepsilon = 3\varepsilon + O(\varepsilon^2)$ and $\alpha_\varepsilon = 1 - \sqrt{3\varepsilon} + O(\varepsilon)$. These expressions give

$$\frac{\varepsilon(2+\alpha_\varepsilon)}{(1+\varepsilon)(1-\varepsilon-\alpha_\varepsilon)} = \sqrt{3\varepsilon} + O(\varepsilon), \qquad -\log\alpha_\varepsilon = \sqrt{3\varepsilon} + O(\varepsilon). \tag{C.9}$$

Combining the two contributions and using $(1-\varepsilon)^{-1} = 1 + O(\varepsilon)$ proves (C.7). □

2. The trace control system and analytic realization

Optimizing $\mathcal{B}_+$ determines the best bound within that comparison argument. To examine sharpness over actual solutions, we must also consider the coupled evolution of the equatorial variables. For $\ell_0 > 0$, we define the slope drift

$$\mathfrak{g}(u, y, s) = \frac{2(1+s)[(1-s) - y(1-u^2 s)]}{2 + y(1-u^2)}. \tag{C.10}$$

Rearranging the exact equatorial identity (V.8) gives $\dot{s} = \mathfrak{g} - Z$. Together with the definitions of s and y, this yields

$$\dot{u} = su, \qquad \dot{y} = (1-s)y, \qquad \dot{s} = \mathfrak{g}(u, y, s) + w, \qquad w = -Z \geq 0, \tag{C.11}$$

subject to

$$0 < u < 1, \qquad 0 < y \leq \frac{2}{1+u^2}, \qquad |s| \leq \varepsilon. \tag{C.12}$$

For a fixed branch, the initial data satisfy $y(0) = \ell_0 R_1/u(0)$. The first two equations give $d(yu)/d\sigma = yu$, so they preserve $yu = \ell_0 R_1 e^{\sigma} = \ell_0 r$. Thus, the control formulation preserves the relation between the state variables and the radial coordinate. Every trace satisfying Assumption C.1 gives a trajectory of (C.11) and its logarithmic radial span is the length of the corresponding σ interval inside the region (C.12). The nonnegative control w keeps track of the magnitude of the negative transverse curvature and can only increase $\dot{s}$ above the drift. Conversely, realizing a formal trajectory as a solution of the full equation requires suitable regularity. The analytic lifting lemma below supplies a sufficient condition. The same control has a useful interpretation in terms of the auxiliary trace. Let us set $X = r^2$ and define the one-variable function $h(X) = \mathcal{F}(\sqrt{X}, \sqrt{X}\, u(\sqrt{X}))$. We reserve $f(r, z)$ for the auxiliary field on the meridional domain. The chain rule gives $4r^2 h''(X) = (d^2/dr^2 - r^{-1} d/dr)\mathcal{F}(r, ru(r))$. Substituting $d(ru)/dr = u(1+s)$ and $d^2(ru)/dr^2 = u(\dot{s} + s + s^2)/r$ into the right-hand side yields

$$\frac{d^2 h}{dX^2} = \frac{u\partial_\eta \mathcal{F}}{4r^3}[\dot{s} - \mathfrak{g}(u, y, s)] = \frac{u\partial_\eta \mathcal{F}}{4r^3} w. \tag{C.13}$$

Since $u\partial_\eta \mathcal{F} > 0$, transverse concavity of a realized trace is equivalent to convexity of h as a function of r^2.

Lemma C.1. *Let us fix $\ell_0 > 0$, $0 \leq \varepsilon < 1$ and u be real analytic on a neighbourhood of a compact radial interval $[R_1, R_2]$ with $0 < R_1 < R_2$. We further define s and y by (V.6). Let us suppose that, throughout the interval, $0 < u < 1$, $0 < y < 2/(1+u^2)$, $|s| \leq \varepsilon$ and $\dot{s} > \mathfrak{g}(u, y, s)$. Then, u is the equatorial trace of a reflection-symmetric real-analytic local solution of the exact constant-ℓ_0 velocity equation. This solution admits a local metric reconstruction with $\rho > 0$ and $0 < v < 1$ on a neighbourhood of the interval and $v_{zz}(r, 0) < 0$ throughout the trace.*

Proof. We pick the reference scale $\eta_* > 0$ in $\mathcal{F}$ and prescribe the analytic Cauchy data $f(r, 0) = \mathcal{F}(r, ru(r))$ and $f_z(r, 0) = 0$ for $f_{zz} = -f_{rr} + r^{-1} f_r$. The coefficients are analytic for $r > 0$ and $z = 0$ is noncharacteristic. The Cauchy–Kowalevski theorem therefore gives an analytic solution near each point of the interval. Let us choose overlapping rectangles centred on $z = 0$, so that their overlaps meet the common Cauchy surface. Analytic uniqueness identifies the solutions on these overlaps and compactness supplies a neighbourhood of the entire interval. On the prescribed trace, $\partial_\eta \mathcal{F} = 2 + y(1-u^2) > 0$. Compactness and the analytic implicit-function theorem therefore give a common inverse branch satisfying $f(r, z) = \mathcal{F}(r, \eta(r, z))$ and $\eta(r, 0) = ru(r)$, after shrinking the neighbourhood if necessary. The identity (II.12) then yields the exact velocity equation for $v = \eta/r$. Uniqueness of the auxiliary solution and its inverse also gives reflection symmetry. Let us shrink to a simply connected collar of the interval. We fix $C > 0$ in H, the additive constant in Ω and one value of μ and apply the local reconstruction of the main text. The meridional quadrature is closed because $\mathcal{L}f = 0$, so it has a single-valued analytic primitive on this collar. On the trace, $Z = \mathfrak{g} - \dot{s} < 0$, giving strict transverse concavity. The strict upper density bound, together with $\ell_0 > 0$ and $\eta_r = u(1+s) > 0$, gives $\rho > 0$. Continuity and compactness preserve positive density and subluminality on a smaller neighbourhood of the whole interval. □

The slope constraint in this lemma may be attained at the endpoints or elsewhere on the trace. The lemma imposes strict margins on the physical inequalities, while $|s| \leq \varepsilon < 1$ already ensures $\eta_r > 0$. This analytic existence result is compatible with the Cauchy instability established in the main text. Continuation to the axis and matching to an exterior require additional arguments.

3. A strict refinement for positive ℓ

The main-text comparison bounds the terms in the drift separately. Minimizing the full expression over the allowed ranges of u, y and s gives a stronger lower bound on $\dot{s}$ when y is small, and hence a shorter upper bound for the span.

Theorem C.1. *Let Assumption C.1 hold with $0 < \varepsilon < 1$ and $\ell_0 > 0$. We set $v_{\min} = \underline{v} = \min_{[R_1,R_2]} u \in (0,1)$ and fix $0 < \alpha < 1 - \varepsilon$. Moreover, we define*

$$c_+(\varepsilon,\alpha,v_{\min}) = \frac{2(1+\varepsilon)[1-\varepsilon-\alpha(1-v_{\min}^2\varepsilon)]}{2+\alpha(1-v_{\min}^2)}, \tag{C.14}$$

$$c_-(\varepsilon,\alpha,v_{\min}) = \begin{cases} \dfrac{2(1-\varepsilon)[1+\varepsilon-\alpha(1+v_{\min}^2\varepsilon)]}{2+\alpha(1-v_{\min}^2)}, & \alpha \leqslant \dfrac{1-\varepsilon}{1+\varepsilon}, \\ (1-\varepsilon^2)(1-\alpha), & \alpha > \dfrac{1-\varepsilon}{1+\varepsilon}, \end{cases} \tag{C.15}$$

$$c_*(\varepsilon,\alpha,v_{\min}) = \min\{c_+, c_-\}. \tag{C.16}$$

Then $c_* > 0$ and

$$\Lambda \leq \mathcal{B}_*(\varepsilon,\alpha,v_{\min}) = \frac{2\varepsilon}{c_*} + \frac{1}{1-\varepsilon}\log\frac{2}{(1+v_{\min}^2)\alpha}. \tag{C.17}$$

With the earlier drift bound written as

$$c_0(\varepsilon,\alpha) = \frac{2(1+\varepsilon)(1-\varepsilon-\alpha)}{2+\alpha}, \tag{C.18}$$

the improvement is strict for every allowed α, i.e. $c_* > c_0$ and $\mathcal{B}_* < \mathcal{B}_+$. In particular, we have

$$\Lambda \leq \inf_{0<\alpha<1-\varepsilon} \mathcal{B}_+(\varepsilon,\alpha,v_{\min}) - \delta(\varepsilon,v_{\min}), \tag{C.19}$$

where

$$\delta(\varepsilon,v_{\min}) = 2\varepsilon\left[\frac{1}{c_0(\varepsilon,\alpha_\varepsilon)} - \frac{1}{c_*(\varepsilon,\alpha_\varepsilon,v_{\min})}\right] > 0. \tag{C.20}$$

Proof. Let us set $x = u^2$. On the part of the trace where $y \leq \alpha$, the control equation gives $\dot{s} \geq \mathfrak{g}(u,y,s)$. To find a lower bound for this drift, we minimize over $v_{\min}^2 \leq x < 1$, $0 < y \leq \alpha$ and $|s| \leq \varepsilon$. In the following derivatives, ∂_x acts on the expression $\mathfrak{g}(\sqrt{x},y,s)$

$$\partial_y\mathfrak{g} = \frac{2(1+s)[(1+s)(1+x)-4]}{[2+y(1-x)]^2}, \qquad \partial_x\mathfrak{g} = \frac{2y(1+s)[(1+s)-y(1-s)]}{[2+y(1-x)]^2}. \tag{C.21}$$

Since $|s| \leq \varepsilon < 1$ and $0 \leq x < 1$, we have $0 < 1+s < 2$ and $1 \leq 1+x < 2$. Hence, $(1+s)(1+x) - 4 < 0$. Therefore, $\partial_y\mathfrak{g} < 0$. Thus, the minimum over y occurs at $y = \alpha$. At this value of y, the denominator is independent of s and the numerator is a strictly concave quadratic in s with quadratic coefficient $2(\alpha x - 1) < 0$. The minimum over $[-\varepsilon,\varepsilon]$ therefore occurs at one of its endpoints. At $s = \varepsilon$, the derivative with respect to x is positive, so the minimum occurs at $x = v_{\min}^2$ and equals c_+. At $s = -\varepsilon$, the derivative has the sign of $(1-\varepsilon) - \alpha(1+\varepsilon)$. If this quantity is nonnegative, the minimum again occurs at $x = v_{\min}^2$. If it is negative, the infimum is approached as $x \uparrow 1$. These are the two cases in (C.15). Both are positive for $0 < \alpha < 1 - \varepsilon$, as is c_+, so $\dot{s} \geq c_* > 0$ wherever $y \leq \alpha$. Since $\dot{y} = (1-s)y > 0$, the part with $y \leq \alpha$ is an initial interval and the remainder is a final interval. Either may be empty. On the initial interval, s increases by at most 2ε, giving logarithmic length at most $2\varepsilon/c_*$. On a nonempty final interval, the initial value of y is at least α and the final value is at most $Y_{\max} = 2/(1+v_{\min}^2)$. Integrating $d\log y/d\sigma \geq 1-\varepsilon$ bounds its logarithmic length by $(1-\varepsilon)^{-1}\log(Y_{\max}/\alpha)$. This quantity is positive because $Y_{\max} > 1 > \alpha$, so the same combined estimate covers an empty portion as well. Adding the two bounds proves (C.17). It remains to compare c_* with c_0. Direct subtraction at the positive-slope endpoint gives

$$c_+ - c_0 = \frac{2\alpha v_{\min}^2(1+\varepsilon)[1+\varepsilon-\alpha(1-\varepsilon)]}{(2+\alpha)[2+\alpha(1-v_{\min}^2)]} > 0. \tag{C.22}$$

If $\alpha \le (1-\varepsilon)/(1+\varepsilon)$, monotonicity in x at $s=-\varepsilon$ gives $c_- \ge \mathfrak{g}(0,\alpha,-\varepsilon)$, using the continuous extension of the drift to $u=0$. Moreover, $\mathfrak{g}(0,\alpha,-\varepsilon)-c_0 = 4\alpha\varepsilon/(2+\alpha) > 0$. In the complementary case, the limiting value at $x=1$ gives

$$c_- - c_0 = \frac{\alpha(1+\varepsilon)[1+\varepsilon-\alpha(1-\varepsilon)]}{2+\alpha} > 0. \tag{C.23}$$

Thus, both c_+ and c_- exceed c_0, proving $c_* > c_0$. The logarithmic terms of $\mathcal{B}_*$ and $\mathcal{B}_+$ agree, while their first terms are $2\varepsilon/c_*$ and $2\varepsilon/c_0$. Evaluating their difference at the minimizer α_ε of $\mathcal{B}_+$ gives the stated positive gap from the earlier optimized bound. □

The gap δ is positive for each fixed $\varepsilon > 0$ and $v_{\min} > 0$ and depends on these parameters. The refinement minimizes the drift over an enclosure in which u, y and s vary independently, before adding two separate length estimates. Determining the sharp span for positive ℓ_0 and nonzero flatness tolerance still requires their coupled evolution. In particular, a segment with $s=\varepsilon$ must satisfy $w = -\mathfrak{g}(u,y,\varepsilon) \ge 0$ and the trajectory must remain below the density boundary $y = 2/(1+u^2)$. Any proposed maximizer must respect these constraints, any prescribed inner-edge data and the regularity needed for realization by the full field equation.

4. Approximation by strictly negative branches

For $\ell_0 < 0$, each nondegenerate trace satisfies a strict span inequality. We now show that this strictness does not reduce the supremum over the negative-parameter class. Analytic solutions approaching the $\ell_0 = 0$ branch have spans approaching the bound attained there.

Proposition C.1. *For every $0 < \varepsilon < 1$, the supremum of the logarithmic span over analytic solutions with $\ell_0 < 0$ satisfying Assumption C.1 is $2\operatorname{artanh}\varepsilon$. Each individual nondegenerate trace has strictly smaller span. A sequence of reflection-symmetric solutions with $v_{zz} = 0$ and $\ell_0 \uparrow 0$ approaches the supremum.*

Proof. Under the stated hypotheses, the density interval and the exact equatorial identity give $\partial_\eta \mathcal{F} > 0$ and

$$\dot{s} - (1-s^2) \ge -\frac{y(1+s)[3-u^2-(1+u^2)s]}{\partial_\eta \mathcal{F}} > 0 \tag{C.24}$$

when $\ell_0 < 0$. Indeed, $y < 0$ and $1+s > 0$, while the bracket exceeds $2(1-u^2) > 0$ because $s < 1$. Dividing by $1-s^2 > 0$ and integrating over the nondegenerate interval gives $\Lambda < \operatorname{artanh} s(R_2) - \operatorname{artanh} s(R_1) \le 2\operatorname{artanh}\varepsilon$. For the limiting construction, we set $T = \operatorname{artanh}\varepsilon$, choose $r_0, b_0 > 0$ and take $0 < a < 1/\cosh(T+b)$. With $\tau = \log(r/r_0)$, the $\ell_0 = 0$ solution is $u_0(\tau) = a\cosh tau$, $s_0(\tau) = \tanh\tau$ and $r = r_0 e^\tau$. The variables τ and σ differ by a constant, so their derivatives agree. The choice of a gives a strict subluminality margin on the larger compact interval $|\tau| \le T + b$. Its auxiliary field, independent of z, is

$$f_0(r) = 2\eta_0(r) = a\left(\frac{r^2}{r_0} + r_0\right). \tag{C.25}$$

Since both 1 and r^2 are $\mathcal{L}$-harmonic, $\mathcal{L} f_0 = 0$. Let us keep f_0 and the reference scale $\eta_* > 0$ fixed while varying ℓ_0. The defining function $\mathcal{F}(r,\eta)$ depends analytically on (r,η,ℓ_0) for $\eta > 0$, and its derivative with respect to η is 2 at $\ell_0 = 0$. The analytic implicit-function theorem, applied uniformly along the compact graph of η_0, therefore solves $\mathcal{F}(r,\eta_{\ell_0}(r)) = f_0(r)$ for all sufficiently small negative ℓ_0. The resulting fields are independent of z and solve the full exact velocity equation by (II.12). Their speeds $u_{\ell_0} = \eta_{\ell_0}/r$ and logarithmic slopes converge with their derivatives to u_0 and s_0 on the compact annulus. At $\ell_0 = 0$, we have $\eta_{0,r} = ar/r_0 > 0$ and density factor $\mathcal{N}_{u_0}(0) = 4u_0^2 > 0$. Compactness gives positive margins for both quantities, which persist for sufficiently small negative ℓ_0. The same holds for $\partial_\eta \mathcal{F}$ and subluminality. Local metric reconstruction on a simply connected meridional neighbourhood therefore gives positive-density dust solutions for these perturbed branches. At the limiting crossings $\tau = \pm T$, the slope equals $\pm\varepsilon$ and $ds_0/d\tau = 1-\varepsilon^2 > 0$. The implicit-function theorem consequently gives nearby crossing locations $\tau_\pm(\ell_0) \to \pm T$ with $s_{\ell_0}(\tau_\pm) = \pm\varepsilon$. Derivative convergence keeps the slope strictly increasing between them. Hence, the intervals with endpoints $R_\pm(\ell_0) = r_0 e^{\tau_\pm(\ell_0)}$ are ε-flat and their logarithmic spans are $\tau_+(\ell_0) - \tau_-(\ell_0) \to 2T$. Every constructed field has $v_{zz} = 0$. Choosing an increasing sequence of negative parameters tending to zero completes the proof. □

Appendix D: Edge calibration and large-radius consistency

In the main body of the present work, we derive asymptotic restrictions directly from the exact metric relations. This appendix compares those results with the near-edge approximation in Section 4 of [6]. We first test its constant-speed calibration in the exact and truncated velocity equations and then state the remainder hypothesis needed to draw a large-radius conclusion from the approximate logarithmic expressions. We consider the calibration

$$\ell_0 = \frac{av_{\rm c}}{R_{\rm G}}, \qquad v_0 = av_{\rm c} = \ell_0 R_{\rm G}, \tag{D.1}$$

where $R_{\rm G} > 0$ is the reference edge radius, $v_{\rm c}$ is a reference speed and a is dimensionless. we assume $0 < v_0 = av_{\rm c} < 1$. For the spatially constant field $v(r, z) \equiv v_0$, all derivative terms vanish in both velocity operators. Substitution therefore gives

$$\mathcal{T}_{\ell_0}[av_{\rm c}] = \mathcal{E}_{\ell_0}[av_{\rm c}] = \frac{2av_{\rm c}}{r^2}\left(\frac{r}{R_{\rm G}} - 1\right). \tag{D.2}$$

The residual vanishes only at $r = R_{\rm G}$. Introducing the dimensionless displacement $\delta = (r - R_{\rm G})/R_{\rm G}$ and expanding for $|\delta| \ll 1$ gives

$$\mathcal{T}_{\ell_0}[av_{\rm c}] = \mathcal{E}_{\ell_0}[av_{\rm c}] = \frac{2av_{\rm c}}{R_{\rm G}^2}\left[\delta - 2\delta^2 + O(\delta^3)\right]. \tag{D.3}$$

Thus, replacing the factors $r/R_{\rm G}$ by one retains the zeroth-order edge relation. An expansion beyond that order must account for the linear residual through spatially dependent corrections. This calculation concerns a field that is constant in both r and z. A flat equatorial trace can still have transverse dependence that supplies the missing derivative terms, as in the local constructions of the main text. Its density restriction under the same calibration is given in Appendix C.

1. The conditional large-radius test

The coordinate angular velocity of the dust, $\Omega = d\phi/dt$, differs from the ZAMO angular velocity $\Omega_{\rm Z} = -g_{t\phi}/g_{\phi\phi}$. To evaluate both in the approximation, we follow Eqs. (4.7)-(4.8) in [6] and take $C = 1$ in $H = -Ce^{\ell_0\eta}$ with $v = v_0$ and constant ℓ_0. For $\ell_0 \neq 0$, the additive constant in the retained expression for Ω can be represented by a fixed reference scale $r_0 > 0$. The resulting expressions are

$$\Omega^{(1)}(r) = -\frac{\ell_0}{2}\log\left(\frac{r}{r_0}\right), \qquad g_{tt}^{(1)} = -1, \quad g_{t\phi}^{(1)} = \frac{\ell_0 r^2}{2}\log\left(\frac{r}{r_0}\right) + v_0 r, \quad g_{\phi\phi}^{(1)} = r^2. \tag{D.4}$$

The superscript denotes the retained approximation, including its zeroth-order terms. Under (D.1), it is organized as a small-v_0 expansion at fixed $r/R_{\rm G}$ and $r_0/R_{\rm G}$. Substitution into the definition of the ZAMO angular velocity gives

$$\Omega_{\rm Z}^{(1)} = -\frac{\ell_0}{2}\log\left(\frac{r}{r_0}\right) - \frac{v_0}{r}. \tag{D.5}$$

In particular, $\Omega^{(1)} - \Omega_{\rm Z}^{(1)} = v_0/r$, in agreement with the speed relation at this order of approximation. These formulas describe the approximation at fixed radial scales. Their use on an unbounded end requires control of the omitted terms. For the following test, ℓ_0 and v_0 may be independent constants. The comparison is made in the stationary and axial coordinates fixed by the asymptotic normalization.

Proposition D.1. *Let us suppose that a stationary, axisymmetric metric satisfies the asymptotically nonrotating equatorial falloff specified in Section VI. Moreover, we assume that, in the same coordinates, its exact mixed component satisfies*

$$g_{t\phi}(r, 0) = \frac{\ell_0 r^2}{2}\log\left(\frac{r}{r_0}\right) + v_0 r + o\left(r^2\log\left(\frac{r}{r_0}\right)\right) \tag{D.6}$$

as $r \to \infty$ with fixed $r_0 > 0$ and finite constants ℓ_0 and v_0. Then, $\ell_0 = 0$.

Proof. We divide (D.6) by $r^2 \log(r/r_0)$. The linear term contributes $v_0/[r\log(r/r_0)] \to 0$ and the normalized remainder tends to zero by hypothesis. Hence, we have

$$\lim_{r\to\infty} \frac{g_{t\phi}(r,0)}{r^2 \log(r/r_0)} = \frac{\ell_0}{2}. \tag{D.7}$$

The azimuthal falloff lemma in the main text gives $g_{t\phi}(r,0) = O(1)$ along the equatorial tail. The same limit is therefore zero, proving $\ell_0 = 0$. □

The same equatorial falloff gives $g_{\phi\phi}/r^2 \to 1$. Hence, the ansatz also implies $\Omega_{\rm Z}/\log(r/r_0) \to -\ell_0/2$, whereas the falloff requires $\Omega_{\rm Z} = O(r^{-2})$. The mixed-component and shift tests thus give the same condition. When ℓ_0 and v_0 are independent, this remainder hypothesis alone places no restriction on v_0. A remainder containing $-v_0 r$ still satisfies $o(r^2\log(r/r_0))$ and can cancel the linear term. If the displayed mixed-component formula is extrapolated without a remainder or with the stronger remainder $o(r)$, the bound $g_{t\phi} = O(1)$ also forces $v_0 = 0$. Under the specific calibration (D.1), this already follows from $v_0 = \ell_0 R_{\rm G}$. The need for an independent remainder estimate can be seen directly from the expansion of H. Evaluating its exact functional dependence on the prescribed constant profile gives, with $C = 1$

$$H = -e^{\ell_0 v_0 r} = -\left[1 + \xi(r) + O(\xi(r)^2)\right], \qquad \xi(r) = \ell_0 v_0 r, \tag{D.8}$$

where the Taylor expansion is taken at $\xi = 0$ and is used for $|\xi| \ll 1$. The parameter ξ is dimensionless and is distinct from the flatness tolerance used in the span estimates. For fixed $\ell_0 v_0 \neq 0$, its magnitude grows without bound as $r \to \infty$. Thus, the small-parameter expansion cannot remain uniformly valid along the constant profile on an unbounded end. The mixed term supplies a further measure of the approximation's range. For an exact metric with $g_{\phi\phi} > 0$, the unit azimuthal vector is $e_{\hat\phi} = m/\sqrt{g_{\phi\phi}}$ and its contraction with the stationary field is $g(k, e_{\hat\phi}) = g_{t\phi}/\sqrt{g_{\phi\phi}}$. At this order of approximation, such a quantity is

$$\frac{g^{(1)}_{t\phi}}{\sqrt{g^{(1)}_{\phi\phi}}} = \frac{\ell_0 r}{2} \log\left(\frac{r}{r_0}\right) + v_0. \tag{D.9}$$

For nonzero ℓ_0, it grows without bound, whereas the equatorial falloff requires the exact contraction to be $O(r^{-1})$. Its growth also shows why smallness of $|\xi|$ alone is insufficient to control all metric corrections. Applying Proposition D.1 therefore requires establishing the stated remainder estimate for the exact mixed component. If omitted terms change its leading large-radius behaviour, the logarithmic formula at this order of approximation does not determine the asymptotics. Theorem 10 derives the restrictions directly from the exact metric relations. Under its regularity and uniform subluminality hypotheses, a nonzero constant-ℓ_0 branch with positive speed on an asymptotically flat equatorial tail must instead have inverse-radius velocity decay.

Appendix E: Additional junction formulas

We keep the boundary identification and normal convention of Section VII. The symbols $-$ and $+$ denote the dust interior and vacuum exterior, respectively. Both unit normals point from the interior towards the exterior and $[Q] = Q^+ - Q^-$ denotes the jump after pulling both quantities back to the common timelike boundary Σ. Equality of the induced metrics is assumed throughout this appendix. In the intrinsic basis $(\partial_t, \partial_\phi, e_s)$, the common Killing block is G_{AB}, with $A, B \in \{t, \phi\}$ and e_s is the unit meridional tangent defined by proper arclength along the boundary curve. Thus, $h(e_s, e_s) = 1$ and e_s is orthogonal to the Killing directions. With $K(X,Y) = g(\nabla_X n, Y)$ and $K = h^{ab}K_{ab}$, Israel's relation is

$$8\pi G_{\rm N} S_{ab} = -[K_{ab}] + h_{ab}[K]. \tag{E.1}$$

Separating the Killing and meridional components gives the surface stresses directly in terms of the mismatched boundary data.

Corollary E.1. *Let us define $\Delta_{AB} = [K_{AB}]$, $\Delta_s = [K_{ss}]$ and $\Delta_G = G^{AB}\Delta_{AB}$, where G^{AB} is the inverse of the common Killing block and $K_{ss} = K(e_s, e_s)$. Then, the surface stress-energy tensor has components $8\pi G_{\rm N} S_{AB} = -\Delta_{AB} + G_{AB}(\Delta_G + \Delta_s)$, $S_{As} = 0$ and $8\pi G_{\rm N} S_{ss} = \Delta_G$. It vanishes if and only if $\Delta_{AB} = 0$ and $\Delta_s = 0$.*

Proof. In the common proper-arclength basis, $h_{AB} = G_{AB}$, $h_{As} = 0$ and $h_{ss} = 1$. Hence, $[K] = \Delta_G + \Delta_s$ and substitution into (E.1) gives the stated Killing and meridional components. The mixed components vanish because $K^{\pm}_{As} = 0$ in the circular geometry. To establish the vanishing criterion, we contract Israel's relation with h^{ab}. Since the boundary is three-dimensional, $h^{ab}h_{ab} = 3$, so $8\pi G_{\mathrm{N}} h^{ab} S_{ab} = -[K] + 3[K] = 2[K]$. Thus, $S_{ab} = 0$ first gives $[K] = 0$ and Israel's relation then gives $[K_{ab}] = 0$. These are precisely $\Delta_{AB} = 0$ and $\Delta_s = 0$, since the mixed jumps already vanish. Conversely, these conditions make every displayed component of S_{ab} zero. □

The physical interpretation of the Killing components requires a normalized observer frame. Let w be the unit future-directed four-velocity of a circular observer on Σ and $e_{(\phi)}$ be a unit spacelike vector in the Killing plane orthogonal to w. The measured surface energy density, azimuthal momentum density and azimuthal stress are $S(w, w)$, $-S(w, e_{(\phi)})$ and $S(e_{(\phi)}, e_{(\phi)})$, respectively. The meridional stress is $S(e_s, e_s) = S_{ss}$. These projections determine the measured quantities from the coordinate components, including S_{tt} and $S_{t\phi}$. A finite jump in the volume density does not by itself imply a surface layer. The surface tensor is determined by the jump in the second fundamental form. We now specialize to a cylindrical boundary portion $r = R_\Sigma > 0$ in the interior. Let us assume that the exterior admits canonical Weyl coordinates (ϖ, z_+) near the boundary with orbit metric $e^{\mu_+}(d\varpi^2 + dz_+^2)$. We write the interior orbit metric as $e^{\mu_-}(dr^2 + dz_-^2)$. Since the common Killing basis is preserved, equality of G_{AB} gives equality of the orbit-area radii and places the exterior boundary at $\varpi = R_\Sigma$. Let us choose the common normal orientation towards increasing canonical radius. The generating curves are straight in their respective meridional coordinate planes, so their Euclidean curvatures vanish. Let ξ be a common intrinsic parameter with regular parameterizations $z_\pm = z_\pm(\xi)$ and $dz_\pm/d\xi \neq 0$. The complete Darmois conditions are

$$G^+_{AB} = G^-_{AB}, \tag{E.2}$$

$$e^{\mu_+}\left(\frac{dz_+}{d\xi}\right)^2 = e^{\mu_-}\left(\frac{dz_-}{d\xi}\right)^2, \tag{E.3}$$

$$e^{-\mu_+/2}\partial_\varpi G^+_{AB} = e^{-\mu_-/2}\partial_r G^-_{AB}, \tag{E.4}$$

$$e^{-\mu_+/2}\partial_\varpi \mu_+ = e^{-\mu_-/2}\partial_r \mu_-. \tag{E.5}$$

All quantities are evaluated at the identified boundary points. Radial derivatives are taken at fixed longitudinal coordinate on the corresponding side before restriction to the boundary. The Killing part of the induced metric gives (E.2), while equality of the meridional line elements gives (E.3). The latter ensures that $d\lambda_{\mathrm{p}} = e^{\mu_\pm/2}|dz_\pm/d\xi|d\xi$ defines the same proper arclength on both sides. The unit normals are $n^- = e^{-\mu_-/2}\partial_r$ and $n^+ = e^{-\mu_+/2}\partial_\varpi$. Hence, the symmetry-reduced formula for the second fundamental form gives

$$K^-_{AB} = \frac{1}{2}e^{-\mu_-/2}\partial_r G^-_{AB}, \qquad K^+_{AB} = \frac{1}{2}e^{-\mu_+/2}\partial_\varpi G^+_{AB}. \tag{E.6}$$

For the unit meridional tangent, the vanishing Euclidean curvature gives

$$K^-_{ss} = \frac{1}{2}e^{-\mu_-/2}\partial_r \mu_-, \qquad K^+_{ss} = \frac{1}{2}e^{-\mu_+/2}\partial_\varpi \mu_+. \tag{E.7}$$

Equality of these components yields (E.4) and (E.5). Since $K^\pm_{As} = 0$, no further components remain to be matched. The full matching conditions also restrict the longitudinal boundary map. Jacobi's formula and the canonical identities $\det G^- = -r^2$ and $\det G^+ = -\varpi^2$ give $G^{AB}\partial_r G^-_{AB} = G^{AB}\partial_\varpi G^+_{AB} = 2/R_\Sigma$ on the boundary. Contracting (E.4) with the common inverse G^{AB} therefore yields $2e^{-\mu_+/2}/R_\Sigma = 2e^{-\mu_-/2}/R_\Sigma$. Thus, $\mu_+ = \mu_-$ follows from the complete cylindrical matching conditions themselves. Hence, (E.3) reduces to $|dz_+/d\xi| = |dz_-/d\xi|$. Regularity makes the relative sign constant on each connected boundary segment, so $z_+(\xi) = \pm z_-(\xi) + z_{\mathrm{c}}$, where z_{c} is constant. After aligning the longitudinal orientations and coordinate origins, a common reparameterization can set $z_+(\xi) = z_-(\xi) = -\xi$. This familiar identification is therefore a convenient coordinate choice once the full matching conditions hold. These formulas specify the complete local geometry required at a cylindrical interface. Constructing a vacuum metric that realizes the boundary data and satisfies the global completion conditions remains a separate problem.

Appendix F: Nonlinear exterior compatibility

The linearized Ernst theorem concerns a fixed smooth bounded domain away from the axis. An isolated exterior requires additional conditions at infinity and on every axis segment together with the periods needed for global metric reconstruction. Its twist periods must also agree with the interior matter current. As allowed by Definition 2, a nonzero twist period can be represented by local twist potentials with constant differences on overlaps, by prescribed jumps across cuts or directly by the closed twist one-form. To compare interior and exterior data, we fix the stationary and axial normalizations, the common normal orientation and compatible additive normalizations for the local twist potentials. Moreover, we pull both sets of boundary data back to a fixed reference hypersurface Σ. If the material boundary is allowed to vary, its embedding and the boundary identification must be included among the configuration variables. This places the matching conditions on a common domain even when the boundary shape changes. Let us suppose that, after these choices and the gauge have been fixed, the interior and exterior solution spaces $\mathcal{M}_{\mathrm{dust}}$ and $\mathcal{M}_{\mathrm{vac}}$ admit Banach-manifold structures. Let d and v denote complete interior and exterior configurations, respectively, and let $\mathbf{D}$ and $\mathbf{V}$ be their boundary maps into a common Banach manifold $\mathcal{B}_\Sigma$ of boundary data. The elements of this space specify the induced metric, the full second fundamental form and the twist periods, subject to their geometric compatibility relations. In particular, the specified twist periods must equal the integrals of the boundary twist one-form around the corresponding cycles. This relation must be incorporated into the definition of the data space. For the interior map $\mathbf{D}$, the periods of a toroidal dust body are determined by the matter-current relation in (VIII.6). Let us assume that a matched configuration (d_0, v_0) is available and that the boundary maps are continuously differentiable near it. We choose one local chart on $\mathcal{B}_\Sigma$ around their common value $\mathbf{D}(d_0) = \mathbf{V}(v_0)$ and use the same symbols for the chart representations of the maps. Matching without a surface layer, together with the period compatibility, is then expressed by $\mathcal{C}(d, v) = \mathbf{D}(d) - \mathbf{V}(v) = 0$ with $(d, v) \in \mathcal{M}_{\mathrm{dust}} \times \mathcal{M}_{\mathrm{vac}}$. Its derivative at the matched configuration is $D\mathcal{C}_{(d_0,v_0)}(\delta d, \delta v) = D\mathbf{D}_{d_0}\ \delta d - D\mathbf{V}_{v_0}\ \delta v$, where $(\delta d, \delta v)$ belongs to the tangent space of $\mathcal{M}_{\mathrm{dust}} \times \mathcal{M}_{\mathrm{vac}}$ at (d_0, v_0). Let us suppose this derivative is Fredholm and surjective. Its kernel is then finite-dimensional and admits a closed complementary subspace. The restriction of the derivative to that complement is a bounded linear isomorphism onto the target Banach space. The implicit-function theorem therefore solves the matching equation locally for the complementary variables as a C^1 function of the kernel variables. Consequently, the nearby matched configurations form a finite-dimensional C^1 manifold with tangent space $\ker D\mathcal{C}_{(d_0,v_0)}$. This manifold may be zero-dimensional. To apply this argument, we must establish the assumed solution spaces and mapping properties for the chosen exterior problem. This includes specifying function spaces that control decay at infinity and regularity at the axis, fixing the gauge and identifying admissible variations of the interior, exterior, and boundary. Variations of the twist periods must remain consistent with the interior current and the boundary data must include the orbit-area and meridional metric information alongside the local Ernst variables. Prescribing both boundary values and normal derivatives does not by itself establish that the resulting matching derivative is Fredholm or surjective. The main text provides a compatibility test for linearized Ernst data about Minkowski space on a bounded domain. The conditional argument above describes how a suitable matched configuration could generate nearby nonlinear solutions once the required mapping properties have been proved. Constructing the first such configuration and establishing these properties remain open for the zero-density boundaries considered here.

Appendix G: Observer dependence and spectroscopic prediction

The frequency-ratio formula in Section IX relates the measured shift to the emitter and observer motions and the connecting null ray. We first show explicitly why the local ZAMO-speed field does not determine this shift. We then describe the additional data and estimates needed to compare a spectroscopic rotation profile with the equatorial trace used in the span theorems.

Proposition G.1. *The ZAMO-speed field v alone does not determine the frequency shift for all admissible choices of circular emitters, circular observers and connecting null rays.*

Proof. We fix the metric, the emitter, a connecting null ray and its reception event x_{o} in a region where the ZAMO frame is defined. Let us choose an orthonormal basis $(e_{\hat{0}}, e_{\hat{\phi}})$ of the Lorentzian Killing plane at x_{o} with $e_{\hat{0}}$ future-directed and $e_{\hat{\phi}}$ spacelike. A circular observer with rapidity β has unit four-velocity

$u(\beta) = \cosh\beta\, e_{\hat{0}} + \sinh\beta\, e_{\hat{\phi}}$. Moreover, we keep the photon wave vector p fixed, including its normalization at emission. The frequency measured by this observer is $\omega_{\mathrm{o}}(\beta) = c_0 \cosh\beta + c_1 \sinh\beta$ with $c_0 = -g(p, e_{\hat{0}}) > 0$ and $c_1 = -g(p, e_{\hat{\phi}})$. Since p is nonzero and future-directed, $\omega_{\mathrm{o}}(\beta) > 0$. Differentiation gives $\omega_{\mathrm{o}}''(\beta) = \omega_{\mathrm{o}}(\beta) > 0$, so the measured frequency cannot be constant on any open interval of rapidities. Changing the receiving observer can therefore change $1 + z_{\mathrm{obs}} = \omega_{\mathrm{e}}/\omega_{\mathrm{o}}$ while leaving v, the emitter and the ray fixed. □

For an emitter comoving with the dust, the local Doppler relation also displays the dependence on emission direction. If the emitted line has fixed rest-frame frequency $\omega_{\mathrm{e}} > 0$, then

$$\omega_{\mathrm{Z}} = \frac{\omega_{\mathrm{e}}}{\Gamma(1 - v\cos\alpha)}. \tag{G.1}$$

Here, $\Gamma = (1 - v^2)^{-1/2}$ and α is the photon propagation angle relative to the positive azimuthal direction in the ZAMO frame. For $0 < v < 1$, the denominator is positive and ω_{Z} varies with α even when v is fixed. The directions actually received are selected by the null-geodesic boundary-value problem for the completed metric and the specified viewing geometry. Although v determines $\eta = rv$ in the canonical coordinates, metric reconstruction also uses H and the integration constants, including the angular velocity normalization. Propagation to a distant observer further requires the exterior geometry. The proposition establishes dependence on these additional physical inputs. Proving nonuniqueness while holding the complete observational setup fixed would require a separate analysis of the global Einstein-dust problem. To specify the asymptotic observer limit, let the reception events x_{o} tend to infinity in an asymptotically flat end with k normalized to the future unit time translation there. Sufficient conditions for circular observers to approach rest in this frame are

$$g(k,k)|_{x_{\mathrm{o}}} \to -1, \qquad \Omega_{\mathrm{o}}^2 g_{\phi\phi}|_{x_{\mathrm{o}}} \to 0, \qquad \Omega_{\mathrm{o}} g_{t\phi}|_{x_{\mathrm{o}}} \to 0. \tag{G.2}$$

Indeed, their normalization satisfies $U_{\mathrm{o}}^{-2} = -\left(g_{tt} + 2\Omega_{\mathrm{o}} g_{t\phi} + \Omega_{\mathrm{o}}^2 g_{\phi\phi}\right)\big|_{x_{\mathrm{o}}} \longrightarrow 1$, so $U_{\mathrm{o}} \to 1$. When the frequency ratio is expressed using $b = L/E$ with $E \neq 0$, the additional condition $\Omega_{\mathrm{o}} b \to 0$ makes the observer factor $U_{\mathrm{o}}(1 - \Omega_{\mathrm{o}} b)$ tend to unity. This condition must be checked for the chosen rays, particularly when b varies along the family of reception events. A complete prediction can be organized as the forward map $(g, u_{\mathrm{e}}, u_{\mathrm{o}}, \mathcal{S}, \iota, \mathcal{P}) \longmapsto \{\gamma_\Theta\} \longmapsto z_{\mathrm{obs}}(\Theta)$. Here, Θ labels detector-screen position, $\mathcal{S}$ specifies the source geometry and emission model and ι is the viewing inclination relative to the symmetry axis. The prescription $\mathcal{P}$ specifies how measured line frequencies are converted into a shift and an inferred rotation speed. The family $\{\gamma_\Theta\}$ consists of the connecting rays and g includes the exterior and any transition region or surface layer. The source model must identify the emission event or specify the emission along each ray. If several emission contributions are unresolved, $\mathcal{P}$ must specify how their spectra are combined. An approximate propagation model must control errors in both the source positions and the inferred frequency shifts. Comparisons of profile slopes also require control of the corresponding derivatives. To compare with the span theorems, suppose the observations determine a positive differentiable profile $v_{\mathrm{spec}}(R_{\mathrm{obs}})$ and a differentiable identification $R_{\mathrm{obs}} = \mathscr{R}(r) > 0$ with an equatorial emission interval. Let us assume that this identification is increasing, with controlled logarithmic distortion

$$0 < c_- \leq \frac{d\log\mathscr{R}(r)}{d\log r} \leq c_+ < \infty. \tag{G.3}$$

A sufficient slope comparison would then have the form

$$\left| \frac{d\log v_{\mathrm{spec}}}{d\log R_{\mathrm{obs}}} - \frac{d\log v(r,0)}{d\log r} \right| \leq \delta, \tag{G.4}$$

where $\delta \geq 0$ and both slopes are evaluated at corresponding radii $R_{\mathrm{obs}} = \mathscr{R}(r)$. In particular, a spectroscopic slope bound $|d\log v_{\mathrm{spec}}/d\log R_{\mathrm{obs}}| \leq \varepsilon_{\mathrm{spec}}$ would imply the intrinsic bound $|d\log v(r,0)/d\log r| \leq \varepsilon_{\mathrm{spec}} + \delta$. The span theorems could then be applied if $\varepsilon_{\mathrm{spec}} + \delta < 1$ and their remaining physical and geometric hypotheses, including nonnegative density and transverse concavity, were verified. Integrating the bounds on $d\log\mathscr{R}/d\log r$ would convert the resulting intrinsic radial-span estimate into an observed one. Establishing these comparison estimates requires the completed geometry and the viewing configuration.

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